

# Notes on Functional Analysis\*

Mengxi Wu  
University of Southern California

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\* Author Email: mengxiwu@usc.edu. The first draft of these notes was written during Math 525b (Spring 2025), taught by Professor Calum Rickard.

# Metric Spaces and Normed Vector Spaces

## Metric Space

A metric on a set  $M$  is a function  $\rho : M \times M \rightarrow [0, \infty)$  such that for all  $x, y, z \in M$ :

1.  $\rho(x, y) = \rho(y, x)$  (Symmetry).
2.  $\rho(x, y) \leq \rho(x, z) + \rho(z, y)$  (Triangle Inequality).
3.  $\rho(x, y) = 0 \iff x = y$  (Identity of Indiscernibles).

## Example

Let  $M = \mathbb{R}^p$ . For  $x = (x_1, \dots, x_p) \in \mathbb{R}^p$  and  $y = (y_1, \dots, y_p) \in \mathbb{R}^p$ , we can define different metrics:

$$\begin{aligned}\rho_2(x, y) &= \sqrt{(x_1 - y_1)^2 + \dots + (x_p - y_p)^2}, \\ \rho_1(x, y) &= |x_1 - y_1| + \dots + |x_p - y_p|, \quad \rho_\infty(x, y) = \max_{1 \leq i \leq p} |x_i - y_i|.\end{aligned}$$

## Convergence

A sequence  $x_n \in M$  converges to  $x \in M$  if  $\lim_{n \rightarrow \infty} \rho(x_n, x) = 0$ . We write  $\lim_{n \rightarrow \infty} x_n = x$  or  $x_n \rightarrow x$  as  $n \rightarrow \infty$ .

**Note.** For the standard metrics on  $\mathbb{R}^p$  ( $\rho_1, \rho_2, \rho_\infty$ ), a sequence of vectors  $x^n = (x_1^n, \dots, x_p^n)$  converges to  $x = (x_1, \dots, x_p)$  if and only if each component converges individually, i.e.,  $x_i^n \rightarrow x_i$  as  $n \rightarrow \infty$  for all  $1 \leq i \leq p$ .

## Example

Let  $M = C([0, 1])$ , the space of continuous functions on  $[0, 1]$ . We can define:

$$\rho_1(x, y) = \int_0^1 |x(t) - y(t)| dt, \quad \rho_\infty(x, y) = \sup_{0 \leq t \leq 1} |x(t) - y(t)|.$$

Let  $K = \rho_\infty(x, y)$ . Then:

$$\rho_1(x, y) = \int_0^1 |x(t) - y(t)| dt \leq \int_0^1 K dt = K = \rho_\infty(x, y).$$

Note that since  $\rho_1(x, y) \leq \rho_\infty(x, y)$ ,  $\rho_\infty$ -convergence (uniform convergence) implies  $\rho_1$ -convergence (convergence in the  $L^1$  norm), but the converse is not necessarily true.

## Cauchy Sequence

A sequence  $x_n \in M$  is a Cauchy sequence if  $\rho(x_n, x_m) \rightarrow 0$  as  $n, m \rightarrow \infty$ .

### Proposition 1

If  $x_n \rightarrow x$  in  $M$ , then  $(x_n)$  is a Cauchy sequence.

**Proof.** By the triangle inequality,  $\rho(x_n, x_m) \leq \rho(x_n, x) + \rho(x_m, x) \rightarrow 0$  as  $n, m \rightarrow \infty$ .  $\square$

### Completeness of Metric Space

A metric space  $(M, \rho)$  is complete if every Cauchy sequence inside it converges to a limit in  $M$ . That is, if  $\rho(x_n, x_m) \rightarrow 0$  as  $n, m \rightarrow \infty$ , then there exists some  $x \in M$  such that  $x_n \rightarrow x$ .

### Example

- $M = \mathbb{R}$  is complete, but the rational numbers  $\mathbb{Q} \subset \mathbb{R}$  (equipped with the subspace metric) are not complete.
- $M = L^1([0, 1])$  with the metric  $\rho_1(x, y) = \int_0^1 |x(t) - y(t)| dt$  is complete.
- The space of continuous functions  $C([0, 1])$  viewed as a subspace of  $L^1([0, 1])$  under the  $\rho_1$  metric is not complete.

**Remark.** For example c, recall that every  $x \in L^1$  can be approximated by a sequence of continuous functions  $x_n \in C([0, 1])$  such that  $\int_0^1 |x_n(t) - x(t)| dt \rightarrow 0$ . Consider the specific discontinuous step function:

$$x(t) = \begin{cases} 0 & \text{if } 0 \leq t < \frac{1}{2} \\ 1 & \text{if } \frac{1}{2} \leq t \leq 1 \end{cases} \in L^1([0, 1]).$$

We can approximate it using the following continuous piecewise linear functions  $x_n(t) \in C([0, 1])$ :

$$x_n(t) = \begin{cases} 0 & \text{if } 0 \leq t < \frac{1}{2} \\ n(t - \frac{1}{2}) & \text{if } \frac{1}{2} \leq t < \frac{1}{2} + \frac{1}{n} \\ 1 & \text{if } \frac{1}{2} + \frac{1}{n} \leq t \leq 1 \end{cases}.$$

Then we can compute the distance:

$$\int_0^1 |x_n(t) - x(t)| dt = \frac{1}{2n} \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

This shows that the sequence  $x_n$  is Cauchy with respect to  $\rho_1$ , but its limit  $x \notin C([0, 1])$ . Thus, the space is not complete.

### Dense Set

A subset  $B \subseteq M$  is dense in  $M$  if every element  $x \in M$  can be approximated by elements from  $B$ . That is, for any  $x \in M$ , there exists a sequence  $x_n \in B$  such that  $x_n \rightarrow x$ .

### Theorem 1 (Completion Theorem)

For any metric space  $(M, \rho)$ , there exists a complete metric space  $(\tilde{M}, \tilde{\rho})$  such that  $M$  is isometric to a dense subset  $B \subseteq \tilde{M}$ . Specifically, there exists a distance-preserving bijection  $h : M \rightarrow B$  satisfying:

$$\rho(x, y) = \tilde{\rho}(h(x), h(y)), \quad \forall x, y \in M.$$

This space  $(\tilde{M}, \tilde{\rho})$  is uniquely defined up to isometry and is called the *completion* of  $(M, \rho)$ .

**Remark.** *Isometric means distance-preserving.* When we say that a map  $h : M \rightarrow B$  is an isometry, it means that the distance between any two points in  $M$  is exactly equal to the distance between their corresponding images in  $B$  under the mapping  $h$ . In Theorem 1 above, because distances are preserved perfectly, both  $h$  and its inverse  $h^{-1} : B \rightarrow M$  are automatically continuous.

### Example

1.  $\mathbb{R}$  is the completion of  $\mathbb{Q}$  under the standard absolute value metric.
2.  $L^1([0, 1])$  is the completion of  $C([0, 1])$  with respect to the integral metric  $\rho_1(x, y) = \int_0^1 |x(t) - y(t)| dt$ .

### Equivalence Relation on Cauchy Sequences

If  $(x_n)$  and  $(y_n)$  are two Cauchy sequences in  $M$ , we say they are equivalent, denoted  $(x_n) \sim (y_n)$ , if:

$$\lim_{n \rightarrow \infty} \rho(x_n, y_n) = 0.$$

### Continuous Functions

Let  $f : (M, \rho) \rightarrow (Y, \nu)$  be a function between two metric spaces.

1.  $f$  is continuous at a point  $x_0 \in M$  if whenever  $x_n \rightarrow x_0$  in  $M$ , we have  $f(x_n) \rightarrow f(x_0)$  in  $Y$ . Equivalently, for any  $\epsilon > 0$ , there exists a  $\delta > 0$  such that:

$$\nu(f(x), f(x_0)) \leq \epsilon \quad \text{whenever} \quad \rho(x, x_0) \leq \delta.$$

We write  $\lim_{x \rightarrow x_0} f(x) = f(x_0)$ .

2.  $f$  is continuous if it is continuous at every point  $x \in M$ .

### Open and Closed Balls

The open ball  $B_\delta(x)$  and closed ball  $B_\delta[x]$  centered at  $x$  with radius  $\delta > 0$  are defined as:

$$B_\delta(x) = \{y \in M \mid \rho(x, y) < \delta\}, \quad B_\delta[x] = \{y \in M \mid \rho(x, y) \leq \delta\}.$$

### Open Set

A subset  $A \subseteq M$  is open if every point in  $A$  has some wiggle room around it inside  $A$ . That is,  $\forall x \in A$ , there exists some  $\delta > 0$  such that  $B_\delta(x) \subseteq A$ .

## Closed Set

A subset  $F \subseteq M$  is closed if its complement  $F^c = M \setminus F$  is an open set.

## Theorem 2

Let  $\mathcal{T}$  be the collection of all open subsets of  $M$ . Then:

1.  $M \in \mathcal{T}$  and  $\emptyset \in \mathcal{T}$ .
2.  $\mathcal{T}$  is closed under arbitrary (finite or infinite) unions.
3.  $\mathcal{T}$  is closed under finite intersections.

**Remark.** Notice that the empty set  $\emptyset$  and the whole space  $M$  satisfy the conditions for being both open and closed at the same time! The collection  $\mathcal{T}$  is formally called the *topology* of  $M$ .

## Corollary 1

Let  $\mathcal{F}$  be the set of all closed subsets of  $M$ . Then  $\mathcal{F}$  contains  $M$  and  $\emptyset$ , and it is closed under arbitrary intersections and finite unions.

## Neighborhood

A subset  $A \subseteq M$  is a neighborhood of a point  $x$  if  $x \in A$  and there is enough room to fit an open ball around  $x$  inside  $A$ , i.e.,  $B_\delta(x) \subseteq A$  for some  $\delta > 0$ .

## Interior Point (Inner Point)

A point  $x \in A$  is called an interior point of  $A$  if there exists an open ball centered at  $x$  completely contained in  $A$ :  $B_\delta(x) \subseteq A$  for some  $\delta > 0$ .

**Fact.** The set of all interior points of  $A$  (denoted  $A^\circ$  or  $\text{int}(A)$ ) is always an open set.

## Closure

Given a subset  $A \subseteq M$ , its closure  $\overline{A}$  is defined as the smallest closed set that contains  $A$ .

## Theorem 3

The closure can also be described sequentially:

$$\overline{A} = \{x \in M \mid \exists \{x_n\} \subseteq A \text{ such that } \lim_{n \rightarrow \infty} x_n = x\}.$$

## Corollary 2

A subset  $F \subseteq M$  is closed if and only if  $\overline{F} = F$ . Equivalently,  $F$  is closed if and only if whenever a sequence  $\{x_n\} \subseteq F$  converges to some  $x \in M$ , the limit  $x$  must also belong to  $F$ .

## Normed Vector Spaces (nvs)

A normed vector space  $E$  is a vector space over a field (usually  $\mathbb{R}$  or  $\mathbb{C}$ ) equipped with a norm function  $\|\cdot\|_E: E \rightarrow [0, \infty)$  satisfying:

1.  $\|\alpha x\|_E = |\alpha| \|x\|_E$  (Absolute Homogeneity).
2.  $\|x + y\|_E \leq \|x\|_E + \|y\|_E$  (Triangle Inequality).
3.  $\|x\|_E = 0 \iff x = 0$  (Definiteness).

**Note.** Every norm automatically induces a metric on the space by defining the distance between two vectors as the norm of their difference:  $\rho(x, y) = \|x - y\|_E$ .

### Triangle Inequality for Induced Metric

$$\|x - y\|_E = \|(x - z) + (z - y)\|_E \leq \|x - z\|_E + \|z - y\|_E.$$

### Example

1. On  $\mathbb{R}^d$ , the standard metrics we saw earlier come directly from these norms: for a vector  $x = (x_1, \dots, x_d) \in \mathbb{R}^d$ ,

$$\|x\|_2 = \sqrt{x_1^2 + x_2^2 + \dots + x_d^2}, \quad \|x\|_1 = |x_1| + \dots + |x_d|, \quad \|x\|_\infty = \max_{1 \leq i \leq d} |x_i|.$$

2. For the space of functions  $E = C([0, 1])$ , the corresponding norms are:

$$\|x\|_1 = \int_0^1 |x(t)| dt, \quad \|x\|_\infty = \sup_{0 \leq t \leq 1} |x(t)|.$$

## Completeness of Normed Vector Space

If a normed vector space  $E$  is complete under its induced metric, it is called a *Banach space*.

### Example

1.  $\mathbb{R}^d$  paired with any of the norms ( $\|x\|_1, \|x\|_2, \|x\|_\infty$ ) is a Banach space.
2.  $C([0, 1])$  is a Banach space when equipped with the supremum norm  $\|x\|_\infty$ , but it is *not* a Banach space under the  $\|x\|_1$  norm.
3.  $L^1([0, 1])$  with the norm  $\|x\|_1 = \int_0^1 |x(t)| dt$  is a complete Banach space, and  $C([0, 1])$  forms a dense subset within it.
4.  $\mathbb{Q}^d$  is not a Banach space because the scalar field  $\mathbb{Q}$  contains gaps.

## Linear Operators

Let  $E$  and  $F$  be two normed vector spaces, and let  $T : E \rightarrow F$  be a linear mapping. The following three conditions are completely equivalent:

- a.  $T$  is continuous at the origin 0.
- b.  $T$  is continuous everywhere on  $E$ .
- c.  $T$  is bounded: There exists a constant  $C > 0$  such that  $\|T(x)\|_F \leq C\|x\|_E$  for all  $x \in E$ .

**Notation.** We denote by  $\mathcal{L}(E, F)$  the set of all bounded linear operators from  $E$  to  $F$ .

**Proof.** The implication  $b \implies a$  is trivial.

For  $a \implies b$ : Let  $x_n \rightarrow x$  in  $E$ . By linearity, this means  $x_n - x \rightarrow 0$  in  $E$ . Since  $T$  is continuous at 0, we have  $T(x_n - x) = T(x_n) - T(x) \rightarrow 0$  as  $n \rightarrow \infty$ , which proves continuity at  $x$ . Thus,  $a \implies b$ .

For  $a \implies c$ : Since  $T$  is continuous at 0, choosing  $\epsilon = 1$ , there must exist some radius  $\delta > 0$  such that:

$$\|T(x)\|_F = \|T(x) - T(0)\|_F \leq 1 \quad \text{whenever} \quad \|x\|_E \leq \delta.$$

Now, pick any non-zero vector  $y \in E$ . We can rescale it to have norm exactly equal to  $\delta$  by considering the vector  $\frac{y}{\|y\|_E} \cdot \delta$ . Since this scaled vector has norm  $\delta$ , applying our property gives:

$$\left\| T \left( \frac{y}{\|y\|_E} \cdot \delta \right) \right\|_F \leq 1 \implies \frac{\delta}{\|y\|_E} \|T(y)\|_F \leq 1 \implies \|T(y)\|_F \leq \frac{1}{\delta} \|y\|_E.$$

Setting  $C = \frac{1}{\delta}$  satisfies condition c.

For  $c \implies a$ : If  $x_n \rightarrow 0$  as  $n \rightarrow \infty$ , then by bounded inequality:

$$\|T(x_n)\|_F \leq C\|x_n\|_E \rightarrow 0,$$

which confirms  $T$  is continuous at 0. □

## Operator Norm

Given a bounded linear operator  $T \in \mathcal{L}(E, F)$ , its operator norm is defined as:

$$\|T\| = \sup_{\|x\|_E \leq 1} \|T(x)\|_F = \sup_{\|x\|_E = 1} \|T(x)\|_F.$$

**Note.**

- i.  $\|T\| \leq C$  for any constant  $C$  that satisfies condition c.
- ii.  $\|T\|$  represents the absolute *smallest* valid constant  $C > 0$  for which condition c holds.
- iii. *Crucial Property.*  $\|\mathbf{T}(\mathbf{x})\|_{\mathbf{F}} \leq \|\mathbf{T}\| \|\mathbf{x}\|_{\mathbf{E}}$ .

## Dual Space

Given a normed vector space  $E$ , its *dual space* (denoted  $E^*$ ) is the space of all continuous linear functionals mapping from  $E$  into its underlying scalar field:

$$E^* = \mathcal{L}(E, \mathbb{R}) \quad (\text{or } E^* = \mathcal{L}(E, \mathbb{C})).$$

Note that because the target fields  $\mathbb{R}$  or  $\mathbb{C}$  are complete,  $E^*$  is always a complete Banach space, even if  $E$  itself is not! For any functional  $\ell \in E^*$  and vector  $x \in E$ , we often write the evaluation using dual pairing notation:

$$\ell(x) = \langle \ell, x \rangle_{E^*, E} = \langle \ell, x \rangle.$$

## Example

1. Let  $E = \mathbb{R}^d$ . Any linear functional  $\ell \in E^* = \mathcal{L}(\mathbb{R}^d, \mathbb{R})$  can be represented as a standard dot product:

$$\ell(x) = a \cdot x = \langle a, x \rangle = \sum_{i=1}^d a_i x_i,$$

where  $a = (a_1, \dots, a_d) \in \mathbb{R}^d$  is a fixed vector uniquely identifying the functional.

2. Let  $E$  and  $F$  be normed vector spaces. We can construct their product space  $E \times F$ , which is also a normed vector space under any of the following natural norms:

$$\|(x, y)\|_2 = \sqrt{\|x\|_E^2 + \|y\|_F^2},$$

$$\|(x, y)\|_1 = \|x\|_E + \|y\|_F,$$

$$\|(x, y)\|_\infty = \max\{\|x\|_E, \|y\|_F\}.$$

Furthermore, if both component spaces  $E$  and  $F$  are complete Banach spaces, then the product space  $E \times F$  is guaranteed to be a Banach space as well.

## Main Theorems of Normed Vector Spaces

### Hahn-Banach Theorem

Let  $E$  be a normed vector space and  $M \subseteq E$  be a vector subspace. Let  $f \in M^* = \mathcal{L}(M, \mathbb{R})$  be a bounded linear functional on  $M$ . Then there exists an extension  $\ell \in E^* = \mathcal{L}(E, \mathbb{R})$  such that  $\ell|_M = f$  and:

$$\|\ell\|_{E^*} = \|f\|_{M^*}.$$

**Remark.** The general proof relies on Zorn's Lemma. However, Zorn's Lemma is not required if the space  $E$  is separable (i.e., possesses a countable dense subset), in which case a sequential induction argument suffices.



### Corollary 1 (Separation from Closed Subspaces)

Let  $E$  be a normed vector space and  $F \subset E$  be a closed vector subspace such that  $F \neq E$ . Then there exists a nonzero functional  $\ell \in E^*$  such that  $\ell|_F = 0$ .

Moreover, if  $x_0 \notin F$ , we can choose  $\ell \in E^*$  such that  $\ell(x_0) = 1$  and  $\ell|_F = 0$ .

**Proof.** Let  $x_0 \notin F$ . We construct the expanded subspace  $M$  spanned by  $x_0$  and  $F$ :

$$M = \mathbb{R}x_0 + F = \{tx_0 + y : t \in \mathbb{R}, y \in F\}.$$

First, we observe that this representation is unique: if  $tx_0 + y = 0$  with  $t \neq 0$ , then  $x_0 = -\frac{y}{t}$ . Since  $F$  is a vector subspace, this implies  $x_0 \in F$ , which directly contradicts our assumption  $x_0 \notin F$ . Thus,  $t = 0$ , which immediately implies  $y = 0$ .

We can therefore safely define a linear functional  $f : M \rightarrow \mathbb{R}$  by:

$$f(tx_0 + y) = t.$$

By construction, this functional satisfies  $f(x_0) = 1$  and  $f(y) = 0$  for all  $y \in F$ . To formally verify it is well-defined, suppose  $t_1x_0 + y_1 = t_2x_0 + y_2$ . Then  $(t_1 - t_2)x_0 + (y_1 - y_2) = 0$ , which by our unique representation requires  $t_1 = t_2$  and  $y_1 = y_2$ . Hence  $f(t_1x_0 + y_1) = f(t_2x_0 + y_2)$ . Linearity follows directly:

$$f(\alpha_1(t_1x_0 + y_1) + \alpha_2(t_2x_0 + y_2)) = \alpha_1t_1 + \alpha_2t_2 = \alpha_1f(t_1x_0 + y_1) + \alpha_2f(t_2x_0 + y_2).$$

To establish that  $f$  is bounded, we define the distance from  $x_0$  to the subspace  $F$ :

$$d = \inf_{z \in F} \|x_0 - z\|_E.$$

Because  $F$  is closed and  $x_0 \notin F$ , it follows that  $d > 0$ . (If  $d = 0$ , there would exist a sequence  $\{z_n\} \subseteq F$  converging to  $x_0$ , forcing  $x_0 \in F = \overline{F}$ , a contradiction).

For any arbitrary element  $tx_0 + y \in M$  with  $t \neq 0$ , we can factor out the scalar:

$$\|tx_0 + y\|_E = |t| \left\| x_0 - \left(-\frac{y}{t}\right) \right\|_E \geq |t|d,$$

since  $-\frac{y}{t} \in F$ . This inequality trivially holds for  $t = 0$  as well. Consequently, if  $\|tx_0 + y\|_E \leq 1$ , we must have  $|t| \leq \frac{1}{d}$ . The norm of  $f$  is therefore bounded by:

$$\|f\|_{M^*} = \sup_{\|tx_0 + y\|_E \leq 1} |f(tx_0 + y)| = \sup_{\|tx_0 + y\|_E \leq 1} |t| \leq \frac{1}{d} < \infty.$$

By the Hahn-Banach Theorem,  $f$  extends to a global linear functional  $\ell \in E^*$  preserving the norm. This extended functional satisfies  $\ell(x_0) = f(x_0) = 1$  and  $\ell(y) = f(y) = 0$  for all  $y \in F$ , completing the proof.  $\square$

Corollary 1 provides a standard mechanism to prove that a vector subspace  $D \subseteq E$  is dense ( $\overline{D} = E$ ) via contradiction:

1. Assume the closure is proper:  $\overline{D} \neq E$ . Note that  $\overline{D}$  remains a valid closed vector subspace.
2. By Corollary 1, there must exist a nonzero functional  $\ell \in E^*$  vanishes entirely on  $\overline{D}$  (i.e.,  $\ell|_{\overline{D}} = 0$ ).
3. If independent structural arguments demonstrate that  $\ell|_D = 0 \implies \ell = 0$  on all of  $E$ , we obtain a contradiction, proving density.

By continuity, if a bounded functional vanishes on  $D$ , then for any  $x_0 \in \overline{D}$  approached by a sequence  $\{x_n\} \subseteq D$ , we have  $\ell(x_0) = \lim_{n \rightarrow \infty} \ell(x_n) = 0$ .

### Corollary 2 (Norm Realization and Alignment)

Let  $E$  be a normed vector space.

- a. For each nonzero  $x \in E$ , there exists a normalized functional  $\ell \in E^*$  ( $\|\ell\|_{E^*} = 1$ ) such that:

$$\ell(x) = \langle \ell, x \rangle = \|\ell\|_{E^*} \|x\|_E.$$

We say that  $\ell$  and  $x$  are *aligned*.

- b. For any  $x \in E$ , the norm can be calculated via dual evaluation:

$$\|x\|_E = \max_{\|\ell\|_{E^*} \leq 1} |\ell(x)|.$$

*Geometric Context.* By definition, the bound  $|\ell(x)| \leq \|\ell\|_{E^*} \|x\|_E$  always holds. Alignment represents the case where this inequality is sharply saturated.

- If  $E = \mathbb{R}^d$  under the Euclidean norm  $\|\cdot\|_2$ , then  $E^* \cong \mathbb{R}^d$  under  $\|\cdot\|_2$ . Alignment means  $\langle a, x \rangle = \|a\|_2 \|x\|_2$ , which occurs if and only if  $a$  and  $x$  are linearly dependent and point in the same direction ( $\cos \alpha = 1$ ).
- If  $E = \mathbb{R}^d$  under the  $\ell^1$ -norm  $\|\cdot\|_1$ , its dual is governed by the  $\ell^\infty$ -norm  $\|\cdot\|_\infty$ . Alignment requires  $\langle a, x \rangle = \|a\|_\infty \|x\|_1$ , meaning  $a_i = \max_j |a_j| \cdot \text{sgn}(x_i)$  for all indices where  $x_i \neq 0$ .

**Proof.** We assume without loss of generality that the scalar field is  $\mathbb{R}$ .

- a. Let  $x_0 \in E$  with  $x_0 \neq 0$ . Define the one-dimensional subspace  $M = \mathbb{R}x_0$ . Construct the linear functional  $f : M \rightarrow \mathbb{R}$  by  $f(tx_0) = t\|x_0\|_E$ . Its operator norm on  $M$  is:

$$\|f\|_{M^*} = \sup_{\|tx_0\|_E \leq 1} |f(tx_0)| = \sup_{|t|\|x_0\|_E \leq 1} |t|\|x_0\|_E = 1.$$

By construction,  $f(x_0) = \|x_0\|_E = \|f\|_{M^*} \|x_0\|_E$ . Applying the Hahn-Banach Theorem yields an extension  $\ell \in E^*$  matching the norm  $\|\ell\|_{E^*} = \|f\|_{M^*} = 1$ , which satisfies  $\ell(x_0) = f(x_0) = \|x_0\|_E = \|\ell\|_{E^*} \|x_0\|_E$ .

- b. For any functional satisfying  $\|\ell\|_{E^*} \leq 1$ , we have  $|\ell(x_0)| \leq \|\ell\|_{E^*} \|x_0\|_E \leq \|x_0\|_E$ , hence  $\sup_{\|\ell\|_{E^*} \leq 1} |\ell(x_0)| \leq \|x_0\|_E$ . Part a guarantees the existence of a specific functional achieving exactly  $\|x_0\|_E$ , turning the supremum into a maximum.  $\square$

## Norm of the Adjoint Operator

Let  $E$  and  $F$  be normed vector spaces, and let  $T \in \mathcal{L}(E, F)$  be a bounded linear operator. The *adjoint operator*  $T' : F^* \rightarrow E^*$  is defined by mapping each linear functional  $\ell \in F^*$  to the functional  $T'\ell \in E^*$  via:

$$(T'\ell)(x) := \ell(T(x)) \quad \forall x \in E.$$

### Proposition 1

The adjoint operator satisfies  $T' \in \mathcal{L}(F^*, E^*)$ , and its operator norm is identical to the original operator:

$$\|T'\| = \|T\|.$$

**Proof.** We expand the definition of the operator norm using nested suprema over the respective unit balls:

$$\begin{aligned} \|T'\| &= \sup_{\|\ell\|_{F^*} \leq 1} \|T'\ell\|_{E^*} = \sup_{\|\ell\|_{F^*} \leq 1} \sup_{\|x\|_E \leq 1} |(T'\ell)(x)| \\ &= \sup_{\|\ell\|_{F^*} \leq 1} \sup_{\|x\|_E \leq 1} |\ell(T(x))| = \sup_{\|x\|_E \leq 1} \sup_{\|\ell\|_{F^*} \leq 1} |\ell(T(x))|. \end{aligned}$$

Applying Corollary 2 b directly to the inner supremum over the element  $T(x) \in F$ :

$$\sup_{\|\ell\|_{F^*} \leq 1} |\ell(T(x))| = \|T(x)\|_F.$$

Substituting this back into the norm expression completes the proof:

$$\|T'\| = \sup_{\|x\|_E \leq 1} \|T(x)\|_F = \|T\|.$$

□

## Canonical Isometric Embedding into the Bidual

Let  $E$  be a normed vector space and  $E^{**} = (E^*)^*$  denote its bidual space. Every element  $x_0 \in E$  induces an evaluation functional  $J_{x_0} \in E^{**}$  defined via:

$$J_{x_0}(\ell) = \ell(x_0) \quad \forall \ell \in E^*.$$

Boundedness follows immediately from standard duality:  $|J_{x_0}(\ell)| = |\ell(x_0)| \leq \|\ell\|_{E^*} \|x_0\|_E$ .

### Proposition 2

The canonical evaluation map  $J : E \rightarrow E^{**}$  defined by  $x_0 \mapsto J_{x_0}$  is a linear isometric embedding. Thus,  $E$  can be viewed as an isometric subspace of  $E^{**}$  (written  $E \subseteq E^{**}$ ).

**Proof.** Linearity is inherited from the vector space operations on  $E^*$ . Utilizing Corollary 2b once more to evaluate the norm of the functional  $J_{x_0}$  on  $E^*$ :

$$\|J_{x_0}\|_{E^{**}} = \sup_{\|\ell\|_{E^*} \leq 1} |J_{x_0}(\ell)| = \sup_{\|\ell\|_{E^*} \leq 1} |\ell(x_0)| = \|x_0\|_E.$$

□

### Baire's Category Theorem (Closed Set Formulation)

Let  $(M, \rho)$  be a complete metric space. If  $M$  is expressed as the countable union of closed subsets  $M = \bigcup_{n=1}^{\infty} M_n$ , then there exists an index  $n_0 \geq 1$  such that its interior is nonempty ( $\overset{\circ}{M}_{n_0} \neq \emptyset$ ). That is,  $M_{n_0}$  contains an open ball  $B_{r_0}(x_0)$ .

**Proof.** We proceed by contradiction. Assume that  $\overset{\circ}{M}_n = \emptyset$  for all  $n \geq 1$ . This implies that the complements:

$$O_n = M \setminus M_n$$

are open and dense in  $M$ . Consequently, every open ball in  $M$  must intersect each  $O_n$ . We aim to show that the intersection  $G = \bigcap_{n=1}^{\infty} O_n$  is dense. However, by De Morgan's laws:

$$G = \bigcap_{n=1}^{\infty} O_n = \bigcap_{n=1}^{\infty} M_n^c = \left( \bigcup_{n=1}^{\infty} M_n \right)^c = M^c = \emptyset,$$

which yields a contradiction since a dense subset of a nonempty complete space cannot be empty.

To prove  $G$  is dense, it suffices to show that an arbitrary open ball  $B_{r_0}(x_0) \subseteq M$  contains at least one point of  $G$ . Since  $O_1$  is open and dense, the intersection  $B_{r_0}(x_0) \cap O_1$  is open and nonempty. We can therefore choose a nested closed ball  $\overline{B}_{r_1}(x_1)$  inside it:

$$\overline{B}_{r_1}(x_1) \subseteq B_{r_0}(x_0) \cap O_1 \quad \text{with } r_1 \leq \frac{r_0}{2}.$$

Next, because  $O_2$  is dense, the open set  $B_{r_1}(x_1) \cap O_2$  is nonempty. We select a smaller nested closed ball:

$$\overline{B}_{r_2}(x_2) \subseteq B_{r_1}(x_1) \cap O_2 \quad \text{with } r_2 \leq \frac{r_1}{2} \leq \frac{r_0}{2^2}.$$

Proceeding inductively, given the open ball  $B_{r_n}(x_n)$ , density of  $O_{n+1}$  allows us to find:

$$\overline{B}_{r_{n+1}}(x_{n+1}) \subseteq B_{r_n}(x_n) \cap O_{n+1} \quad \text{with } r_{n+1} \leq \frac{r_0}{2^{n+1}}.$$

The constructed sequence of centers  $\{x_n\}_{n=1}^{\infty}$  is Cauchy. Indeed, for any indices  $j, j' \geq n$ , both points belong to the ball  $B_{r_n}(x_n)$ , so:

$$\rho(x_j, x_{j'}) \leq \rho(x_j, x_n) + \rho(x_{j'}, x_n) < 2r_n \leq \frac{2r_0}{2^n} \longrightarrow 0 \quad \text{as } n \rightarrow \infty.$$

Since  $M$  is complete, the limit  $x = \lim_{n \rightarrow \infty} x_n$  exists. Because the balls are nested, for any fixed index  $n$ , the tail of the sequence lies completely inside the closed ball  $\overline{B}_{r_n}(x_n)$ , forcing the limit to remain inside it:

$$x \in \overline{B}_{r_n}(x_n) \subseteq B_{r_{n-1}}(x_{n-1}) \cap O_n \subseteq O_n.$$

Since this holds for every index,  $x \in \bigcap_{m=1}^{\infty} O_m = G$ . Thus  $B_{r_0}(x_0) \cap G \neq \emptyset$ , proving density and completing the contradiction.  $\square$

### Banach-Steinhaus Theorem (Uniform Boundedness Principle)

Let  $E$  be a Banach space and  $F$  be a normed vector space. Let  $\{T_i\}_{i \in I} \subset \mathcal{L}(E, F)$  be a family of bounded linear operators. Suppose that the family is pointwise uniformly bounded, meaning:

$$\sup_{i \in I} \|T_i(x)\|_F < \infty \quad \forall x \in E.$$

Then the operators are uniformly bounded in operator norm:

$$\sup_{i \in I} \|T_i\| < \infty.$$

**Proof.** For each integer  $n \geq 1$ , we define the sets:

$$M_n = \left\{ x \in E : \sup_{i \in I} \|T_i(x)\|_F \leq n \right\} = \bigcap_{i \in I} \{x \in E : \|T_i(x)\|_F \leq n\}.$$

Because each operator  $T_i$  is continuous, the set  $\{x \in E : \|T_i(x)\|_F \leq n\}$  is the inverse image of a closed set, hence closed. The intersection of any family of closed sets remains closed, so each  $M_n$  is closed. The pointwise uniform boundedness assumption guarantees that every element  $x \in E$  belongs to  $M_n$  for a sufficiently large  $n$ , yielding the countable decomposition  $E = \bigcup_{n=1}^{\infty} M_n$ .

Since  $E$  is a complete Banach space, Baire's Category Theorem dictates that there exists an index  $n_0 \geq 1$  such that  $M_{n_0} \neq \emptyset$ . Thus, there exists a closed ball centered at some  $x_0 \in E$  completely contained within this set:

$$\overline{B}_{r_0}(x_0) \subseteq M_{n_0} \implies \sup_{i \in I} \|T_i(x)\|_F \leq n_0 \quad \forall x \in \overline{B}_{r_0}(x_0).$$

We now estimate the operator norm by evaluating perturbations around  $x_0$ . Let  $y \in E$  be an arbitrary vector satisfying  $\|y\|_E \leq r_0$ . The shifted vector  $x_0 + y$  belongs to  $\overline{B}_{r_0}(x_0)$ . By linearity, we can isolate  $y$ :

$$\|T_i(y)\|_F = \|T_i(y + x_0 - x_0)\|_F \leq \|T_i(x_0 + y)\|_F + \|T_i(x_0)\|_F.$$

Since both  $x_0$  and  $x_0 + y$  reside within  $M_{n_0}$ , their images under any  $T_i$  are bounded by  $n_0$ :

$$\|T_i(y)\|_F \leq n_0 + n_0 = 2n_0 \quad \forall i \in I.$$

Taking the supremum over the ball of radius  $r_0$  allows us to calculate the operator norm bound:

$$\|T_i\| = \frac{1}{r_0} \sup_{\|y\|_E \leq r_0} \|T_i(y)\|_F \leq \frac{2n_0}{r_0} < \infty.$$

As this bound is entirely independent of the index  $i$ , we conclude  $\sup_{i \in I} \|T_i\| \leq \frac{2n_0}{r_0}$ .  $\square$

**Finite-Dimensional Example.** Let  $T_i \in \mathcal{L}(\mathbb{R}^d, \mathbb{R}^d)$  be represented by matrices  $(a_{k\ell}^i)$  for  $1 \leq k, \ell \leq d$ . Pointwise boundedness  $\sup_i \|T_i(x)\| < \infty$  for all  $x \in \mathbb{R}^d$  implies uniform boundedness of all matrix entries:  $\sup_i |a_{k\ell}^i| < \infty$ . This can be checked by testing the standard basis vectors  $e_\ell$ :

$$\sup_i |a_{k\ell}^i| = \sup_i |e_k \cdot (T_i(e_\ell))| \leq \sup_i \|e_k\|_2 \|T_i(e_\ell)\|_2 = \sup_i \|T_i(e_\ell)\|_2 < \infty.$$

## The Open Mapping Theorem

### Theorem 3

Let  $E$  and  $F$  be Banach spaces and let  $T \in \mathcal{L}(E, F)$  be a surjective (onto) bounded linear operator. Then there exists a scaling constant  $c > 0$  such that the image of the open unit ball contains an open ball centered at the origin in  $F$ :

$$B_c^F \subseteq T(B_1^E).$$

**Proof.** By surjectivity, the images of expanding balls cover the entire target space:  $F = \bigcup_{n=1}^{\infty} T(B_n^E)$ . To invoke Baire's Category Theorem, we look at their closures:  $F = \bigcup_{n=1}^{\infty} \overline{T(B_n^E)}$ . Complete continuity of  $F$  guarantees that there exists an index  $n_0$  containing an open neighborhood:

$$B_{r_0}^F(x_0) \subseteq \overline{T(B_{n_0}^E)}.$$

By linearity, the image is symmetric about the origin ( $y \in T(B_{n_0}^E) \iff -y \in T(B_{n_0}^E)$ ), which implies the closure is symmetric as well, so  $B_{r_0}^F(-x_0) \subseteq \overline{T(B_{n_0}^E)}$ . Let  $y \in B_{r_0}^F$  be an arbitrary vector centered at the origin. We can decompose it convexly:

$$y = \frac{1}{2}(x_0 + y) + \frac{1}{2}(-x_0 + y).$$

Since  $x_0 + y \in B_{r_0}^F(x_0)$  and  $-x_0 + y \in B_{r_0}^F(-x_0)$ , both belong to the convex set  $\overline{T(B_{n_0}^E)}$ . Summing them shows that the centered ball is absorbed:  $B_{r_0}^F \subseteq \overline{T(B_{n_0}^E)}$ . By linear scaling, dividing by  $n_0$  yields:

$$B_{\delta}^F \subseteq \overline{T(B_1^E)} \quad \text{where } \delta = \frac{r_0}{n_0}.$$

We now show that this closure can be swallowed by a slightly larger open ball, specifically  $\overline{T(B_1^E)} \subseteq T(B_2^E)$ . Let  $z \in \overline{T(B_1^E)}$ . By scaling the absorption property,  $B_{\delta/2}^F \subseteq \overline{T(B_{1/2}^E)}$ . This allows us to approximate  $z$  closely: there exists an element  $x_1 \in B_1^E$  such that:

$$z - T(x_1) \in B_{\delta/2}^F \subseteq \overline{T(B_{1/2}^E)}.$$

Iterating this approximation property at scale  $1/2$ , there must exist an  $x_2 \in B_{1/2}^E$  satisfying:

$$z - T(x_1) - T(x_2) \in B_{\delta/4}^F \subseteq \overline{T(B_{1/4}^E)}.$$

Inductively, for step  $n$ , we can find a correction vector  $x_n \in B_{1/2^{n-1}}^E$  such that:

$$z - T(x_1 + \cdots + x_n) \in B_{\delta/2^n}^F \subseteq \overline{T(B_{1/2^n}^E)},$$

which means  $\|z - T(\sum_{k=1}^n x_k)\|_F \leq \frac{\delta}{2^n} \rightarrow 0$  as  $n \rightarrow \infty$ . The series  $\sum_{k=1}^{\infty} x_k$  converges absolutely within the Banach space  $E$  because:

$$\sum_{k=1}^{\infty} \|x_k\|_E < 1 + \sum_{j=1}^{\infty} \frac{1}{2^j} = 2.$$

Completeness implies the limit  $x = \sum_{k=1}^{\infty} x_k$  exists and satisfies  $\|x\|_E < 2$ . By continuity of  $T$ :

$$T(x) = \lim_{n \rightarrow \infty} T\left(\sum_{k=1}^n x_k\right) = z.$$

Since  $\|x\|_E < 2$ , it follows that  $z \in T(B_2^E)$ , which proves  $\overline{T(B_1^E)} \subseteq T(B_2^E)$ . Combining our results:

$$B_\delta^F \subseteq \overline{T(B_1^E)} \subseteq T(B_2^E) \implies B_{\delta/2}^F \subseteq T(B_1^E).$$

Setting  $c = \delta/2$  completes the proof.  $\square$

### Corollary 3 (Openness of Operator Images)

Let  $E$  and  $F$  be Banach spaces and let  $T \in \mathcal{L}(E, F)$  be surjective. If  $U \subseteq E$  is an open set, then its image  $T(U) \subseteq F$  is open.

**Proof.** Let  $y_0 \in T(U)$ . By definition, there exists an  $x_0 \in U$  such that  $T(x_0) = y_0$ . Since  $U$  is open, we can find a radius  $\delta > 0$  such that  $B_\delta^E(x_0) = x_0 + B_\delta^E \subseteq U$ . By Theorem 3, there exists a constant  $c > 0$  such that  $B_c^F \subseteq T(B_1^E)$ , which scales to  $B_{c\delta}^F \subseteq T(B_\delta^E)$ . Using linearity to translate the sets:

$$y_0 + B_{c\delta}^F \subseteq T(x_0) + T(B_\delta^E) = T(x_0 + B_\delta^E) \subseteq T(U).$$

Thus,  $T(U)$  contains an open ball around each of its points, meaning it is open.  $\square$

### Inverse Mapping Theorem

Let  $E$  and  $F$  be Banach spaces. If a bounded linear operator  $T \in \mathcal{L}(E, F)$  is bijective, then its set-theoretic inverse is automatically bounded:  $T^{-1} \in \mathcal{L}(F, E)$ .

**Proof.** To establish continuity of  $T^{-1}$ , it suffices to show that the preimage of any open set  $U \subseteq E$  under  $T^{-1}$  is open in  $F$ . Notice that:

$$(T^{-1})^{-1}(U) = T(U).$$

By the Open Mapping Theorem (Corollary 3), since  $T$  is surjective, the image  $T(U)$  is guaranteed to be open. Thus  $T^{-1}$  is continuous, which is equivalent to being bounded.  $\square$

### Closed Graph Theorem

Let  $E$  and  $F$  be Banach spaces and let  $T : E \rightarrow F$  be a linear operator. If the graph of the operator:

$$\Gamma = \{(x, T(x)) \in E \times F : x \in E\}$$

is a closed subspace of the product space  $E \times F$ , then  $T$  is bounded ( $T \in \mathcal{L}(E, F)$ ).

**Proof.** The product space  $E \times F$  equipped with the norm  $\|(x, y)\|_{E \times F} = \sqrt{\|x\|_E^2 + \|y\|_F^2}$  is a Banach space. Because  $\Gamma$  is assumed to be a closed subspace, it is a complete Banach

space in its own right under this induced norm. We consider the two standard canonical coordinate projections restricted to  $\Gamma$ :

$$\pi_1 : \Gamma \rightarrow E, \quad \pi_1(x, T(x)) = x,$$

$$\pi_2 : \Gamma \rightarrow F, \quad \pi_2(x, T(x)) = T(x).$$

The projection  $\pi_1$  is bounded because  $\|\pi_1(x, T(x))\|_E = \|x\|_E \leq \|(x, T(x))\|_{E \times F}$ . Furthermore,  $\pi_1$  is clearly a bijection between  $\Gamma$  and  $E$ . Applying the Inverse Mapping Theorem, the inverse operator  $\pi_1^{-1} : E \rightarrow \Gamma$  is automatically bounded.

We can now express the original operator as a composition of these two projections:

$$T = \pi_2 \circ \pi_1^{-1}.$$

Since  $\pi_2$  is obviously bounded ( $\|T(x)\|_F \leq \|(x, T(x))\|_{E \times F}$ ),  $T$  is the composition of two bounded linear operators, making it bounded.  $\square$

**Operational Summary.** The Closed Graph Theorem simplifies proofs of continuity by relaxing the convergent sequence condition:

- *Standard Proof Method.* Assume  $x_n \rightarrow x$  in  $E$ . One must simultaneously prove that the limit  $\lim_{n \rightarrow \infty} T(x_n) = y$  exists, and that  $y = T(x)$ .
- *Closed Graph Method.* Assume  $x_n \rightarrow x$  in  $E$  and assume that  $T(x_n) \rightarrow y$  in  $F$ . One only needs to show the structural identity  $y = T(x)$ .

## Hilbert Spaces

### Inner Product

Let  $H$  be a complex vector space. An inner product on  $H$  is a  $\mathbb{C}$ -valued function denoted by  $\langle x, y \rangle$ , for  $x, y \in H$ , such that:

- i.  $\langle x, x \rangle \geq 0 \ \forall x \in H$ ;  $\langle x, x \rangle = 0$  if and only if  $x = 0$
- ii.  $\forall x, y, z \in H$ , and  $\alpha, \beta \in \mathbb{C}$ :

$$\langle \alpha x + \beta y, z \rangle = \alpha \langle x, z \rangle + \beta \langle y, z \rangle,$$

$$\langle z, \alpha x + \beta y \rangle = \overline{\alpha} \langle z, x \rangle + \overline{\beta} \langle z, y \rangle,$$

$$\langle x, y \rangle = \overline{\langle y, x \rangle}.$$

### Examples

1.  $H = \mathbb{C}^d$ ,  $x = (x_1, \dots, x_d)$ ,  $y = (y_1, \dots, y_d) \in \mathbb{C}^d$ :

$$\langle x, y \rangle = \sum_{j=1}^d x_j \overline{y_j}.$$

Then,  $\langle x, x \rangle = \sum_{j=1}^d |x_j|^2$ .



2.  $\ell^2$  is the space of all sequences  $(x_1, x_2, \dots)$  such that

$$\sum_{n=1}^{\infty} |x_n|^2 < \infty,$$

with the inner product  $\langle x, y \rangle = \sum_{n=1}^{\infty} x_n \bar{y}_n$ .

3.  $\mathcal{C} = C([a, b])$  is the space of continuous complex-valued functions on  $[a, b]$  with

$$\langle f, g \rangle = \int_a^b f(t) \overline{g(t)} dt.$$

Then,  $\langle f, f \rangle = \int_a^b |f(t)|^2 dt$ .

4.  $L^2(X, \mu)$  is the space of complex-valued square-integrable functions (equivalence classes) on a measure space  $(X, \mathcal{X}, \mu)$  with

$$\langle f, g \rangle = \int f(x) \overline{g(x)} d\mu.$$

In fact, Examples 1 and 2 are just specific cases of Example 4 using the counting measure.

## Inner Product Space (Pre-Hilbert Space)

A complex vector space equipped with an inner product is called an inner product space or pre-Hilbert space.

## Orthogonality

Let  $H$  be a pre-Hilbert space. We say  $x \perp y$  in  $H$  if  $\langle x, y \rangle = 0$ . The induced norm is defined as:

$$\|x\|_H := \sqrt{\langle x, x \rangle}, \quad x \in H.$$

## Pythagorean Theorem

If  $x \perp y$ , then

$$\|x + y\|_H^2 = \|x\|_H^2 + \|y\|_H^2.$$

**Proof.** Let  $\langle x, y \rangle = \langle y, x \rangle = 0$ . Then,

$$\begin{aligned} \|x + y\|_H^2 &= \langle x + y, x + y \rangle \\ &= \langle x, x \rangle + \langle x, y \rangle + \langle y, x \rangle + \langle y, y \rangle \\ &= \|x\|_H^2 + \|y\|_H^2 \end{aligned}$$

## Orthonormal System

A subset  $\{e_j : j = 1, \dots, N\}$  is an orthonormal system (ONS) if

$$\langle e_k, e_\ell \rangle = \begin{cases} 1 & \text{if } k = \ell \\ 0 & \text{if } k \neq \ell \end{cases}.$$

Note that  $\|e_k\|_H^2 = \langle e_k, e_k \rangle = 1$ .

### Theorem 2

Let  $\{e_j : j = 1, \dots, N\}$  be an ONS. Then for any  $x \in H$ :

- i.  $\sum_{j=1}^N \langle x, e_j \rangle e_j$  and  $x - \sum_{j=1}^N \langle x, e_j \rangle e_j$  are orthogonal.
- ii.  $\|x\|_H^2 = \|x - \sum_{j=1}^N \langle x, e_j \rangle e_j\|_H^2 + \|\sum_{j=1}^N \langle x, e_j \rangle e_j\|_H^2$ .
- iii. *Bessel's Inequality*.

$$\sum_{j=1}^N |\langle x, e_j \rangle|^2 \leq \|x\|_H^2.$$

**Proof.** We first prove i by showing that  $x - \sum_{j=1}^N \langle x, e_j \rangle e_j \perp e_k$  for all  $k = 1, \dots, N$ :

$$\langle x - \sum_{j=1}^N \langle x, e_j \rangle e_j, e_k \rangle = \langle x, e_k \rangle - \langle x, e_k \rangle \langle e_k, e_k \rangle = \langle x, e_k \rangle - \langle x, e_k \rangle = 0.$$

Next, ii follows directly from i and the Pythagorean Theorem, and iii clearly follows by dropping the non-negative first term from the right-hand side of ii.

**Remark.** If  $\{e_j : j = 1, \dots, N\}$  is an ONS, then  $\{e_1, \dots, e_N\}$  is linearly independent. Indeed,

$$0 = \sum_{j=1}^N \alpha_j e_j \implies \alpha_k = 0 \quad \forall k$$

which is shown by taking the inner product of both sides with each  $e_k$ .

## Cauchy-Schwarz Inequality

For any  $x, y \in H$ ,

$$|\langle x, y \rangle| \leq \|x\|_H \|y\|_H.$$

**Proof.** Let  $y \neq 0$ . Consider the single-element ONS  $\{\frac{y}{\|y\|_H}\}$ . By Bessel's inequality:

$$\sum_{j=1}^1 |\langle x, e_j \rangle|^2 \leq \|x\|_H^2 \implies \left| \left\langle x, \frac{y}{\|y\|_H} \right\rangle \right|^2 \leq \|x\|_H^2 \implies |\langle x, y \rangle|^2 \leq \|x\|_H^2 \|y\|_H^2.$$

### Corollary 1

$\|x\|_H = \sqrt{\langle x, x \rangle}$ ,  $x \in H$ , is a well-defined norm.

**Proof.** We prove the triangle inequality. For any  $x, y \in H$ :

$$\begin{aligned}\|x + y\|_H^2 &= \langle x + y, x + y \rangle \\ &= \langle x, x \rangle + \langle x, y \rangle + \langle y, x \rangle + \langle y, y \rangle \\ &\leq \langle x, x \rangle + |\langle x, y \rangle| + |\langle y, x \rangle| + \langle y, y \rangle \\ &\leq \|x\|_H^2 + \|y\|_H^2 + 2\|x\|_H\|y\|_H \quad (\text{by Cauchy-Schwarz}) \\ &= (\|x\|_H + \|y\|_H)^2.\end{aligned}$$

### Hilbert Space

A pre-Hilbert space  $H$  is called a *Hilbert space* if it is complete with respect to the induced norm  $\|\cdot\|_H$ .

### Projection Theorem

Let  $H$  be a pre-Hilbert space. Given a vector subspace  $M \subseteq H$ , its orthogonal complement is defined as:

$$M^\perp = \{x \in H : \langle x, m \rangle = 0 \quad \forall m \in M\}.$$

We write  $x \perp M$  if  $x \in M^\perp$ . Note that  $M^\perp$  is always a closed vector subspace of  $H$  due to the continuity of the inner product.

### Theorem 3

Let  $H$  be a Hilbert space and  $M \subseteq H$  be a closed vector subspace.

- i.  $\forall x \in H$ , there exists a unique  $m_0 \in M$  such that

$$d := \inf_{m \in M} \|x - m\|_H = \|x - m_0\|_H$$

Moreover,  $(x - m_0) \perp M$ . The vector  $m_0$  is called the *orthogonal projection* of  $x$  onto  $M$ , denoted by  $P_M x = m_0$ .

- ii. Let  $m_1 \in M$  be such that  $(x - m_1) \perp M$ . Then  $m_1 = m_0$ .

**Remark.** Theorem 3 implies that  $\forall x \in H$ , there exists a unique decomposition  $x = m_0 + w$  where  $m_0 \in M$  and  $w \in M^\perp$ . Thus,  $H = M \oplus M^\perp$ .

**Proof.**

- i. Let  $x \in H$ . By the definition of infimum, there exists a sequence  $\{m_k\} \subseteq M$  such that  $\|x - m_k\|_H \rightarrow d$  as  $k \rightarrow \infty$ . We show that  $\{m_k\}$  is a Cauchy sequence using the Parallelogram Law:

$$\|m_k - m_\ell\|_H^2 = \|(x - m_\ell) - (x - m_k)\|_H^2 = 2\|x - m_\ell\|_H^2 + 2\|x - m_k\|_H^2 - \|2x - (m_\ell + m_k)\|_H^2.$$

Since  $M$  is a vector subspace,  $\frac{m_\ell + m_k}{2} \in M$ , which implies  $\|x - (\frac{m_\ell + m_k}{2})\|_H \geq d$ . Thus,

$$\|2x - (m_\ell + m_k)\|_H^2 = 4 \left\| x - \left( \frac{m_\ell + m_k}{2} \right) \right\|_H^2 \geq 4d^2.$$

Substituting this back gives:

$$\|m_k - m_\ell\|_H^2 \leq 2\|x - m_\ell\|_H^2 + 2\|x - m_k\|_H^2 - 4d^2 \xrightarrow{k, \ell \rightarrow \infty} 2d^2 + 2d^2 - 4d^2 = 0.$$

Since  $H$  is complete,  $m_0 := \lim_{k \rightarrow \infty} m_k$  exists, and by closure of  $M$ ,  $m_0 \in M$  with  $\|x - m_0\|_H = d$ .

For uniqueness, if  $\tilde{m}_0 \in M$  also achieves the distance  $d$ , applying the same inequality with  $m_k = m_0$  and  $m_\ell = \tilde{m}_0$  yields:

$$\|m_0 - \tilde{m}_0\|_H^2 \leq 2d^2 + 2d^2 - 4d^2 = 0 \implies m_0 = \tilde{m}_0.$$

To prove that  $(x - m_0) \perp M$ , let  $m \in M$  and  $t \in \mathbb{R}$ . Expanding the norm using inner product properties:

$$\|(x - m_0) - tm\|_H^2 = \|x - m_0\|_H^2 - 2t\operatorname{Re}\{\langle x - m_0, m \rangle\} + t^2\|m\|_H^2.$$

Since  $m_0$  minimizes the distance, we have  $d^2 \leq \|(x - m_0) - tm\|_H^2$ , which simplifies to:

$$0 \leq t^2\|m\|_H^2 - 2t\operatorname{Re}\{\langle x - m_0, m \rangle\} \quad \forall t \in \mathbb{R}.$$

For this quadratic function to remain non-negative for all  $t$ , the linear coefficient must vanish:

$$\operatorname{Re}\{\langle x - m_0, m \rangle\} = 0.$$

Replacing  $t$  by  $it$  in the original formulation yields  $\operatorname{Im}\{\langle x - m_0, m \rangle\} = 0$ . Since both parts are zero,  $\langle x - m_0, m \rangle = 0$ , proving orthogonality.

- ii. Let  $m_1 \in M$  satisfy  $(x - m_1) \perp M$ . For any  $m \in M$ , since  $(m_1 - m) \in M$ , we have  $(x - m_1) \perp (m_1 - m)$ . By the Pythagorean Theorem:

$$\|x - m\|_H^2 = \|(x - m_1) + (m_1 - m)\|_H^2 = \|x - m_1\|_H^2 + \|m_1 - m\|_H^2 \geq \|x - m_1\|_H^2.$$

Taking the infimum over all  $m \in M$  implies  $\|x - m_1\|_H^2 = d^2$ . By the uniqueness proven in part i,  $m_1 = m_0$ .

## Corollary 2

Let  $\{e_1, \dots, e_N\}$  be an ONS in a Hilbert space  $H$ , and let  $M = \operatorname{span}\{e_1, \dots, e_N\}$ . Then for any  $x \in H$ :

$$P_M(x) = \sum_{j=1}^N \langle x, e_j \rangle e_j.$$

**Proof.** Theorem 2 established that  $(x - \sum_{j=1}^N \langle x, e_j \rangle e_j) \perp e_k$  for all  $k$ , which implies it is orthogonal to the entire span  $M$ . Applying Theorem 3 ii completes the proof.

## Hilbert Space Basis

Let  $H$  be a Hilbert space. An orthonormal system  $S \subseteq H$  is called a *complete orthonormal system* (CONS) or a *Hilbert basis* if for any  $z \in H$ :

$$\langle z, e_\alpha \rangle = 0 \quad \forall e_\alpha \in S \implies z = 0.$$

### Example

Let  $\mu$  be the counting measure on  $[0, 1]$ . Consider  $H = L^2([0, 1], \mu)$ , the space of functions  $f : [0, 1] \rightarrow \mathbb{R}$  such that

$$\sum_{t \in [0, 1]} [f(t)]^2 = \int_{[0, 1]} [f(t)]^2 d\mu < \infty.$$

The inner product is given by  $\langle f, g \rangle = \sum_{t \in [0, 1]} f(t)g(t)$ . For each  $\alpha \in [0, 1]$ , defining the standard indicator functions  $e_\alpha(t) = 1$  if  $\alpha = t$  and 0 otherwise forms a non-countable CONS where  $f(t) = \sum_{\alpha \in [0, 1]} \langle f, e_\alpha \rangle e_\alpha(t)$ .

### Theorem 4

Let  $H$  be a Hilbert space with a CONS  $\{e_\alpha : \alpha \in A\}$ . For any  $x \in H$ :

- i. The set  $\mathcal{A} := \{\alpha \in A : \langle x, e_\alpha \rangle \neq 0\}$  is at most countable:  $\mathcal{A} = \{\alpha_1, \alpha_2, \dots\}$ .
- ii.  $x = \sum_{\alpha \in A} \langle x, e_\alpha \rangle e_\alpha = \sum_{k=1}^{\infty} \langle x, e_{\alpha_k} \rangle e_{\alpha_k} = \lim_{n \rightarrow \infty} \sum_{k=1}^n \langle x, e_{\alpha_k} \rangle e_{\alpha_k}$ .
- iii. *Parseval's Identity*.

$$\|x\|_H^2 = \sum_{\alpha \in A} |\langle x, e_\alpha \rangle|^2 = \sum_{k=1}^{\infty} |\langle x, e_{\alpha_k} \rangle|^2.$$

### Proof.

- i. For any finite subset  $\{\beta_1, \dots, \beta_n\} \subseteq A$ , Bessel's inequality states  $\sum_{k=1}^n |\langle x, e_{\beta_k} \rangle|^2 \leq \|x\|_H^2 < \infty$ . For each  $n \geq 1$ , define  $A_n = \{\alpha \in A : |\langle x, e_\alpha \rangle| > \frac{1}{n}\}$ . If any  $A_n$  were infinite, it would contain a countable sequence contradicting Bessel's bounded sum. Hence, each  $A_n$  is finite, making  $\mathcal{A} = \bigcup_{n=1}^{\infty} A_n$  a countable union of finite sets, which is countable.
- ii. Let  $\{\alpha_k\}_{k=1}^{\infty}$  be an enumeration of  $\mathcal{A}$  and define the partial sums  $s_n = \sum_{k=1}^n \langle x, e_{\alpha_k} \rangle e_{\alpha_k}$ . For  $m > n$ , by the Pythagorean theorem:

$$\|s_m - s_n\|_H^2 = \sum_{k=n+1}^m |\langle x, e_{\alpha_k} \rangle|^2.$$

As  $n, m \rightarrow \infty$ , this goes to 0 because the total series  $\sum_{k=1}^{\infty} |\langle x, e_{\alpha_k} \rangle|^2$  converges. Thus  $\{s_n\}$  is Cauchy, and since  $H$  is complete, it converges to some  $y \in H$ . To show  $y = x$ , we look at the inner product for any  $\alpha \in A$ :

$$\langle x - y, e_\alpha \rangle = \langle x, e_\alpha \rangle - \lim_{n \rightarrow \infty} \sum_{k=1}^n \langle x, e_{\alpha_k} \rangle \langle e_{\alpha_k}, e_\alpha \rangle.$$

If  $\alpha \notin \mathcal{A}$ , both terms are 0. If  $\alpha \in \mathcal{A}$ , then  $\alpha = \alpha_n$  for some  $n$ , yielding  $\langle x, e_{\alpha_n} \rangle - \langle x, e_{\alpha_n} \rangle = 0$ . Since  $\langle x - y, e_{\alpha} \rangle = 0$  for all  $\alpha$ , by the completeness of the system,  $x - y = 0 \implies x = y$ .

iii. Applying Theorem 2 ii to the finite partial sums yields:

$$\|x\|_H^2 = \left\| x - \sum_{k=1}^n \langle x, e_{\alpha_k} \rangle e_{\alpha_k} \right\|_H^2 + \sum_{k=1}^n |\langle x, e_{\alpha_k} \rangle|^2.$$

Taking  $n \rightarrow \infty$ , the first norm term vanishes due to part ii, leaving Parseval's Identity.

## Theorem 5

A Hilbert space  $H$  has a countable CONS if and only if  $H$  is separable (contains a countable dense subset).

**Proof.** ( $\implies$ ) Let  $\{e_n\}_{n \geq 1}$  be a countable CONS. The set of all finite linear combinations  $\sum_{k=1}^n r_k e_k$  where  $r_k \in \mathbb{Q} + i\mathbb{Q}$  forms a countable dense subset in  $H$ . ( $\impliedby$ ) Let  $\{y_1, y_2, \dots\}$  be a countable dense subset. We can filter out linearly dependent vectors to get a linearly independent sequence  $\{x_n\}$  spanning the same subspace. Applying the standard Gram-Schmidt procedure to  $\{x_n\}$  constructs a countable orthonormal system  $\{e_n\}$  whose span is dense in  $H$ , hence forming a CONS.

## Proposition 2

Let  $H$  be a pre-Hilbert space. For a fixed  $z \in H$ , define  $\ell_z(x) = \langle x, z \rangle$ . Then  $\ell_z \in H^*$  and  $\|\ell_z\|_{H^*} = \|z\|_H$ .

**Proof.** Boundedness follows from Cauchy-Schwarz:  $|\ell_z(x)| = |\langle x, z \rangle| \leq \|z\|_H \|x\|_H$ , showing  $\|\ell_z\|_{H^*} \leq \|z\|_H$ . Testing the functional on the unit vector  $x = \frac{z}{\|z\|_H}$  (for  $z \neq 0$ ) yields exactly  $\|z\|_H$ , establishing equality.

## Riesz Representation Theorem

Let  $H$  be a Hilbert space. For every bounded linear functional  $\ell \in H^*$ , there exists a unique  $z_\ell \in H$  such that

$$\ell(x) = \langle x, z_\ell \rangle, \quad \forall x \in H.$$

Moreover,  $\|\ell\|_{H^*} = \|z_\ell\|_H$ .

**Proof.** If  $\ell = 0$ , choosing  $z_\ell = 0$  works. Assume  $\ell \neq 0$ , and let  $M = \ker(\ell)$ . Since  $\ell$  is continuous,  $M$  is a closed subspace, and  $M \neq H$ . By the Projection Theorem,  $M^\perp \neq \{0\}$ . Pick a nonzero  $x_0 \in M^\perp$  scaled such that  $\ell(x_0) = 1$ . For any  $x \in H$ , the vector  $x - \ell(x)x_0$  belongs to  $M$  because  $\ell(x - \ell(x)x_0) = \ell(x) - \ell(x) = 0$ . Since  $x_0 \in M^\perp$ , we have:

$$\langle x - \ell(x)x_0, x_0 \rangle = 0 \implies \langle x, x_0 \rangle = \ell(x) \|x_0\|_H^2 \implies \ell(x) = \left\langle x, \frac{x_0}{\|x_0\|_H^2} \right\rangle.$$

Setting  $z_\ell = \frac{x_0}{\|x_0\|_H^2}$  proves existence. Uniqueness follows since if  $\langle x, z_1 \rangle = \langle x, z_2 \rangle$  for all  $x$ , then  $\langle x, z_1 - z_2 \rangle = 0 \implies z_1 - z_2 = 0$ .

## Adjoint Operator

Let  $H$  be a Hilbert space. For each  $T \in \mathcal{L}(H)$ , there exists a unique  $T^* \in \mathcal{L}(H)$  such that

$$\langle Tx, y \rangle = \langle x, T^*y \rangle \quad \forall x, y \in H.$$

Moreover,  $\|T^*\| = \|T\|$ .

**Proof.** For a fixed  $y \in H$ ,  $x \mapsto \langle Tx, y \rangle$  is a bounded linear functional on  $H$ . By the Riesz Representation Theorem, there exists a unique vector, which we denote as  $T^*y$ , such that  $\langle Tx, y \rangle = \langle x, T^*y \rangle$ . To verify that  $T^*$  is linear on a complex Hilbert space, consider  $y_1, y_2 \in H$  and  $\alpha, \beta \in \mathbb{C}$ . For any  $x \in H$ :

$$\begin{aligned} \langle x, T^*(\alpha y_1 + \beta y_2) \rangle &= \langle Tx, \alpha y_1 + \beta y_2 \rangle \\ &= \overline{\alpha} \langle Tx, y_1 \rangle + \overline{\beta} \langle Tx, y_2 \rangle \quad (\text{conjugate-linear in 2nd slot}) \\ &= \overline{\alpha} \langle x, T^*y_1 \rangle + \overline{\beta} \langle x, T^*y_2 \rangle \\ &= \langle x, \alpha T^*y_1 + \beta T^*y_2 \rangle \quad (\text{bringing scalars back inside}) \end{aligned}$$

Since this holds for all  $x \in H$ ,  $T^*(\alpha y_1 + \beta y_2) = \alpha T^*y_1 + \beta T^*y_2$ . The norm equality follows from:

$$\|T^*\| = \sup_{\|y\| \leq 1} \sup_{\|x\| \leq 1} |\langle x, T^*y \rangle| = \sup_{\|y\| \leq 1} \sup_{\|x\| \leq 1} |\langle Tx, y \rangle| = \sup_{\|x\| \leq 1} \|Tx\|_H = \|T\|.$$

## Self-Adjoint Operator

An operator  $T \in \mathcal{L}(H)$  is called *self-adjoint* if  $T^* = T$ .

### Lemma

Let  $H$  be a Hilbert space and  $T \in \mathcal{L}(H)$ . Then,

$$\|T^*T\| = \|T\|^2 = \|T^*\|^2.$$

**Proof.** For any  $x \in H$  with  $\|x\|_H \leq 1$ :

$$\|Tx\|_H^2 = \langle Tx, Tx \rangle = \langle x, T^*Tx \rangle \leq \|x\|_H \|T^*Tx\|_H \leq \|T^*T\|.$$

Taking the supremum over all such  $x$  yields  $\|T\|^2 \leq \|T^*T\|$ . Combined with the standard norm inequality  $\|T^*T\| \leq \|T^*\| \|T\| = \|T\|^2$ , we get equality.

### Remark

Let  $H$  be a Hilbert space. Then:

1.  $\mathcal{A} := \mathcal{L}(H)$  is a  $C^*$ -algebra.
2.  $\mathcal{A}$  is a Banach algebra satisfying  $\|ST\| \leq \|S\| \|T\|$ .
3. The adjoint operation  $T \mapsto T^*$  is an involution satisfying:

- $(\alpha T + \beta S)^* = \overline{\alpha} T^* + \overline{\beta} S^*.$
- $(ST)^* = T^* S^*.$
- $\|T^* T\| = \|T\|^2.$

## $L^p$ Spaces

Let  $(X, \mathcal{X}, \mu)$  be a measure space. For  $0 < p \leq \infty$ , we define the space  $L^p(X, \mathcal{X}, \mu)$  (abbreviated as  $L^p(\mu)$  or  $L^p$ ) as:

$$L^p = \left\{ f : X \rightarrow \mathbb{C} \text{ is measurable} : \int_X |f|^p d\mu < \infty \right\}.$$

We identify functions that are equal almost everywhere (a.e.) to form equivalence classes. Additionally, we define  $L^\infty$  to be the space of equivalence classes of measurable functions  $f : X \rightarrow \mathbb{C}$  that are essentially bounded:

$$|f| \leq C \text{ a.e. for some } C > 0.$$

**Remark.**  $L^p$  is a vector space. For instance, if  $p \in (0, \infty)$  and  $a, b > 0$ , the inequality  $(a + b)^p \leq 2^p(a^p + b^p)$  holds because:

$$(a + b)^p \leq (2 \max\{a, b\})^p = 2^p \max\{a^p, b^p\} \leq 2^p(a^p + b^p).$$

For  $f \in L^p$  where  $p \in (0, \infty)$ , we define the  $L^p$  norm (or quasi-norm if  $p < 1$ ) as:

$$\|f\|_{L^p} = \|f\|_p = \left( \int_X |f|^p d\mu \right)^{1/p}.$$

For  $p = \infty$ , the essential supremum norm is defined as:

$$\|f\|_{L^\infty} = \|f\|_\infty = \inf\{C \geq 0 : |f| \leq C \text{ a.e.}\}.$$

*Note.* By definition,  $|f| \leq \|f\|_\infty$  a.e.

## Young's Inequality

For  $a, b \geq 0$ ,  $p \in (1, \infty)$ , and  $\frac{1}{p} + \frac{1}{q} = 1$ ,

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q}.$$

**Proof.** Let  $a, b > 0$  (the case where  $a = 0$  or  $b = 0$  is trivial). Since the natural logarithm function  $\ln$  is concave, we can apply Jensen's inequality:

$$\ln\left(\frac{a^p}{p} + \frac{b^q}{q}\right) \geq \frac{1}{p} \ln(a^p) + \frac{1}{q} \ln(b^q) = \ln(a) + \ln(b) = \ln(ab).$$

Taking the exponential (exp) of both sides yields the desired result.



## Hölder's Inequality

Let  $p \in [1, \infty]$  and let  $\frac{1}{p} + \frac{1}{q} = 1$  (with  $\frac{1}{\infty} = 0$ ). If  $f \in L^p$  and  $g \in L^q$ , then  $fg \in L^1$  and:

$$\int_X |fg| d\mu \leq \|f\|_p \|g\|_q$$

**Proof.** For  $p = 1$  or  $p = \infty$ , the proof is straightforward. Let  $p \in (1, \infty)$ ,  $f \neq 0$ , and  $g \neq 0$  a.e. By Young's inequality, for any parameter  $t > 0$ , we have:

$$\int_X |fg| d\mu = \int_X |(tf)(t^{-1}g)| d\mu \leq \frac{t^p}{p} \int_X |f|^p d\mu + \frac{t^{-q}}{q} \int_X |g|^q d\mu$$

Let  $A = \int_X |f|^p d\mu$  and  $B = \int_X |g|^q d\mu$ . Define the auxiliary function  $L(t)$  for  $t > 0$ :

$$L(t) = \frac{t^p}{p} A + \frac{t^{-q}}{q} B$$

To minimize  $L(t)$ , we find its derivative and set it to zero:

$$L'(t) = t^{p-1} A - t^{-q-1} B = 0 \implies t^{p+q} = \frac{B}{A} \implies t = \left(\frac{B}{A}\right)^{\frac{1}{pq}}$$

Evaluating  $L(t)$  at this critical point gives:

$$\inf_{t>0} L(t) = \frac{1}{p} \left(\frac{B}{A}\right)^{1/q} A + \frac{1}{q} \left(\frac{B}{A}\right)^{-1/p} B = \frac{1}{p} A^{1/p} B^{1/q} + \frac{1}{q} A^{1/p} B^{1/q} = A^{1/p} B^{1/q}$$

Substituting  $A^{1/p} = \|f\|_p$  and  $B^{1/q} = \|g\|_q$  completes the proof.

**Remark.** If  $f \in L^p$  and  $\frac{1}{p} + \frac{1}{q} = 1$ , then  $|f|^{p-1} \in L^q$ .

## Proposition 1 (Minkowski's Inequality)

For  $p \in [1, \infty]$ ,  $\|\cdot\|_p$  satisfies the triangle inequality and forms a norm on  $L^p$ .

**Proof.** For  $p = 1$  or  $p = \infty$ , the proof is left as an exercise. Let  $p \in (1, \infty)$  and  $f, g \in L^p$ . We assume  $f + g \neq 0$  dynamically. Splitting the integrand yields:

$$\int_X |f + g|^p d\mu = \int_X |f + g|^{p-1} |f + g| d\mu \leq \int_X |f + g|^{p-1} |f| d\mu + \int_X |f + g|^{p-1} |g| d\mu$$

Applying Hölder's inequality to both integrals on the right-hand side using conjugate exponents  $p$  and  $q = \frac{p}{p-1}$ , we get:

$$\int_X |f + g|^p d\mu \leq \left( \int_X |f + g|^{(p-1)q} d\mu \right)^{1/q} \|f\|_p + \left( \int_X |f + g|^{(p-1)q} d\mu \right)^{1/q} \|g\|_p$$

Since  $(p-1)q = p$  and  $1 - \frac{1}{q} = \frac{1}{p}$ , this simplifies directly to:

$$\int_X |f + g|^p d\mu \leq \left( \int_X |f + g|^p d\mu \right)^{1-\frac{1}{p}} (\|f\|_p + \|g\|_p)$$

Dividing both sides by  $\left( \int_X |f + g|^p d\mu \right)^{1-\frac{1}{p}}$  yields:

$$\|f + g\|_p \leq \|f\|_p + \|g\|_p$$

**Remark.** Let  $f_n, f \in L^\infty$ . Then  $\|f_n - f\|_\infty \rightarrow 0$  if and only if there exist representatives  $\tilde{f}_n, \tilde{f}$  in their respective equivalence classes such that  $\sup_{x \in X} |\tilde{f}_n(x) - \tilde{f}(x)| \rightarrow 0$  as  $n \rightarrow \infty$ .

### Chebyshev's Inequality

Let  $p \in (0, \infty)$  and  $f \in L^p$ . Then, for any  $\epsilon > 0$ :

$$\mu(\{x \in X : |f(x)| > \epsilon\}) \leq \frac{\int_X |f|^p d\mu}{\epsilon^p}$$

**Proof.** Let  $\epsilon > 0$ . By utilizing monotonic sets, we observe:

$$\int_X |f|^p d\mu \geq \int_{\{|f| > \epsilon\}} |f|^p d\mu \geq \epsilon^p \mu(\{x \in X : |f(x)| > \epsilon\})$$

### Theorem 2

$L^p$  is a Banach space for any  $p \in [1, \infty]$ .

**Proof.** The case  $p = \infty$  is left as an exercise. Let  $p \in [1, \infty)$  and let  $(f_n)$  be a Cauchy sequence in  $L^p$ . By Chebyshev's Inequality, for any  $\epsilon > 0$ :

$$\mu(\{|f_n - f_m| > \epsilon\}) \leq \frac{\int_X |f_n - f_m|^p d\mu}{\epsilon^p} \xrightarrow{n, m \rightarrow \infty} 0$$

Hence,  $(f_n)$  is Cauchy in measure. This implies there exists a measurable function  $f$  and a subsequence  $(f_{n_k})$  such that  $f_{n_k} \rightarrow f$  a.e. as  $k \rightarrow \infty$ .

Fix  $\epsilon > 0$ . Choose  $N$  such that  $\int_X |f_n - f_m|^p d\mu \leq \epsilon$  for all  $n, m \geq N$ . Letting  $m = n_l \rightarrow \infty$ , Fatou's Lemma implies:

$$\int_X |f_n - f|^p d\mu = \int_X \lim_{l \rightarrow \infty} |f_n - f_{n_l}|^p d\mu \leq \liminf_{l \rightarrow \infty} \int_X |f_n - f_{n_l}|^p d\mu \leq \epsilon$$

for all  $n \geq N$ . Thus,  $f - f_n \in L^p$ , which means  $f = (f - f_n) + f_n \in L^p$ , and  $f_n \rightarrow f$  in  $L^p$ .

## Comparison of $L^p$ Spaces

In general, spaces are not nested: neither  $L^p \subseteq L^q$  nor  $L^q \subseteq L^p$  if  $p \neq q$  on arbitrary measure spaces.

**Proof.** Let  $X = (0, \infty)$  equipped with Lebesgue measure  $\mu$ . Consider the function:

$$f(x) = x^{-\alpha} \chi_{(0,1)}(x) + x^{-\beta} \chi_{[1,\infty)}(x) \quad (\alpha, \beta > 0)$$

Calculating the  $L^r$  integral gives  $\int_X |f(x)|^r d\mu = \int_0^1 x^{-\alpha r} dx + \int_1^\infty x^{-\beta r} dx$ . This converges if and only if  $\alpha < \frac{1}{r} < \beta$ . Thus, if we select exponents such that  $\frac{1}{q} < \alpha < \frac{1}{p} < \beta$ , then  $f \in L^p$  but  $f \notin L^q$ . Conversely, choosing  $\alpha < \frac{1}{q} < \beta < \frac{1}{p}$  yields  $f \in L^q$  but  $f \notin L^p$ .

## Proposition 3

Let  $\mu(X) < \infty$  and  $0 < p < q \leq \infty$ . Then  $L^q \subseteq L^p$  and:

$$\|f\|_p \leq \|f\|_q \mu(X)^{\frac{1}{p} - \frac{1}{q}}$$

**Proof.** Applying Hölder's inequality with conjugate exponents  $\frac{q}{p}$  and  $\frac{q}{q-p}$  yields:

$$\int_X |f|^p \cdot 1 d\mu \leq \left( \int_X |f|^q d\mu \right)^{\frac{p}{q}} \left( \int_X 1 d\mu \right)^{1 - \frac{p}{q}} = \|f\|_q^p \mu(X)^{1 - \frac{p}{q}}$$

Taking the  $p$ -th root of both sides proves the assertion.

## Proposition 4

Let  $0 < p < q < r \leq \infty$ . Then  $L^q \subseteq L^p + L^r$ .

**Proof.** Let  $f \in L^q$ . We decompose  $f = f \chi_{\{|f| \leq 1\}} + f \chi_{\{|f| > 1\}} =: k + h$ . Then  $|k| \leq |f|$  and  $|k| \leq 1$ , so  $|k|^r \leq |f|^q \in L^1$ , implying  $k \in L^r$ . Similarly,  $|h| \leq |f|$  and on the support of  $h$ ,  $|f| > 1$ , meaning  $|h|^p \leq |f|^q \in L^1$ , implying  $h \in L^p$ .

## Proposition 5

Let  $0 < p < q < r \leq \infty$ . Then  $L^p \cap L^r \subseteq L^q$  and  $\|f\|_q \leq \|f\|_p^t \|f\|_r^{1-t}$  where  $\frac{1}{q} = \frac{t}{p} + \frac{1-t}{r}$ .

**Proof.** Let  $f \in L^p \cap L^r$ .

- i. If  $r = \infty$ , then  $\int_X |f|^q d\mu = \int_X |f|^{q-p} |f|^p d\mu \leq \|f\|_\infty^{q-p} \|f\|_p^p$ . Taking the  $q$ -th root gives the result with  $t = p/q$ .
- ii. If  $r < \infty$ , rewrite  $|f|^q = |f|^{tq} |f|^{(1-t)q}$ . Apply Hölder's inequality with conjugate exponents  $\frac{p}{tq}$  and  $\frac{r}{(1-t)q}$ . The conditions require  $\frac{tq}{p} + \frac{(1-t)q}{r} = 1$ , which aligns perfectly with the definition of  $t$ .

### Proposition 6 (Sequence Spaces)

If  $X = \mathbb{N}$  and  $\mu$  is the counting measure, we denote  $\ell^p = L^p(\mathbb{N}, \mu)$ . If  $0 < p < q \leq \infty$ , then  $\ell^p \subseteq \ell^q$  and  $\|x\|_q \leq \|x\|_p$ .

**Proof.** Let  $x \in \ell^p$ . For  $q = \infty$ ,  $\|x\|_\infty^p = \sup_n |x_n|^p \leq \sum_n |x_n|^p = \|x\|_p^p \implies \|x\|_\infty \leq \|x\|_p$ . For  $q < \infty$ , by interpolation (Proposition 5),  $\|x\|_q \leq \|x\|_p^{p/q} \|x\|_\infty^{1-p/q} \leq \|x\|_p$ .

### Simple Function Spaces

Define  $\Sigma = \{f \in \mathcal{E} : f \text{ vanishes outside a set of finite measure}\}$ , where  $\mathcal{E}$  is the space of all complex-valued simple functions:

$$\mathcal{E} = \left\{ f = \sum_{j=1}^n z_j \chi_{E_j} : n \geq 1, z_j \in \mathbb{C}, E_j \in \mathcal{X} \text{ are disjoint} \right\}$$

**Remark.**  $\mathcal{E} \subseteq L^\infty$  and  $\Sigma \subseteq L^p$  for any  $p \in (0, \infty]$ .

### Theorem 2.10 [Folland]

Let  $(X, \mathcal{X})$  be a measurable space. If  $f : X \rightarrow \mathbb{C}$  is measurable, there exists a sequence  $\{f_n\} \subseteq \mathcal{E}$  such that  $0 \leq |f_1| \leq |f_2| \leq \dots \leq |f|$ ,  $f_n \rightarrow f$  pointwise, and  $f_n \rightarrow f$  uniformly on any set where  $f$  is bounded.

### Proposition 7

- i.  $\mathcal{E}$  is dense in  $L^\infty$ .
- ii.  $\Sigma$  is dense in  $L^p$  if  $p \in [1, \infty)$ .

**Proof.** Apply Theorem 2.10. For i, since  $f \in L^\infty$  is bounded outside a null set,  $f_n \rightarrow f$  uniformly a.e., meaning  $\|f_n - f\|_\infty \rightarrow 0$ . For ii, if  $f \in L^p$ ,  $|f_n - f|^p \leq 2^{p+1}|f|^p \in L^1$ . By the Dominated Convergence Theorem,  $\int_X |f_n - f|^p d\mu \rightarrow 0$ .

### Generalized Dominated Convergence Theorem

Let  $\{f_n\}$  be a sequence of measurable functions such that  $f_n \rightarrow f$  a.e. Suppose there exists a sequence of nonnegative integrable functions  $\{g_n\}$  such that  $|f_n| \leq g_n$  a.e.,  $g_n \rightarrow g$  a.e., and  $\int_X g_n d\mu \rightarrow \int_X g d\mu < \infty$ . Then  $f_n \rightarrow f$  in  $L^1$ .

**Proof.** Consider the sequence of nonnegative functions  $h_n = g_n + g - |f_n - f|$ . Since  $|f_n - f| \leq g_n + g$  a.e.,  $h_n \geq 0$  a.e. Applying Fatou's Lemma to  $\{h_n\}$  yields:

$$\int_X \liminf_{n \rightarrow \infty} h_n d\mu \leq \liminf_{n \rightarrow \infty} \int_X h_n d\mu$$

Since  $f_n \rightarrow f$  and  $g_n \rightarrow g$  a.e.,  $\liminf h_n = 2g$  a.e. Thus:

$$\int_X 2g d\mu \leq \int_X g d\mu + \int_X g d\mu - \limsup_{n \rightarrow \infty} \int_X |f_n - f| d\mu$$

Subtracting  $2 \int_X g d\mu < \infty$  from both sides results in  $\limsup_{n \rightarrow \infty} \int_X |f_n - f| d\mu \leq 0$ , which ensures  $\lim_{n \rightarrow \infty} \|f_n - f\|_1 = 0$ .

### The Dual of $L^p$

Let  $p \in (1, \infty]$  and  $q \in [1, \infty)$  with  $\frac{1}{p} + \frac{1}{q} = 1$ . For any  $h \in L^q$ , define the linear functional  $T_h$  on  $L^p$  by:

$$T_h(f) = \int_X fh d\mu, \quad f \in L^p$$

Then  $T_h \in (L^p)^*$  and  $\|T_h\|_{(L^p)^*} = \|h\|_q = \sup\{|\int_X fh d\mu| : \|f\|_p \leq 1\}$ .

### Theorem 4 (Riesz Representation Theorem for $L^p$ )

Let  $p \in (1, \infty)$ . For each bounded linear functional  $T \in (L^p)^*$ , there exists a unique  $h \in L^q$  such that  $T(f) = \int_X hf d\mu$  for all  $f \in L^p$ , and  $\|T\|_{(L^p)^*} = \|h\|_q$ . If  $\mu$  is  $\sigma$ -finite, this also holds for  $p = 1$  and  $q = \infty$ .

### Weak $L^p$

Let  $p \in (0, \infty)$ . A measurable function  $f$  belongs to *weak  $L^p$*  if its distribution function  $\lambda_f(\alpha) = \mu(\{x \in X : |f(x)| > \alpha\})$  satisfies:

$$[f]_p := \left( \sup_{\alpha > 0} \alpha^p \lambda_f(\alpha) \right)^{1/p} < \infty$$

1. By Chebyshev's Inequality,  $L^p \subseteq \text{weak } L^p$  and  $[f]_p \leq \|f\|_p$ .
2. Using layer-cake integration by parts, if  $f \in L^p$ , its norm can be computed via:

$$\|f\|_p^p = \int_X |f|^p d\mu = p \int_0^\infty \alpha^{p-1} \lambda_f(\alpha) d\alpha$$

## Convolution and Smoothing

Let  $X = \mathbb{R}^d$ ,  $\mathcal{X} = \mathcal{L}$  (the Lebesgue  $\sigma$ -algebra), and let  $\mu = m$  denote the standard Lebesgue measure, so  $d\mu = dm = dx$ .

### Lemma 1 (Commutativity of Integrability)

Let  $f, g : \mathbb{R}^d \rightarrow \mathbb{C}$  be measurable functions, and fix  $x_0 \in \mathbb{R}^d$ . Then,

$$f(x_0 - \cdot)g(\cdot) \in L^1(\mathbb{R}^d) \iff g(x_0 - \cdot)f(\cdot) \in L^1(\mathbb{R}^d) \quad (1)$$

and whenever the compatibility condition (1) holds, we have the identity:

$$\int_{\mathbb{R}^d} f(x_0 - y)g(y) dy = \int_{\mathbb{R}^d} g(x_0 - y)f(y) dy.$$

**Proof.** Let  $G : \mathbb{R}^d \rightarrow \mathbb{R}^d$  be a global diffeomorphism (bijective with  $G, G^{-1} \in C^1(\mathbb{R}^d)$ ), defining the coordinate transformation  $x = G(y) = (g_1(y), \dots, g_d(y))$ . By the multi-dimensional change-of-variables formula:

$$\int_{\mathbb{R}^d} f(x) dx = \int_{\mathbb{R}^d} f(G(y)) |\det \nabla G(y)| dy,$$

where the Jacobian matrix  $\nabla G(y)$  is given by:

$$\nabla G(y) = \begin{bmatrix} \frac{\partial g_1(y)}{\partial y_1} & \dots & \frac{\partial g_1(y)}{\partial y_d} \\ \vdots & \ddots & \vdots \\ \frac{\partial g_d(y)}{\partial y_1} & \dots & \frac{\partial g_d(y)}{\partial y_d} \end{bmatrix}.$$

In our specific framework, we set the translation-reflection map  $G(y) = x_0 - y = z$ . The corresponding Jacobian is:

$$\nabla G(y) = \begin{bmatrix} -1 & 0 & \dots & 0 \\ 0 & -1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & -1 \end{bmatrix} = -I_d \implies |\det \nabla G(y)| = |(-1)^d| = 1.$$

Applying this transformation preserves the integration measure ( $dy = dz$ ) and yields:

$$\int_{\mathbb{R}^d} f(x_0 - y)g(y) dy = \int_{\mathbb{R}^d} f(G(y))g(x_0 - G(y)) |\det \nabla G(y)| dy = \int_{\mathbb{R}^d} g(x_0 - z)f(z) dz.$$

□

## Convolution

Let  $f, g : \mathbb{R}^d \rightarrow \mathbb{C}$  be measurable functions such that the integrability condition  $f(x - \cdot)g(\cdot) \in L^1(\mathbb{R}^d)$  holds for  $m$ -almost every  $x \in \mathbb{R}^d$ . The *convolution* of  $f$  and  $g$ , denoted by  $f * g$ , is the well-defined function given by:

$$(f * g)(x) = \int_{\mathbb{R}^d} f(x - y)g(y) dy = \int_{\mathbb{R}^d} g(x - y)f(y) dy = (g * f)(x).$$

We say that  $(f * g)(x)$  is defined at any point  $x \in \mathbb{R}^d$  where the absolute integrand converges:

$$\int_{\mathbb{R}^d} |f(x - y)||g(y)| dy < \infty.$$

## Support

Given a function  $f : \mathbb{R}^d \rightarrow \mathbb{C}$ , the *support* of  $f$ , denoted  $\text{supp}(f)$ , is the topological closure of its non-vanishing set:

$$\text{supp}(f) = \overline{\{x \in \mathbb{R}^d : f(x) \neq 0\}}.$$

We denote by  $C_c(\mathbb{R}^d)$  the vector space of continuous functions with compact support.

## Lemma 2 (Convolution of Compactly Supported Functions)

If  $f, g \in C_c(\mathbb{R}^d)$ , then their convolution  $f * g \in C_c(\mathbb{R}^d)$ . Moreover, its support satisfies:

$$\text{supp}(f * g) \subseteq \text{supp}(f) + \text{supp}(g),$$

where  $A + B = \{x + y : x \in A, y \in B\}$  denotes the algebraic Minkowski sum.

**Proof.** Because  $f, g \in C_c(\mathbb{R}^d)$ , both functions are continuous on compact sets and are thus uniformly bounded. Consequently, the integral converges absolutely for every  $x \in \mathbb{R}^d$ :

$$\int_{\mathbb{R}^d} |f(x - y)| |g(y)| dy < \infty \quad \forall x \in \mathbb{R}^d.$$

Hence,  $f * g$  is pointwise well-defined across all of  $\mathbb{R}^d$ .

To establish continuity, let  $\{x_n\}_{n=1}^\infty$  be a sequence in  $\mathbb{R}^d$  converging to  $x_0$ . Define the auxiliary sequences  $h_n(y) = [f(x_n - y) - f(x_0 - y)]g(y)$  and  $h(y) = 0$ . Since  $f$  is continuous,  $f(x_n - y) \rightarrow f(x_0 - y)$  pointwise for each fixed  $y$ , meaning  $h_n(y) \rightarrow h(y) = 0$  pointwise everywhere.

To invoke the Dominated Convergence Theorem, let  $M = \|f\|_\infty < \infty$ . We find a uniform  $L^1$ -dominating function  $\phi$ :

$$|h_n(y)| = |[f(x_n - y) - f(x_0 - y)]g(y)| \leq (|f(x_n - y)| + |f(x_0 - y)|)|g(y)| \leq 2M|g(y)|.$$

Since  $g \in C_c(\mathbb{R}^d)$ , the function  $\phi(y) = 2M|g(y)|$  is continuous with compact support and is therefore integrable ( $\phi \in L^1(\mathbb{R}^d)$ ). Applying the DCT:

$$\lim_{n \rightarrow \infty} ((f * g)(x_n) - (f * g)(x_0)) = \lim_{n \rightarrow \infty} \int_{\mathbb{R}^d} h_n(y) dy = \int_{\mathbb{R}^d} \lim_{n \rightarrow \infty} h_n(y) dy = 0.$$

This proves  $f * g$  is continuous.

Now consider the algebraic restriction on non-zero values. If  $(f * g)(x) \neq 0$ , there must exist a point  $y \in \mathbb{R}^d$  such that  $f(x - y)g(y) \neq 0$ . This implies:

$$x - y \in \text{supp}(f) \quad \text{and} \quad y \in \text{supp}(g),$$

which yields  $x = (x - y) + y \in \text{supp}(f) + \text{supp}(g)$ , and thus:

$$\{x \in \mathbb{R}^d : (f * g)(x) \neq 0\} \subseteq \text{supp}(f) + \text{supp}(g).$$

Taking the topological closure on both sides gives:

$$\text{supp}(f * g) = \overline{\{x \in \mathbb{R}^d : (f * g)(x) \neq 0\}} \subseteq \overline{\text{supp}(f) + \text{supp}(g)}.$$

Since  $\text{supp}(f)$  and  $\text{supp}(g)$  are compact subsets of  $\mathbb{R}^d$ , their Minkowski sum  $\text{supp}(f) + \text{supp}(g)$  is also compact, which implies it is closed. Therefore, the closure bar can be dropped:

$$\overline{\text{supp}(f) + \text{supp}(g)} = \text{supp}(f) + \text{supp}(g),$$

which confirms that  $\text{supp}(f * g) \subseteq \text{supp}(f) + \text{supp}(g)$ . Because  $\text{supp}(f * g)$  is a closed subset of a compact set, it is compact, meaning  $f * g \in C_c(\mathbb{R}^d)$ .  $\square$

**Remark.**

1. Any continuous function with compact support is automatically bounded ( $C_c(\mathbb{R}^d) \subseteq C_b(\mathbb{R}^d)$ ) by the Extreme Value Theorem, since it achieves finite absolute maxima on its compact support domain and vanishes identically outside it.
2. The standard Lebesgue measure space  $(\mathbb{R}^d, \mathcal{L}, m)$  is  $\sigma$ -finite, as it can be exhaustively partitioned into a countable union of compact balls  $B_n(0) = \{x \in \mathbb{R}^d : |x| \leq n\}$ , each possessing finite volume:

$$m(B_n(0)) = \frac{\pi^{d/2}}{\Gamma(\frac{d}{2} + 1)} n^d < \infty.$$

**Proposition 1 (Convolution  $L^p \times L^q$ )**

Let  $p, q \in [1, \infty]$  be conjugate exponents satisfying  $\frac{1}{p} + \frac{1}{q} = 1$ . If  $f \in L^p(\mathbb{R}^d)$  and  $h \in L^q(\mathbb{R}^d)$ , then  $(f * h)(x)$  is explicitly defined for all  $x \in \mathbb{R}^d$  and satisfies the uniform bound:

$$\|f * h\|_\infty = \sup_{x \in \mathbb{R}^d} |(f * h)(x)| \leq \|f\|_p \|h\|_q.$$

Furthermore,  $f * h$  is uniformly continuous on  $\mathbb{R}^d$ , and if  $p \in (1, \infty)$ , it vanishes at infinity:  $\lim_{|x| \rightarrow \infty} (f * h)(x) = 0$ .

**Proof.** For any fixed  $x \in \mathbb{R}^d$ , the shift-invariance of the Lebesgue measure implies  $\|f(x - \cdot)\|_p = \|f\|_p$ . Applying Hölder's inequality:

$$\int_{\mathbb{R}^d} |f(x - y)| |h(y)| dy \leq \|f(x - \cdot)\|_p \|h\|_q = \|f\|_p \|h\|_q < \infty.$$

Thus, the convolution integral converges absolutely for every  $x$ , and the triangle inequality yields:

$$|(f * h)(x)| \leq \int_{\mathbb{R}^d} |f(x - y)| |h(y)| dy \leq \|f\|_p \|h\|_q \implies \sup_{x \in \mathbb{R}^d} |(f * h)(x)| \leq \|f\|_p \|h\|_q.$$

*Continuity Analysis:*

- i. Let  $p \in (1, \infty)$ , which implies  $q \in (1, \infty)$ . By the density of smooth, compactly supported functions in  $L^p$  spaces, there exist sequences  $\{f_n\}_{n=1}^\infty, \{h_n\}_{n=1}^\infty \subseteq C_c(\mathbb{R}^d)$  such that  $\|f_n - f\|_p \rightarrow 0$  and  $\|h_n - h\|_q \rightarrow 0$ . By Lemma 2, each  $f_n * h_n \in C_c(\mathbb{R}^d)$ . Expanding the difference using a telescoping sum gives:

$$\begin{aligned} \sup_{x \in \mathbb{R}^d} |(f_n * h_n)(x) - (f * h)(x)| &\leq \sup_{x \in \mathbb{R}^d} |((f_n - f) * h_n)(x)| + \sup_{x \in \mathbb{R}^d} |(f * (h_n - h))(x)| \\ &\leq \|f_n - f\|_p \|h_n\|_q + \|f\|_p \|h_n - h\|_q \\ &\leq \left( \sup_{k \in \mathbb{N}} \|h_k\|_q \right) \|f_n - f\|_p + \|f\|_p \|h_n - h\|_q \xrightarrow{n \rightarrow \infty} 0. \end{aligned}$$

Because the sequence of continuous, compactly supported functions  $f_n * h_n$  converges uniformly to  $f * h$ , the limit  $f * h$  is uniformly continuous. Furthermore, since each  $f_n * h_n$  vanishes identically outside a sufficiently large ball, uniform convergence preserves this behavior at infinity, so  $\lim_{|x| \rightarrow \infty} (f * h)(x) = 0$ .



- ii. Let  $p = 1$ , implying  $q = \infty$ , so  $f \in L^1(\mathbb{R}^d)$  and  $h \in L^\infty(\mathbb{R}^d)$ . Let  $f_n \in C_c(\mathbb{R}^d)$  be a sequence approximating  $f$  in  $L^1$  norm ( $\|f_n - f\|_1 \rightarrow 0$ ). Then:

$$\sup_{x \in \mathbb{R}^d} |(f_n * h)(x) - (f * h)(x)| \leq \|f_n - f\|_1 \|h\|_\infty \xrightarrow{n \rightarrow \infty} 0.$$

Thus,  $f_n * h \rightarrow f * h$  uniformly. To check the continuity of each  $f_n * h$ , we invoke the DCT on the shifted functions. Since  $f_n \in C_c(\mathbb{R}^d)$ , it is uniformly continuous, so  $\lim_{x \rightarrow x_0} [f_n(x - y) - f_n(x_0 - y)] = 0$  pointwise everywhere. Since the integrand is dominated by  $2\|f_n\|_\infty |h(y)| \chi_{\text{supp}(f_n)}(x - y)$ , the DCT gives:

$$\lim_{x \rightarrow x_0} ((f_n * h)(x) - (f_n * h)(x_0)) = \int_{\mathbb{R}^d} \lim_{x \rightarrow x_0} [f_n(x - y) - f_n(x_0 - y)] h(y) dy = 0.$$

The uniform limit of these continuous functions guarantees that  $f * h$  is continuous.  $\square$

### Proposition 2 (Young's Inequality for Convolutions - Base Case)

Let  $p \in [1, \infty)$ ,  $f \in L^p(\mathbb{R}^d)$ , and  $h \in L^1(\mathbb{R}^d)$ . Then  $f * h$  is well-defined  $m$ -almost everywhere,  $f * h \in L^p(\mathbb{R}^d)$ , and satisfies:

$$\|f * h\|_p \leq \|f\|_p \|h\|_1.$$

**Proof.** We use Minkowski's inequality for integrals (the continuous integral version of the triangle inequality):

$$\begin{aligned} \|f * h\|_p &= \left( \int_{\mathbb{R}^d} \left| \int_{\mathbb{R}^d} f(x - y) h(y) dy \right|^p dx \right)^{1/p} \leq \int_{\mathbb{R}^d} \left( \int_{\mathbb{R}^d} |f(x - y)|^p dx \right)^{1/p} |h(y)| dy \\ &= \int_{\mathbb{R}^d} \|f\|_p |h(y)| dy \\ &= \|f\|_p \|h\|_1 < \infty. \end{aligned}$$

Fubini's theorem implies the absolute integral  $\int_{\mathbb{R}^d} |f(x - y)| |h(y)| dy$  is finite for  $m$ -almost every  $x$ , confirming that  $f * h \in L^p(\mathbb{R}^d)$  is well-defined with  $\|f * h\|_p \leq \|f\|_p \|h\|_1$ .  $\square$

### Continuity of Shifts

For a measurable function  $f$  on  $\mathbb{R}^d$  and a translation vector  $z \in \mathbb{R}^d$ , the translation operator  $\tau_z$  is defined as:

$$\tau_z f(x) = f(x + z), \quad x \in \mathbb{R}^d.$$

By the translation invariance of Lebesgue measure,  $\|\tau_z f\|_p = \|f\|_p$  for any  $p \in (0, \infty]$ . Note that  $\tau_0$  is the identity operator.

### Lemma 3 ( $L^p$ Continuity of Translation)

Let  $p \in [1, \infty)$  and  $f \in L^p(\mathbb{R}^d)$ . Then:

1.  $\lim_{z \rightarrow 0} \|\tau_z f - f\|_p = 0$ , which means  $\int_{\mathbb{R}^d} |f(x+z) - f(x)|^p dx \xrightarrow{z \rightarrow 0} 0$ .
2.  $\lim_{z \rightarrow z_0} \|\tau_z f - \tau_{z_0} f\|_p = \lim_{z \rightarrow z_0} \|\tau_{z-z_0} f - f\|_p = 0$ .

**Proof.** First, let  $g \in C_c(\mathbb{R}^d)$ . Fix a compact neighborhood of the translation vector, say  $|z| \leq 1$ . If  $\text{supp}(g) \subseteq B_R(0)$ , then for all  $|z| \leq 1$ , the shifted support satisfies  $\text{supp}(\tau_z g) \subseteq B_{R+1}(0)$ . We can bound the shifted difference using a uniform  $L^p$  envelope:

$$|g(x+z) - g(x)|^p \leq (|g(x+z)| + |g(x)|)^p \leq (2\|g\|_\infty \chi_{B_{R+1}(0)}(x))^p \in L^1(\mathbb{R}^d).$$

Since  $g$  is continuous,  $\lim_{z \rightarrow 0} |g(x+z) - g(x)|^p = 0$  pointwise everywhere. Applying the DCT directly yields  $\lim_{z \rightarrow 0} \|\tau_z g - g\|_p = 0$ .

Now let  $f \in L^p(\mathbb{R}^d)$  be arbitrary. For any test function  $g \in C_c(\mathbb{R}^d)$ , we use the triangle inequality to insert  $g$ :

$$\begin{aligned} \|\tau_z f - f\|_p &\leq \|\tau_z f - \tau_z g\|_p + \|\tau_z g - g\|_p + \|g - f\|_p \\ &= \|\tau_z(f - g)\|_p + \|\tau_z g - g\|_p + \|g - f\|_p \\ &= 2\|f - g\|_p + \|\tau_z g - g\|_p. \end{aligned}$$

Taking the limit superior as  $z \rightarrow 0$  gives:

$$\limsup_{z \rightarrow 0} \|\tau_z f - f\|_p \leq 2\|f - g\|_p + \limsup_{z \rightarrow 0} \|\tau_z g - g\|_p = 2\|f - g\|_p.$$

Since  $C_c(\mathbb{R}^d)$  is dense in  $L^p(\mathbb{R}^d)$  for  $p < \infty$ , for any  $\epsilon > 0$  we can choose a function  $g \in C_c(\mathbb{R}^d)$  such that  $\|f - g\|_p < \frac{\epsilon}{2}$ . This means:

$$\limsup_{z \rightarrow 0} \|\tau_z f - f\|_p \leq 2 \left( \frac{\epsilon}{2} \right) = \epsilon.$$

Because  $\epsilon > 0$  is arbitrary, we have  $\limsup_{z \rightarrow 0} \|\tau_z f - f\|_p = 0$ , which implies  $\lim_{z \rightarrow 0} \|\tau_z f - f\|_p = 0$ .  $\square$

### Standard Mollifiers

Let  $\ell : \mathbb{R} \rightarrow \mathbb{R}$  be the standard smooth bump generator, and define the radial profile  $\tilde{w}(x) := \ell(|x|_2^2)$  for  $x \in \mathbb{R}^d$ , where  $|x|_2 = \sqrt{x_1^2 + \dots + x_d^2}$ . This construction ensures that:

$$\tilde{w} \in C_c^\infty(\mathbb{R}^d) \quad \text{and} \quad \text{supp}(\tilde{w}) = \overline{B_1(0)}.$$

We define the normalized profile  $w(x) = c\tilde{w}(x)$ , picking the scaling constant  $c > 0$  such that  $\int_{\mathbb{R}^d} w(x) dx = 1$ . For any scale parameter  $\epsilon > 0$ , we define the *standard mollifier*  $w_\epsilon$  by:

$$w_\epsilon(x) := \epsilon^{-d} w\left(\frac{x}{\epsilon}\right), \quad x \in \mathbb{R}^d.$$

By construction,  $w_\varepsilon \in C_c^\infty(\mathbb{R}^d)$ ,  $\text{supp}(w_\varepsilon) = \overline{B_\varepsilon(0)}$ ,  $w_\varepsilon(x) \geq 0$ , and  $\int_{\mathbb{R}^d} w_\varepsilon(x) dx = 1$  for all  $\varepsilon > 0$ .

*Probabilistic Interpretation.* The mollifier  $w$  can be viewed as the probability density function of a random variable  $\xi \in \mathbb{R}^d$  that satisfies  $|\xi|_2 \leq 1$  almost surely. The rescaled function  $w_\varepsilon$  then acts as the density of the scaled variable  $\varepsilon\xi$ . Expectation computations align directly with scaling transformations:

$$\mathbb{E}[f(x - \varepsilon\xi)] = \int_{\mathbb{R}^d} f(x - \varepsilon y) w(y) dy = \int_{\mathbb{R}^d} f(x - z) w_\varepsilon(z) dz = (f * w_\varepsilon)(x).$$

### Smoothing Theorem

Let  $p \in [1, \infty)$  and  $f \in L^p(\mathbb{R}^d)$ . We define the mollified approximation  $f_\varepsilon$  by:

$$f_\varepsilon(x) = (f * w_\varepsilon)(x) = \int_{\mathbb{R}^d} f(x - y) w_\varepsilon(y) dy = \mathbb{E}[f(x - \varepsilon\xi)], \quad x \in \mathbb{R}^d.$$

Then the following properties hold:

1.  $f_\varepsilon \in L^p(\mathbb{R}^d)$ , with  $\|f_\varepsilon\|_p \leq \|f\|_p$ , and  $f_\varepsilon \rightarrow f$  in  $L^p$  norm:  $\lim_{\varepsilon \rightarrow 0} \|f_\varepsilon - f\|_p = 0$ .
2.  $f_\varepsilon \in C_b^\infty(\mathbb{R}^d)$ , meaning  $f_\varepsilon$  is infinitely differentiable with bounded derivatives of all orders.
3. For any multi-index  $\gamma = (\gamma_1, \dots, \gamma_d)$  with order  $|\gamma| = \gamma_1 + \dots + \gamma_d$ , the differential operator  $D^\gamma = \frac{\partial^{|\gamma|}}{\partial x_1^{\gamma_1} \dots \partial x_d^{\gamma_d}}$  distributes onto the mollifier:

$$D^\gamma f_\varepsilon(x) = \int_{\mathbb{R}^d} D^\gamma w_\varepsilon(x - y) f(y) dy = ((D^\gamma w_\varepsilon) * f)(x), \quad x \in \mathbb{R}^d.$$

### Proof.

1. Applying Proposition 2 with  $\|w_\varepsilon\|_1 = 1$  gives  $\|f_\varepsilon\|_p = \|f * w_\varepsilon\|_p \leq \|f\|_p \|w_\varepsilon\|_1 = \|f\|_p$ . To prove norm convergence, we exploit the identity  $\int_{\mathbb{R}^d} w_\varepsilon(y) dy = 1$  to rewrite the difference:

$$|f_\varepsilon(x) - f(x)| = \left| \int_{\mathbb{R}^d} [f(x - y) - f(x)] w_\varepsilon(y) dy \right| \leq \int_{\mathbb{R}^d} |\tau_{-y} f(x) - f(x)| w_\varepsilon(y) dy.$$

Taking the  $L^p$  norm and applying Minkowski's inequality for integrals yields:

$$\|f_\varepsilon - f\|_p \leq \int_{\mathbb{R}^d} \|\tau_{-y} f - f\|_p w_\varepsilon(y) dy.$$

Since  $w_\varepsilon(y)$  vanishes outside  $B_\varepsilon(0)$ , we can restrict the integration domain and bound the integrand by its supremum over that ball:

$$\|f_\varepsilon - f\|_p \leq \int_{B_\varepsilon(0)} \left( \sup_{|y| \leq \varepsilon} \|\tau_{-y} f - f\|_p \right) w_\varepsilon(y) dy = \sup_{|y| \leq \varepsilon} \|\tau_{-y} f - f\|_p.$$

By Lemma 3,  $\lim_{\varepsilon \rightarrow 0} (\sup_{|y| \leq \varepsilon} \|\tau_{-y} f - f\|_p) = 0$ , which proves that  $\lim_{\varepsilon \rightarrow 0} \|f_\varepsilon - f\|_p = 0$ .

2. The infinite differentiability follows from Leibniz's rule for differentiating under the integral sign, which is justified because  $w_\varepsilon \in C_c^\infty(\mathbb{R}^d)$  has bounded derivatives of all orders.
3. Let  $q \in [1, \infty]$  be the conjugate exponent of  $p$ . Because  $w_\varepsilon \in C_c^\infty(\mathbb{R}^d)$ , taking any partial derivative preserves smoothness and compact support, so  $D^\gamma w_\varepsilon \in C_c^\infty(\mathbb{R}^d) \subset L^q(\mathbb{R}^d)$ . Applying Proposition 1:

$$\sup_{x \in \mathbb{R}^d} |D^\gamma f_\varepsilon(x)| = \sup_{x \in \mathbb{R}^d} |((D^\gamma w_\varepsilon) * f)(x)| \leq \|f\|_p \|D^\gamma w_\varepsilon\|_q < \infty.$$

This confirms that every derivative is uniformly bounded, so  $f_\varepsilon \in C_b^\infty(\mathbb{R}^d)$ . □

## Representations of Positive Linear Functionals

A linear functional  $\ell : C(X) \rightarrow \mathbb{R}$  is *positive* if  $\ell(f) \geq 0$  whenever  $f \geq 0$ .

**Remark.** If  $f \geq g$ , then  $\ell(f) \geq \ell(g)$  for a positive functional  $\ell$ . In fact,  $\ell$  is positive if and only if this property holds.

### Proposition 2 (Boundedness of Positive Linear Functionals)

If  $\ell$  is positive, then  $\ell \in [C(X)]^*$ .

**Proof.** Let  $f \in C(X)$ . Then we can bound  $f$  by its uniform norm:

$$-\|f\|_u \leq f \leq \|f\|_u \implies -\|f\|_u \ell(1) \leq \ell(f) \leq \|f\|_u \ell(1) \implies |\ell(f)| \leq \ell(1) \|f\|_u.$$

This immediately implies continuity with respect to the supremum norm. □

### Definition

Let  $\mu$  be a Borel measure on a metric space  $X$ , and let  $A \in \mathcal{B}$  (the Borel  $\sigma$ -algebra). We say  $\mu$  is:

- i. *Outer regular* on  $A$  if

$$\mu(A) = \inf\{\mu(U) : U \supseteq A, U \text{ is open}\}$$

- ii. *Inner regular* on  $A$  if

$$\mu(A) = \sup\{\mu(K) : K \subseteq A, K \text{ is compact}\}$$

- iii. *Regular* if  $\mu$  is both inner and outer regular for all  $E \in \mathcal{B}$ .

**Theorem 1 (Riesz Representation Theorem for Positive Linear Functionals)**

Let  $\ell : C(X) \rightarrow \mathbb{R}$  be a positive linear functional. Then, there exists a unique finite Borel regular measure  $\mu$  on  $X$  such that:

$$\ell(f) = \int_X f \, d\mu, \quad \forall f \in C(X)$$

i. For any open set  $U \subseteq X$ ,

$$\mu(U) = \sup\{\ell(f) : f \prec U\} \leq \ell(1) = \mu(X)$$

and consequently,

$$\ell(1) = \int_X 1 \, d\mu = \mu(X).$$

ii. For any compact set  $K \subseteq X$ ,

$$\mu(K) = \inf\{\ell(f) : f \geq \chi_K, f \in C(X)\}.$$

**Note.** For an open set  $U \subseteq X$ , its indicator function satisfies:

$$\chi_U(x) = \sup\{f(x) : f \prec U\}, \quad \forall x \in X.$$

**Comment.** Statement i implies statement ii: if  $K$  is compact, then its complement  $K^c$  is open, so we can write:

$$\begin{aligned} \mu(K) &= \mu(X) - \mu(K^c) \\ &= \ell(1) - \sup\{\ell(f) : f \prec K^c\} \\ &= \ell(1) - \sup\{\ell(f) : 0 \leq f \leq 1, \text{supp}(f) \subseteq K^c\} \\ &= \inf\{\ell(1 - f) : 0 \leq f \leq 1, \text{supp}(f) \subseteq K^c\} \\ &\geq \inf\{\ell(g) : g \geq \chi_K, g \in C(X)\}. \end{aligned}$$

For  $f$  with  $\text{supp}(f) \subseteq K^c$ ,  $f$  is zero on  $K$  since  $K$  and  $\text{supp}(f)$  are disjoint. Thus,  $1 - f \geq 1$  on  $K$  because  $f = 0$  on  $K$ , and  $1 - f \geq 0$  everywhere since  $f \leq 1$ . Therefore,  $1 - f \geq \chi_K$ , where  $\chi_K$  is the indicator function of  $K$ .

**Proof.** *Uniqueness.* Let  $\mu$  be a finite regular measure satisfying  $\ell(f) = \int f \, d\mu$ . Let  $U \subseteq X$  be open. By the note above,

$$\mu(U) = \int_U 1 \, d\mu = \sup\{\ell(f) : f \prec U\}$$

Let  $K \subseteq U$  be compact. By Urysohn's Lemma (Proposition 1), there exists  $f \prec U$  such that  $f|_K = 1$ . Then,

$$\mu(K) \leq \int_X f \, d\mu = \ell(f)$$

$$\begin{aligned} \implies \mu(U) &= \sup\{\mu(K') : K' \subseteq U, K' \text{ compact}\} \leq \sup\{\ell(f) : f \prec U\} \\ \implies \mu(U) &= \sup\{\ell(f) : f \prec U\}, \text{ which is exactly i.} \end{aligned}$$

Why does it imply uniqueness? Suppose there were two measures  $\mu_1$  and  $\mu_2$  representing  $\ell$ :

$$\ell(f) = \int_X f \, d\mu_1 = \int_X f \, d\mu_2 \quad \forall f \in C(X).$$

For any open set  $U$ , both  $\mu_1(U)$  and  $\mu_2(U)$  must satisfy:

$$\mu_i(U) = \sup\{\ell(f) : f \prec U\}, \quad i = 1, 2.$$

Since the right-hand side depends only on  $\ell$ , we must have:

$$\mu_1(U) = \mu_2(U).$$

Thus,  $\mu_1$  and  $\mu_2$  agree on all open sets. The measure  $\mu$  is outer regular, meaning for any Borel set  $E \subseteq X$ ,

$$\mu(E) = \inf\{\mu(U) : U \supseteq E, U \text{ open}\}.$$

Since  $\mu_1$  and  $\mu_2$  agree on open sets, they must assign the same measure to any Borel set  $E$ :

$$\mu_1(E) = \inf\{\mu_1(U) : U \supseteq E\} = \inf\{\mu_2(U) : U \supseteq E\} = \mu_2(E).$$

Hence,  $\mu_1 = \mu_2$  on the entire Borel  $\sigma$ -algebra. In short, open sets generate the topology of  $X$ , and their measures are uniquely specified by  $\ell$ . Since  $\mu$  is outer regular, its values on arbitrary sets are determined by its values on open sets.

*Existence.* We begin by defining the measure on open sets and then constructing an outer measure:

$$\begin{aligned} \mu(U) &:= \sup\{\ell(f) : f \prec U\} \text{ for } U \subseteq X \text{ open.} \\ \mu^*(E) &:= \inf\{\mu(U) : U \supseteq E, U \text{ open}\} \text{ for } E \subseteq X. \end{aligned}$$

*Claim 1.*  $\mu^*$  is an outer measure.

An outer measure must satisfy:

1.  $\mu^*(\emptyset) = 0$ .
2.  $A \subseteq B \implies \mu^*(A) \leq \mu^*(B)$ .
3.  $\mu^*(\bigcup_{n=1}^{\infty} E_n) \leq \sum_{n=1}^{\infty} \mu^*(E_n)$ .

$\mu^*(\emptyset) = 0$  because  $\emptyset$  is covered by itself, and  $\mu(\emptyset) = 0$ . If  $A \subseteq B$ , any open cover of  $B$  also covers  $A$ , so the infimum for  $A$  is over a larger set of covers, so  $\mu^*(A) \leq \mu^*(B)$ . Thus, it is enough to prove that

$$\mu^*\left(\bigcup_{n=1}^{\infty} U_n\right) \leq \sum_{n=1}^{\infty} \mu^*(U_n) \text{ for } U_n \text{ open.}$$

The key observation is that open sets can be used to approximate arbitrary sets when defining the outer measure. There exists  $f \prec \bigcup_{n=1}^{\infty} U_n = U$  such that  $\ell(f) > \mu(\bigcup_{n=1}^{\infty} U_n) - \epsilon$  for

a given  $\epsilon > 0$  since  $\mu(U) = \sup\{\ell(f) : f \prec U\}$ . Since every open cover of a compact set has a finite subcover, there exists  $N$  such that  $\text{supp}(f) \subseteq \bigcup_{n=1}^N U_n$ . By a partition of unity argument (Corollary 1), there exists  $f_n \prec U_n$  such that

$$\left( \sum_{n=1}^N f_n \right) \Big|_{\text{supp}(f)} = 1, \quad \text{and thus } f = \sum_{n=1}^N (f_n f).$$

Applying the functional gives:

$$\mu \left( \bigcup_{n=1}^{\infty} U_n \right) \leq \ell(f) + \epsilon = \left[ \sum_{n=1}^N \ell(f f_n) \right] + \epsilon \leq \left[ \sum_{n=1}^N \mu(U_n) \right] + \epsilon.$$

Hence  $\mu^*$  is a valid outer measure.

*Claim 2.*  $\mu$  has a finite regular Borel extension.

By Carathéodory's theorem,  $\mu^*$  is a measure on the  $\sigma$ -algebra  $\mathcal{F}$  of all  $A \subseteq X$  satisfying the condition:

$$\mu^*(E) \geq \mu^*(E \cap A) + \mu^*(E \setminus A) \text{ for all } E \subseteq X. \quad (2)$$

We now show  $\mathcal{F}$  contains all the open sets of  $X$ , i.e.,  $U \in \mathcal{F}$  if  $U \subseteq X$  is open. If this is the case,  $\mathcal{F}$  contains  $\mathcal{B}$  and  $\mu$  can be extended to a measure on  $\mathcal{B}$ .

Let  $U \subseteq X$  be open. Equation (2) holds if  $E$  itself is open. In this case,  $E \cap U$  is open, so there exists  $f \prec E \cap U$  such that  $\ell(f) > \mu(E \cap U) - \frac{\epsilon}{2}$  where  $\epsilon > 0$ . Similarly, there exists  $g \prec E \setminus \text{supp}(f)$  such that:

$$\ell(g) > \mu(E \setminus \text{supp}(f)) - \frac{\epsilon}{2}.$$

Then,  $f + g \prec E$  since  $\text{supp}(f), \text{supp}(g) \subseteq E$ . Therefore:

$$\begin{aligned} \mu(E) &\geq \ell(f + g) = \ell(f) + \ell(g) \\ &> \mu(E \cap U) - \frac{\epsilon}{2} + \mu(E \setminus \text{supp}(f)) - \frac{\epsilon}{2} \\ &\geq \mu(E \cap U) + \mu^*(E \setminus U) - \epsilon. \end{aligned}$$

Now, we prove Equation (2) holds for an arbitrary set  $E \subseteq X$ . For all  $\epsilon > 0$ , there exists an open set  $\tilde{E} \supset E$  such that:

$$\mu^*(E) \geq \mu(\tilde{E}) - \epsilon$$

which implies:

$$\mu^*(E) \geq \mu(\tilde{E}) - \epsilon \geq \mu^*(\tilde{E} \cap U) + \mu^*(\tilde{E} \setminus U) - \epsilon \geq \mu^*(E \cap U) + \mu^*(E \setminus U) - \epsilon.$$

Since  $\mu(U) \leq \ell(1) < \infty$  for all open  $U \subseteq X$ ,  $\mu$  is guaranteed to be finite.

We define for any subset  $E \subseteq X$  by:

$$\mu^*(E) = \inf\{\mu(U) : U \supseteq E, U \text{ open}\}.$$

This ensures  $\mu^*$  is outer regular.

Why Outer Regularity on the Borel  $\sigma$ -Algebra Implies Inner Regularity (in a Compact Space)? Let  $A \in \mathcal{X}$  be a Borel set. Since  $\mu$  is outer regular, for the complement  $A^c$ , there exists an open set  $U \supseteq A^c$  such that:

$$\mu(U) \leq \mu(A^c) + \epsilon \text{ for any } \epsilon > 0.$$

Since  $U \supseteq A^c$ , its complement  $K = U^c$  satisfies  $K \subseteq A$ . In a compact space, closed subsets are compact, so  $K$  is compact. From  $\mu(X) = \mu(A) + \mu(A^c)$  and  $\mu(U) \leq \mu(A^c) + \epsilon$ , we have:

$$\mu(K) = \mu(U^c) = \mu(X) - \mu(U) \geq \mu(X) - (\mu(A^c) + \epsilon) = \mu(A) - \epsilon.$$

Since  $K \subseteq A$ , this shows:

$$\mu(A) \leq \mu(K) + \epsilon \leq \sup\{\mu(K') : K' \subseteq A, K' \text{ compact}\} + \epsilon.$$

Taking  $\epsilon \rightarrow 0$  gives:

$$\mu(A) \leq \sup\{\mu(K) : K \subseteq A, K \text{ compact}\}.$$

The reverse inequality ( $\geq$ ) is trivial since  $\mu(K) \leq \mu(A)$  for  $K \subseteq A$ . Thus,

$$\mu(A) = \sup\{\mu(K) : K \subseteq A, K \text{ compact}\}.$$

We conclude  $\mu$  is inner regular. Since  $\mu$  is both outer and inner regular,  $\mu$  is regular.

*Claim 3.*  $\int_X f d\mu = \ell(f)$  for all  $f \in C(X)$ .

Let  $0 \leq f \leq 1$ ,  $f \in C(X)$ . Let  $N > 1$ . We can partition the range of  $f$ :

$$\begin{aligned} f &= \sum_{n=1}^N f \chi_{\{\frac{n-1}{N} \leq f < \frac{n}{N}\}} \\ &= \sum_{n=1}^N \left[ \left( f - \frac{n-1}{N} \right) \chi_{\{\frac{n-1}{N} \leq f < \frac{n}{N}\}} + \frac{n-1}{N} \chi_{\{\frac{n-1}{N} \leq f < \frac{n}{N}\}} \right] \end{aligned}$$

which yields:

$$f = \sum_{n=1}^N \left[ \left( f - \frac{n-1}{N} \right) \chi_{\{\frac{n-1}{N} \leq f < \frac{n}{N}\}} + \frac{1}{N} \chi_{\{\frac{n}{N} \leq f\}} \right] = \sum_{n=1}^N f_n$$

where each  $f_n \in C(X)$ .

Let us unpack the simple function approximation:

$$\begin{aligned} &\sum_{n=1}^N \frac{n-1}{N} (\chi_{\{\frac{n-1}{N} \leq f\}} - \chi_{\{\frac{n}{N} \leq f\}}) \\ &= \frac{1}{N} \sum_{n=1}^N (n-1) (\chi_{\{\frac{n-1}{N} \leq f\}} - \chi_{\{\frac{n}{N} \leq f\}}) \\ &= \frac{1}{N} [1 \cdot (\chi_{\{\frac{1}{N} \leq f\}} - \chi_{\{\frac{2}{N} \leq f\}}) + \dots + (N-1) \cdot (\chi_{\{\frac{N-1}{N} \leq f\}} - \chi_{\{\frac{N}{N} \leq f\}})] \\ &= \frac{1}{N} [\chi_{\{\frac{1}{N} \leq f\}} - \chi_{\{\frac{2}{N} \leq f\}} + 2\chi_{\{\frac{2}{N} \leq f\}} - 2\chi_{\{\frac{3}{N} \leq f\}} + \dots - (N-1)\chi_{\{\frac{N}{N} \leq f\}}] \\ &= \frac{1}{N} [\chi_{\{\frac{1}{N} \leq f\}} + \chi_{\{\frac{2}{N} \leq f\}} + \dots + \chi_{\{\frac{N-1}{N} \leq f\}} - (N-1) \cdot \chi_{\{\frac{N}{N} \leq f\}}]. \end{aligned}$$



Since  $(N-1) \cdot \chi_{\{\frac{N}{N} \leq f\}} = 0$  is handled carefully across the sets, we have:

$$\sum_{n=1}^N \frac{n-1}{N} \chi_{\{\frac{n-1}{N} \leq f < \frac{n}{N}\}} = \sum_{n=1}^N \frac{n-1}{N} (\chi_{\{\frac{n-1}{N} \leq f\}} - \chi_{\{\frac{n}{N} \leq f\}}) = \frac{1}{N} \sum_{n=1}^N \chi_{\{\frac{n}{N} \leq f\}}.$$

For each  $n$ , the continuous components  $f_n$  satisfy:

$$f_n = \left(f - \frac{n-1}{N}\right) \chi_{\{\frac{n-1}{N} \leq f \leq \frac{n}{N}\}} + \frac{1}{N} \chi_{\{\frac{n}{N} \leq f\}} \implies \frac{1}{N} \chi_{\{\frac{n}{N} \leq f\}} \leq f_n \leq \frac{1}{N} \chi_{\{\frac{n-1}{N} \leq f\}}.$$

By using statement ii, we can bound the measure of these sets:

$$\mu\left(\left\{x : \frac{n}{N} \leq f(x)\right\}\right) \leq N \ell(f_n) \leq \mu\left(\left\{x : \frac{n-1}{N} \leq f(x)\right\}\right)$$

Summing from 1 to  $N$  gives:

$$\frac{1}{N} \sum_{n=1}^N \mu\left(\frac{n}{N} \leq f\right) \leq \sum_{n=1}^N \ell(f_n) \leq \frac{1}{N} \sum_{n=1}^N \mu\left(\frac{n-1}{N} \leq f\right).$$

Focus on the right-hand side (RHS):

$$\ell(f) \leq \sum_{n=1}^N \frac{1}{N} \mu\left(\frac{n-1}{N} \leq f\right).$$

We can expand each term sequentially:

$$\mu\left(\frac{n-1}{N} \leq f\right) = \sum_{k=n}^N \mu\left(\frac{k-1}{N} \leq f < \frac{k}{N}\right) + \mu(f \geq 1).$$

Since  $f \leq 1$ , the boundary term vanishes, so:

$$\mu\left(\frac{n-1}{N} \leq f\right) = \sum_{k=n}^N \mu\left(\frac{k-1}{N} \leq f < \frac{k}{N}\right).$$

Substitute this back into the RHS sum:

$$\ell(f) \leq \sum_{n=1}^N \frac{1}{N} \sum_{k=n}^N \mu\left(\frac{k-1}{N} \leq f < \frac{k}{N}\right).$$

Interchanging the order of summation yields:

$$\sum_{n=1}^N \sum_{k=n}^N \frac{1}{N} \mu\left(\frac{k-1}{N} \leq f < \frac{k}{N}\right) = \sum_{k=1}^N \sum_{n=1}^k \frac{1}{N} \mu\left(\frac{k-1}{N} \leq f < \frac{k}{N}\right)$$

$$\begin{aligned} \Rightarrow \sum_{n=1}^k \frac{1}{N} \mu \left( \frac{k-1}{N} \leq f < \frac{k}{N} \right) &= \frac{k}{N} \mu \left( \frac{k-1}{N} \leq f < \frac{k}{N} \right) \\ \Rightarrow \ell(f) &\leq \sum_{k=1}^N \frac{k}{N} \mu \left( \frac{k-1}{N} \leq f < \frac{k}{N} \right). \end{aligned}$$

Writing  $\frac{k}{N} = \frac{k-1}{N} + \frac{1}{N}$ :

$$\begin{aligned} \ell(f) &\leq \sum_{k=1}^N \left( \frac{k-1}{N} + \frac{1}{N} \right) \mu \left( \frac{k-1}{N} \leq f < \frac{k}{N} \right) \\ \Rightarrow \ell(f) &\leq \sum_{k=1}^N \frac{k-1}{N} \mu \left( \frac{k-1}{N} \leq f < \frac{k}{N} \right) + \frac{1}{N} \sum_{k=1}^N \mu \left( \frac{k-1}{N} \leq f < \frac{k}{N} \right). \end{aligned}$$

The second sum represents the full support:

$$\frac{1}{N} \sum_{k=1}^N \mu \left( \frac{k-1}{N} \leq f < \frac{k}{N} \right) = \frac{1}{N} \mu(f > 0)$$

because the sets  $\frac{k-1}{N} \leq f < \frac{k}{N}$  form a disjoint partition of  $\{f > 0\}$ .

Combining both bounds, we trap the functional value:

$$\sum_{n=1}^N \frac{n-1}{N} \mu \left( \frac{n-1}{N} \leq f < \frac{n}{N} \right) \leq \ell(f) \leq \sum_{n=1}^N \frac{n-1}{N} \mu \left( \frac{n-1}{N} \leq f < \frac{n}{N} \right) + \frac{1}{N} \mu(f > 0).$$

The left-hand side is the lower Riemann sum and the right-hand side is the upper Riemann sum. As we integrate via Riemann sums and let  $N \rightarrow \infty$ , both sides converge to the same integral:

$$\int_X f \, d\mu = \ell(f) = \int_X f \, d\mu.$$

Therefore,  $\int_X f \, d\mu = \ell(f)$  holds for all  $f \in C(X)$ . □

## Corollary 2 (Regularity of Finite Borel Measures on Compact Metric Spaces)

Let  $X$  be a compact metric space. Then, any finite Borel measure  $\gamma$  is regular.

**Proof.** Let  $\ell(f) = \int_X f \, d\gamma$  for  $f \in C(X)$ . Since  $\gamma$  is a standard measure,  $\ell$  is positive. By Theorem 1, there exists a regular finite measure  $\mu$  such that:

$$\ell(f) = \int_X f \, d\mu = \int_X f \, d\gamma \quad \forall f \in C(X).$$

Let  $K \subset X$  be compact and define the closed sets:

$$F_n = \left\{ x \in X : \rho(x, K) \geq \frac{1}{n} \right\} \quad \forall n \geq 1.$$

We show that the complement  $F_n^c = \{x \in X : \rho(x, K) < 1/n\}$  is open: Take any  $x \in F_n^c$ , meaning  $\rho(x, K) = d < 1/n$ . By definition of infimum, there exists  $y \in K$  such that  $\rho(x, y) < (1/n + d)/2$ . Now, consider the open ball  $B(x, r)$  where  $r = (1/n - d)/2 > 0$ . For any  $z \in B(x, r)$ , by the triangle inequality:

$$\rho(z, K) \leq \rho(z, y) \leq \rho(z, x) + \rho(x, y) < r + \frac{1/n + d}{2} = \frac{1/n - d}{2} + \frac{1/n + d}{2} = 1/n.$$

Thus,  $\rho(z, K) < 1/n$ , so  $z \in F_n^c$ . Since every  $x \in F_n^c$  has an open neighborhood entirely contained in  $F_n^c$ ,  $F_n^c$  is open, and hence  $F_n$  is closed.

By Urysohn's lemma, there exists continuous functions  $f_n : X \rightarrow [0, 1]$  such that  $f_n|_K = 1$  and  $f_n|_{F_n} = 0$ . Note that  $f_n(x) \rightarrow \chi_K(x)$  pointwise for all  $x \in X$ . By the Dominated Convergence Theorem:

$$\int_X f_n d\mu = \int_X f_n d\gamma \implies \mu(K) = \gamma(K).$$

Therefore,  $\gamma$  matches the regular measure  $\mu$  on compact sets. For an open set  $U$ , complementation gives:

$$\gamma(U) = \gamma(X) - \gamma(U^c) = \mu(X) - \mu(U^c) = \mu(U).$$

Let  $A \in \mathcal{B}$  and  $\epsilon > 0$ . Since  $\mu$  is regular, there exists open and compact bounds  $U_\epsilon \supseteq A \supseteq K_\epsilon$  such that:

$$\mu(U_\epsilon \setminus K_\epsilon) < \epsilon \implies \gamma(U_\epsilon) \leq \gamma(K_\epsilon) + \epsilon \leq \gamma(A) + \epsilon.$$

Hence,  $\gamma$  is outer regular on  $A$ . Similarly,

$$\mu(K_\epsilon) \geq \mu(A) - \epsilon \implies \gamma(K_\epsilon) \geq \gamma(A) - \epsilon.$$

Thus,  $\gamma$  is inner regular. We conclude  $\gamma$  is fully regular, meaning  $\gamma = \mu$  identically across the entire  $\sigma$ -algebra.  $\square$

## Lemma 2 (Jordan Decomposition of Linear Functionals)

Let  $\ell \in [C(X)]^*$ . Then,  $\ell = \ell_1 - \ell_2$  where  $\ell_1$  and  $\ell_2$  are positive linear functionals.

**Proof.** Define the positive cone  $P = \{f \in C(X) : f \geq 0\}$ . For any  $f \in P$ , define:

$$\ell_1(f) := \sup\{\ell(g) : 0 \leq g \leq f, g \in C(X)\}.$$

Note that  $\ell_1(f) \geq 0$  since we can choose  $g = 0$  in the supremum. We first show that  $\ell_1$  is linear on  $P$ . For any  $f, f' \in P$  and  $c \geq 0$ , we check:

$$1. \text{ Additivity. } \ell_1(f + f') = \ell_1(f) + \ell_1(f').$$

For any  $0 \leq g \leq f$  and  $0 \leq g' \leq f'$ , we have  $0 \leq g + g' \leq f + f'$ , and thus:

$$\ell(g) + \ell(g') = \ell(g + g') \leq \ell_1(f + f').$$

Taking suprema over  $g$  and  $g'$  independently gives  $\ell_1(f) + \ell_1(f') \leq \ell_1(f + f')$ . For the reverse direction, given any  $0 \leq h \leq f + f'$ , we can decompose  $h = u + u'$  where  $0 \leq u \leq f$  and  $0 \leq u' \leq f'$  by setting  $u = \min(h, f)$  and  $u' = h - u$ . Then,

$$\ell(h) = \ell(u) + \ell(u') \leq \ell_1(f) + \ell_1(f').$$

Taking the supremum over  $h$  gives  $\ell_1(f + f') \leq \ell_1(f) + \ell_1(f')$ .

2. *Homogeneity.*  $\ell_1(cf) = c\ell_1(f)$ .

$$\ell_1(cf) = \sup\{\ell(g) : 0 \leq g \leq cf\} = \sup\{\ell(ch) : 0 \leq h \leq f\} = c \sup\{\ell(h) : 0 \leq h \leq f\} = c\ell_1(f).$$

We now extend  $\ell_1$  linearly to the entire space  $C(X)$ . Any  $f \in C(X)$  can be split into positive and negative parts:  $f = f_1 - f_2$  where  $f_1, f_2 \in P$ . We define:

$$\ell_1(f) := \ell_1(f_1) - \ell_1(f_2).$$

If  $f = f_1 - f_2 = f'_1 - f'_2$ , then rearranging gives  $f_1 + f'_2 = f'_1 + f_2$ . By additivity on  $P$ :

$$\ell_1(f_1) + \ell_1(f'_2) = \ell_1(f'_1) + \ell_1(f_2) \implies \ell_1(f_1) - \ell_1(f_2) = \ell_1(f'_1) - \ell_1(f'_2).$$

Thus, the extension is well-defined. Finally, define  $\ell_2 := \ell_1 - \ell$ . To see that  $\ell_2$  is positive, observe that for any  $f \geq 0$ :

$$\ell_2(f) = \ell_1(f) - \ell(f) = \sup\{\ell(g) : 0 \leq g \leq f\} - \ell(f) \geq \ell(f) - \ell(f) = 0.$$

Thus,  $\ell_1$  and  $\ell_2$  are both positive linear functionals satisfying  $\ell = \ell_1 - \ell_2$ . □

#### Proposition 4 (Density of Continuous Functions in $L^p$ Spaces)

Let  $X$  be a compact metric space,  $\mu$  be a finite regular Borel measure, and  $p \in [1, \infty)$ . Then,  $C(X)$  is dense in  $L^p(X, \mu)$ .

**Proof.** Since simple functions ( $\Sigma$ ) are dense in  $L^p$  and the measure is finite, it suffices to approximate the indicator function  $\chi_E$  of an arbitrary Borel set  $E \in \mathcal{B}$  by continuous functions.

Let  $\epsilon > 0$ . Since  $\mu$  is regular, there exist a compact set  $K_\epsilon \subseteq E$  and an open set  $U_\epsilon \supseteq E$  such that:

$$\mu(U_\epsilon) < \mu(E) + \frac{\epsilon}{2}, \quad \mu(K_\epsilon) > \mu(E) - \frac{\epsilon}{2}.$$

Subtracting these inequalities gives:

$$\mu(U_\epsilon \setminus K_\epsilon) = \mu(U_\epsilon) - \mu(K_\epsilon) < \epsilon.$$

Since  $K_\epsilon \subseteq E \subseteq U_\epsilon$ , Urysohn's Lemma allows us to construct a continuous function  $f$  such that  $f = 1$  on  $K_\epsilon$ ,  $f = 0$  outside  $U_\epsilon$ , and  $0 \leq f \leq 1$  everywhere. The difference  $|\chi_E - f|$  is nonzero only on the symmetric difference region  $U_\epsilon \setminus K_\epsilon$ , because:

- On  $K_\epsilon$ ,  $\chi_E = 1$  and  $f = 1$ , so  $|\chi_E - f| = 0$ .

- Outside  $U_\epsilon$ ,  $\chi_E = 0$  and  $f = 0$ , so  $|\chi_E - f| = 0$ .
- On  $U_\epsilon \setminus K_\epsilon$ ,  $|\chi_E - f| \leq 1$  since  $0 \leq f \leq 1$  and  $\chi_E \in \{0, 1\}$ .

Therefore:

$$\int_X |\chi_E - f|^p d\mu \leq \int_{U_\epsilon \setminus K_\epsilon} 1 d\mu = \mu(U_\epsilon \setminus K_\epsilon) < \epsilon.$$

Since simple functions are dense in  $L^p$ , and we can approximate any simple function component by continuous functions, we can approximate any general  $L^p$ -function by continuous functions. Thus,  $C(X)$  is dense in  $L^p(X, \mu)$ .  $\square$

### Lusin's Theorem

Let  $X$  be a compact metric space and  $\mu$  be a finite measure. Let  $f$  be a bounded measurable function on  $X$  ( $\|f\|_\infty \leq C$ ). Then, for all  $\epsilon > 0$ , there exists  $g \in C(X)$  such that  $\mu(\{f \neq g\}) \leq \epsilon$ . Moreover,  $\|g\|_\infty \leq \sup_x |f(x)|$ , and  $0 \leq g \leq \sup_x |f(x)|$  if  $f \geq 0$ .

**Proof.** By Proposition 4, since  $C(X)$  is dense in  $L^1(X, \mu)$ , there exists a sequence of continuous functions  $\{f_n\} \subset C(X)$  such that  $f_n \rightarrow f$  in  $L^1$ , which implies a subsequence converges to  $f$  almost everywhere. Since  $f_n \rightarrow f$  almost everywhere and  $\mu(X) < \infty$ , Egoroff's Theorem guarantees that for any  $\epsilon > 0$ , there exists a measurable set  $A \subset X$  such that:

$$\mu(X \setminus A) \leq \frac{\epsilon}{2},$$

and  $f_n \rightarrow f$  uniformly on  $A$ . Since  $\mu$  is regular from Corollary 2, for this set  $A$ , there exists a compact subset  $K \subseteq A$  such that:

$$\mu(A \setminus K) \leq \frac{\epsilon}{2} \implies \mu(X \setminus K) = \mu(X \setminus A) + \mu(A \setminus K) \leq \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon.$$

Since  $f_n \rightarrow f$  uniformly on  $A$ , and  $K \subseteq A$ , the restriction  $f|_K$  is continuous as a uniform limit of continuous functions. By the Tietze Extension Theorem, since  $K$  is closed in  $X$  and  $X$  is a compact metric space, there exists a continuous function  $g \in C(X)$  extending it over the whole space:

$$g|_K = f|_K, \quad \sup_X g = \sup_K f, \quad \inf_X g = \inf_K f.$$

By construction,  $f = g$  on  $K$  and  $\mu(X \setminus K) \leq \epsilon$ . Therefore:

$$\{x \in X : f(x) \neq g(x)\} \subseteq X \setminus K \implies \mu(\{f \neq g\}) \leq \mu(X \setminus K) \leq \epsilon.$$

Since  $g$  is an extension of  $f|_K$ , and  $f$  is bounded by  $C$ , we obtain:

$$|g(x)| \leq \sup_K |f| \leq \sup_X |f| = C.$$

If  $f \geq 0$ , then  $g \geq 0$  holds everywhere since  $\inf_X g \geq \inf_K f \geq 0$ .  $\square$

### Proposition 5

Let  $X$  be a compact metric space, and  $\mu$  be a finite signed measure. Let  $\ell_\mu(f) = \int_X f \, d\mu$  for  $f \in C(X)$ . Then,  $\ell_\mu \in [C(X)]^*$  and  $\|\ell_\mu\|_{[C(X)]^*} = |\mu|(X) = \|\mu\|$ .

**Proof.** By standard integration inequalities, since  $\mu$  is a finite signed measure, we know  $\ell_\mu \in [C(X)]^*$  and  $\|\ell_\mu\| \leq \|\mu\|$ .

Let  $D^+, D^-$  be the disjoint components of the Hahn decomposition such that  $d|\mu| = \chi_{D^+} d\mu - \chi_{D^-} d\mu$  and  $d\mu = (\chi_{D^+} - \chi_{D^-}) d|\mu|$ . Let  $\rho = \chi_{D^+} - \chi_{D^-}$ , noting that  $|\rho| = \rho^2 = 1$ .

Let  $\epsilon > 0$ . By Lusin's Theorem, there exists  $g \in C(X)$  such that  $|\mu|(\{\rho \neq g\}) \leq \epsilon$  and  $\|g\|_\infty \leq 1$ . We compute the action:

$$\begin{aligned} \ell_\mu(g) &= \int_X g \, d\mu = \int_X g\rho \, d|\mu| \\ &= \int_{\{g=\rho\}} g\rho \, d|\mu| + \int_{\{g \neq \rho\}} g\rho \, d|\mu| \\ &= \int_X \rho^2 \, d|\mu| - \int_{\{g \neq \rho\}} \rho^2 \, d|\mu| + \int_{\{g \neq \rho\}} g\rho \, d|\mu| \\ &\geq |\mu|(X) - \epsilon - \epsilon = |\mu|(X) - 2\epsilon. \end{aligned}$$

This follows from the fact that:

$$0 \leq \int_{\{g \neq \rho\}} \rho^2 \, d|\mu| = |\mu|(\{g \neq \rho\}) \leq \epsilon$$

and

$$\left| \int_{\{g \neq \rho\}} g\rho \, d|\mu| \right| \leq \int_{\{g \neq \rho\}} |g\rho| \, d|\mu| \leq \int_{\{g \neq \rho\}} d|\mu| = |\mu|(\{g \neq \rho\}) \leq \epsilon \implies \int_{\{g \neq \rho\}} g\rho \, d|\mu| \geq -\epsilon.$$

Therefore,

$$\|\mu\| \geq |\ell_\mu(g)| \geq \ell_\mu(g) \geq |\mu|(X) - 2\epsilon.$$

Taking  $\epsilon \rightarrow 0$ , we confirm  $\|\ell_\mu\| = \|\mu\|$ . □

**Notation.** Let  $M(X)$  be the space of finite signed Borel measures equipped with the total variation norm  $\|\mu\| = |\mu|(X)$ .

### Theorem 3 (Riesz-Markov-Kakutani Representation Theorem)

Let  $X$  be a compact metric space. Then, for every  $\ell \in [C(X)]^*$ , there exists a unique finite signed measure  $\mu \in M(X)$  such that:

$$\ell(f) = \int_X f \, d\mu, \quad \forall f \in C(X).$$

Moreover,  $\|\ell\|_{[C(X)]^*} = \|\mu\|_{M(X)}$ .

**Proof.** Let  $\ell \in [C(X)]^*$ . By Lemma 2, we split the functional into positive components:  $\ell = \ell_1 - \ell_2$ . Applying Theorem 1 to each component gives unique finite regular positive measures  $\mu_1, \mu_2$  such that for all  $f \in C(X)$ :

$$\ell_1(f) = \int_X f \, d\mu_1, \quad \text{and } \ell_2(f) = \int_X f \, d\mu_2.$$

Taking the difference:

$$\ell(f) = \int_X f \, d(\mu_1 - \mu_2) = \int_X f \, d\mu, \quad \forall f \in C(X)$$

where  $\mu = \mu_1 - \mu_2$ . The fact that  $\mu$  is a finite regular signed measure and is unique follows directly by adapting the uniqueness proof from Corollary 2. Finally, Proposition 5 ensures that  $\|\ell\| = \|\mu\|$ .  $\square$

## Schwartz Space of Functions with Rapid Decrease

### From Fourier Series to Fourier Transform

1. Let  $T > 0$ . Then, the family  $\left\{ \frac{1}{\sqrt{2\pi}} e^{\frac{in\pi}{T}x} : n \in \mathbb{Z} \right\}$  is  $2T$ -periodic and forms a complete orthonormal system (CONS) in  $L^2([-T, T])$  (for complex-valued functions).
2. If  $h \in L^2([-T, T])$ , then its Fourier series expansion converges in  $L^2([-T, T])$ :

$$h(x) = \sum_{n=-\infty}^{\infty} c_n e^{\frac{in\pi}{T}x} \quad \text{with} \quad c_n = \frac{1}{2T} \int_{-T}^T h(x) e^{-\frac{in\pi}{T}x} \, dx.$$

Moreover, Parseval's identity holds:

$$\int_{-T}^T |h(x)|^2 \, dx = 2T \sum_{n=-\infty}^{\infty} |c_n|^2.$$

*Proof.* First, we compute the  $L^2$  norm squared of  $h$ :

$$\int_{-T}^T |h(x)|^2 \, dx = \int_{-T}^T h(x) \overline{h(x)} \, dx,$$

Substituting the Fourier series representations for  $h(x)$  and  $\overline{h(x)}$  yields:

$$\int_{-T}^T \left( \sum_{n=-\infty}^{\infty} c_n e^{\frac{in\pi}{T}x} \right) \left( \sum_{m=-\infty}^{\infty} \overline{c_m} e^{-\frac{im\pi}{T}x} \right) \, dx,$$

Rearranging the terms, we obtain:

$$\int_{-T}^T \sum_{n=-\infty}^{\infty} \sum_{m=-\infty}^{\infty} c_n \overline{c_m} e^{\frac{i(n-m)\pi}{T}x} \, dx,$$

By Fubini's theorem, we can interchange the summation and integration:

$$\sum_{n=-\infty}^{\infty} \sum_{m=-\infty}^{\infty} c_n \overline{c_m} \int_{-T}^T e^{\frac{i(n-m)\pi}{T}x} dx,$$

The exponential functions form an orthogonal system. When  $n \neq m$ , we have:

$$\int_{-T}^T e^{\frac{i(n-m)\pi}{T}x} dx = \left[ \frac{T}{i\pi(n-m)} e^{\frac{i(n-m)\pi}{T}x} \right]_{-T}^T = 0.$$

When  $n = m$ , the integral becomes:

$$\int_{-T}^T e^0 dx = \int_{-T}^T 1 dx = 2T.$$

Therefore, only the diagonal terms survive, and we get:

$$\sum_{n=-\infty}^{\infty} c_n \overline{c_n} \cdot 2T = 2T \sum_{n=-\infty}^{\infty} |c_n|^2.$$

□

3. Let  $f \in C_c^\infty(\mathbb{R})$  such that  $f(x) = 0$  for  $|x| > R > 0$ . Then, for any  $T > R + 1$  and  $|x| < T$ , we can view  $f$  as a  $2T$ -periodic function and expand it as:

$$f(x) = \sum_{n=-\infty}^{\infty} \frac{1}{2T} \int_{-T}^T f(y) e^{-\frac{in\pi}{T}y} dy e^{\frac{in\pi}{T}x} = \sum_{n=-\infty}^{\infty} \frac{1}{2T} \int_{-\infty}^{\infty} f(y) e^{-i\frac{n}{2T}2\pi y} dy e^{i\frac{n}{2T}2\pi x}.$$

Let  $\hat{f}(\xi) = \int_{-\infty}^{\infty} f(y) e^{-i2\pi\xi y} dy$  for  $\xi \in \mathbb{R}$ . Note that  $\hat{f} \in C^\infty(\mathbb{R})$ . Hence, we can write:

$$f(x) = \sum_{n=-\infty}^{\infty} \frac{1}{2T} \hat{f}\left(\frac{n}{2T}\right) e^{i2\pi\frac{n}{2T}x}, \quad \text{for } |x| < T \text{ and } T > R + 1.$$

By Parseval's identity applied to the interval  $[-T, T]$ :

$$\int_{-\infty}^{\infty} |f(x)|^2 dx = \int_{-T}^T |f(x)|^2 dx = \frac{1}{2T} \sum_{n=-\infty}^{\infty} \left| \hat{f}\left(\frac{n}{2T}\right) \right|^2 \quad \text{for } T > R + 1.$$

For any fixed  $x \in \mathbb{R}$ , taking the limit as  $T \rightarrow \infty$  yields:

$$f(x) = \lim_{T \rightarrow \infty} \sum_{n=-\infty}^{\infty} \frac{1}{2T} \hat{f}\left(\frac{n}{2T}\right) e^{i2\pi\frac{n}{2T}x} = \int_{-\infty}^{\infty} \hat{f}(\xi) e^{i2\pi\xi x} d\xi.$$

Similarly, for the  $L^2$  norm, we consider the sum for finite  $T > R + 1$ :

$$\int_{-T}^T |f(x)|^2 dx = \frac{1}{2T} \sum_{n=-\infty}^{\infty} \left| \hat{f}\left(\frac{n}{2T}\right) \right|^2.$$



Since  $f$  has compact support within  $[-R, R]$ , whenever  $T > R + 1$ , we have:

$$\int_{-\infty}^{\infty} |f(x)|^2 dx = \int_{-T}^T |f(x)|^2 dx.$$

Setting  $\xi_n = \frac{n}{2T}$  and  $\Delta\xi = \frac{1}{2T}$ , the right-hand side can be rewritten as:

$$\sum_{n=-\infty}^{\infty} \left| \hat{f}\left(\frac{n}{2T}\right) \right|^2 \cdot \frac{1}{2T} = \sum_{n=-\infty}^{\infty} |\hat{f}(\xi_n)|^2 \cdot \Delta\xi.$$

This expression is precisely a Riemann sum for the function  $|\hat{f}(\xi)|^2$  over the entire real line. As  $T \rightarrow \infty$ , the grid spacing  $\Delta\xi = \frac{1}{2T} \rightarrow 0$ , rendering the partition infinitely fine. Under these conditions (which  $\hat{f}$  naturally satisfies since it is the Fourier transform of a smoothly compactly supported function  $f \in C_c^\infty(\mathbb{R})$ ), the Riemann sum converges to the integral:

$$\lim_{T \rightarrow \infty} \sum_{n=-\infty}^{\infty} |\hat{f}(\xi_n)|^2 \cdot \Delta\xi = \int_{-\infty}^{\infty} |\hat{f}(\xi)|^2 d\xi.$$

Thus, we establish the inversion formula and Plancherel's theorem for  $C_c^\infty(\mathbb{R})$ :

$$f(x) = \int_{-\infty}^{\infty} \hat{f}(\xi) e^{i2\pi\xi x} d\xi, \quad x \in \mathbb{R} \quad \text{and} \quad \int_{-\infty}^{\infty} |f(x)|^2 dx = \int_{-\infty}^{\infty} |\hat{f}(\xi)|^2 d\xi.$$

## Fourier Transform

The *Fourier transform* of  $f \in C_c^\infty(\mathbb{R})$  is defined as:

$$\hat{f}(\xi) = (\mathcal{F}f)(\xi) = \int_{-\infty}^{\infty} f(x) e^{-i2\pi\xi x} dx, \quad \xi \in \mathbb{R}.$$

The *inverse Fourier transform* of  $g \in C_c^\infty(\mathbb{R})$  is defined as:

$$(\mathcal{F}^{-1}g)(x) = \int_{-\infty}^{\infty} g(\xi) e^{i2\pi\xi x} d\xi, \quad x \in \mathbb{R}.$$

Consequently, we have  $\mathcal{F}^{-1}\mathcal{F}f = f$ . Note that  $(\mathcal{F}^{-1}f)(x) = \hat{f}(-x)$  for any  $x \in \mathbb{R}$ , which shows that  $\mathcal{F}$  is a unitary operator. It turns out that  $\mathcal{F}(C_c^\infty(\mathbb{R})) \subseteq S(\mathbb{R})$ , where  $S(\mathbb{R})$  denotes the Schwartz space of rapidly decreasing functions, and  $\mathcal{F} : S(\mathbb{R}) \rightarrow S(\mathbb{R})$  is a bijection.

## Space of Functions of Rapid Decrease

On  $\mathbb{R}$ , the Schwartz space is given by:

$$S(\mathbb{R}) = \left\{ f \in C^\infty(\mathbb{R}) : \sup_{x \in \mathbb{R}} (1 + |x|)^N |f^{(n)}(x)| < \infty \text{ for all } N, n \in \mathbb{N}_0 \right\}.$$

To generalize this to higher dimensions, let  $f, g : \mathbb{R}^d \rightarrow \mathbb{C}$ . For a multi-index  $\alpha = (\alpha_1, \dots, \alpha_d) \in \mathbb{N}_0^d$ , we define:

- $|\alpha| = \alpha_1 + \cdots + \alpha_d$  (the total order of differentiation),
- $\partial^\alpha = \frac{\partial^{|\alpha|}}{\partial x_1^{\alpha_1} \cdots \partial x_d^{\alpha_d}}$  (the total derivative operator),
- $\alpha! = \alpha_1! \cdots \alpha_d!$  with  $0! = 1$ ,
- $x^\alpha = x_1^{\alpha_1} \cdots x_d^{\alpha_d}$  for  $x = (x_1, \dots, x_d) \in \mathbb{R}^d$ .

### Product Rule (Leibniz's Formula)

$$\partial^\alpha(fg) = \sum_{\beta \leq \alpha} \frac{\alpha!}{\beta!(\alpha - \beta)!} \partial^\beta f \partial^{\alpha - \beta} g,$$

where  $\beta \leq \alpha$  signifies that  $\beta_k \leq \alpha_k$  for each  $k = 1, \dots, d$ .

### Schwartz Space

For  $\alpha \in \mathbb{N}_0^d$  and  $N \in \mathbb{N}_0$ , we introduce the family of seminorms:

$$|f|_{N,\alpha} = \sup_{x \in \mathbb{R}^d} (1 + |x|)^N |\partial^\alpha f(x)|.$$

The Schwartz space of rapidly decreasing functions on  $\mathbb{R}^d$  is then defined as:

$$S = S(\mathbb{R}^d) = \{f \in C^\infty(\mathbb{R}^d) : |f|_{N,\alpha} < \infty \text{ for all } N \in \mathbb{N}_0, \alpha \in \mathbb{N}_0^d\}.$$

### Basic Properties

1. The condition  $|f|_{N,\alpha} < \infty$  is true if and only if there exists a constant  $C_{\alpha,N} > 0$  such that:

$$|\partial^\alpha f(x)| \leq \frac{C_{\alpha,N}}{(1 + |x|)^N} \quad \text{for all } x \in \mathbb{R}^d.$$

2.  $C_c^\infty(\mathbb{R}^d) \subseteq S(\mathbb{R}^d)$ .
3. As an example, the Gaussian function  $e^{-|x|^2} \in S(\mathbb{R}^d)$ , but  $e^{-|x|^2} \notin C_c^\infty(\mathbb{R}^d)$  because its support is not compact.

### Seminorms and Convergence

1. Each  $|\cdot|_{N,\alpha}$  constitutes a seminorm on  $S$ :

$$|f + g|_{N,\alpha} \leq |f|_{N,\alpha} + |g|_{N,\alpha}, \quad |af|_{N,\alpha} = |a| |f|_{N,\alpha}.$$

Note that  $|f|_{N,\alpha} = 0$  does not individually imply  $f = 0$ .

2. A sequence  $f_k \in S$  converges to  $f \in S$  (denoted  $f_k \rightarrow f$  in  $S$ ) if:

$$|f_k - f|_{N,\alpha} \rightarrow 0 \quad \text{as } k \rightarrow \infty \quad \text{for all } N \in \mathbb{N}_0, \alpha \in \mathbb{N}_0^d.$$

3. For  $N \in \mathbb{N}_0$ , we can define a directed system of norms on  $S$ :

$$|f|_{(N)} = \max_{|\alpha| \leq N} |f|_{N,\alpha} = \max_{|\alpha| \leq N} \sup_{x \in \mathbb{R}^d} (1 + |x|)^N |\partial^\alpha f(x)|. \text{profile}$$

These norms are monotonically increasing in  $N$ , meaning  $|f|_{(N)} \leq |f|_{(N+1)}$ .

4. The topology of  $S(\mathbb{R}^d)$  can be fully structuralized via the translation-invariant metric:

$$\rho(f, g) = \sum_{N=1}^{\infty} 2^{-N} \min(|f - g|_{(N)}, 1).$$

### Structural Properties of $S(\mathbb{R}^d)$

1. If  $f \in S$ , then  $\partial^\alpha f \in S$  for any multi-index  $\alpha$ .
2. If  $f, g \in S$ , then their pointwise product satisfies  $fg \in S$ .
3. If  $f \in S$ , then  $Pf \in S$  for any polynomial function  $P(x)$ .
4. If  $f \in S$ , then  $f \in L^p(\mathbb{R}^d)$  for any  $p > 0$ .

**Proof.** For  $p = \infty$ , the statement follows directly from the definition of the space. For  $0 < p < \infty$ , we write:

$$\int_{\mathbb{R}^d} |f(x)|^p dx = \int_{\mathbb{R}^d} \left( |f(x)| (1 + |x|)^{\frac{d+1}{p}} \right)^p \frac{dx}{(1 + |x|)^{d+1}} \leq |f|_{[\frac{d+1}{p}], 0}^p \int_{\mathbb{R}^d} \frac{dx}{(1 + |x|)^{d+1}} < \infty.$$

□

5. If  $f, g \in S$ , then their convolution  $f * g \in S$ .

**Proof.** By the differentiating properties of convolutions, we have:

$$\partial^\alpha (f * g)(x) = \int_{\mathbb{R}^d} \partial^\alpha f(x - y) g(y) dy.$$

Using the submultiplicative inequality  $1 + |x| \leq 1 + |x - y| + |y| \leq (1 + |x - y|)(1 + |y|)$ , we find:

$$\begin{aligned} (1 + |x|)^N |\partial^\alpha (f * g)(x)| &\leq \int_{\mathbb{R}^d} (1 + |x - y|)^N |\partial^\alpha f(x - y)| (1 + |y|)^N |g(y)| dy \\ &\leq |f|'_{N,\alpha} \int_{\mathbb{R}^d} (1 + |y|)^{N+d+1} |g(y)| \frac{dy}{(1 + |y|)^{d+1}} \\ &\leq |f|_{N,\alpha} |g|'_{N+d+1,0} \int_{\mathbb{R}^d} \frac{dy}{(1 + |y|)^{d+1}} < \infty. \end{aligned}$$

□

## Continuous Linear Functionals on $S$

A linear functional  $T : S(\mathbb{R}^d) \rightarrow \mathbb{C}$  is continuous with respect to the metric  $\rho$  if and only if:

There exist constants  $C > 0$  and  $N \in \mathbb{N}_0$  such that  $|T(f)| \leq C|f|_{(N)}$  for all  $f \in S$ .

**Proof.** ( $\Rightarrow$ ) Assume  $T$  is continuous. Since  $T$  is linear and continuous at zero, for any  $\epsilon = 1$ , there exists a  $\delta > 0$  such that  $|T(f)| \leq 1$  whenever  $\rho(f, 0) \leq 2\delta$ . We observe that:

$$\rho(f, 0) = \sum_{N=1}^{\infty} 2^{-N} \min\{1, |f|_{(N)}\} \leq \sum_{N=1}^{N_0} 2^{-N} |f|_{(N_0)} + \sum_{N=N_0+1}^{\infty} 2^{-N} \leq |f|_{(N_0)} + 2^{-N_0}.$$

Choose  $N_0 \in \mathbb{N}$  large enough so that  $2^{-N_0} \leq \delta$ . Now, pick any non-zero  $g \in S$ , and scale it by defining  $f = \frac{\delta}{|g|_{(N_0)}} g$ . Then  $|f|_{(N_0)} = \delta$ , which implies:

$$\rho(f, 0) \leq |f|_{(N_0)} + 2^{-N_0} \leq \delta + \delta = 2\delta \implies |T(f)| \leq 1.$$

By substituting back the relation between  $f$  and  $g$  using linearity, we conclude:

$$\left| T\left(\frac{\delta}{|g|_{(N_0)}} g\right) \right| \leq 1 \implies |T(g)| \leq \frac{1}{\delta} |g|_{(N_0)}.$$

Setting  $C = \frac{1}{\delta}$  completes this direction.

( $\Leftarrow$ ) Conversely, suppose there exist  $C > 0$  and  $N_0 \in \mathbb{N}_0$  such that  $|T(f)| \leq C|f|_{(N_0)}$  for all  $f \in S$ . Let  $f_n \rightarrow f$  in  $S$ . Then by definition,  $|f_n - f|_{(N_0)} \rightarrow 0$  as  $n \rightarrow \infty$ . By linearity, we have:

$$|T(f_n) - T(f)| = |T(f_n - f)| \leq C|f_n - f|_{(N_0)} \rightarrow 0.$$

Thus,  $T(f_n) \rightarrow T(f)$ , proving continuity.  $\square$

## Corollary 0

A linear functional  $T : S \rightarrow \mathbb{C}$  is continuous if and only if there exist a constant  $C > 0$ , finite collections of multi-indices  $\alpha_1, \dots, \alpha_m \in \mathbb{N}_0^d$ , and integers  $N_1, \dots, N_m \in \mathbb{N}_0$  such that:

$$|T(f)| \leq C \sum_{k=1}^m |f|'_{N_k, \alpha_k} \quad \text{for all } f \in S.$$

## Fourier Transform on $S(\mathbb{R}^d)$

i. Given  $f \in L^1(\mathbb{R}^d)$ , its *Fourier transform*  $\mathcal{F}f = \hat{f}$  is defined by:

$$\mathcal{F}f(\xi) = \hat{f}(\xi) = \int_{\mathbb{R}^d} f(x) e^{-i2\pi x \cdot \xi} dx, \quad \xi \in \mathbb{R}^d,$$

where  $x \cdot \xi = x_1 \xi_1 + \dots + x_d \xi_d$ .

ii. Given  $g \in L^1(\mathbb{R}^d)$ , its *inverse Fourier transform*  $\mathcal{F}^{-1}g$  is defined by:

$$\mathcal{F}^{-1}g(x) = \int_{\mathbb{R}^d} g(\xi) e^{i2\pi x \cdot \xi} d\xi, \quad x \in \mathbb{R}^d.$$

## Basic Properties of $\mathcal{F}$ and $\mathcal{F}^{-1}$

- a.  $\mathcal{F}f(\xi)$  and  $\mathcal{F}^{-1}g(x)$  are bounded, uniformly continuous functions.
- b.  $\mathcal{F}$  and  $\mathcal{F}^{-1}$  are linear operators.
- c.  $\|\mathcal{F}f\|_{L^\infty} \leq \|f\|_{L^1}$  because  $|\mathcal{F}f(\xi)| \leq \int_{\mathbb{R}^d} |f(x)| dx$ .
- d.  $\|\mathcal{F}^{-1}g\|_{L^\infty} \leq \|g\|_{L^1}$ .
- e.  $\mathcal{F}^{-1}g(x) = \mathcal{F}g(-x)$  for all  $x \in \mathbb{R}^d$ .
- f. Distributional derivatives transform as  $\widehat{\partial^\alpha f}(\xi) = (i2\pi\xi)^\alpha \hat{f}(\xi)$ .

## Simple Properties of Fourier Transform

### Property 1

If  $f \in S$ , then for any multi-index  $\alpha \in \mathbb{N}_0^d$ :

$$\widehat{\partial^\alpha f}(\xi) = (i2\pi\xi)^\alpha \hat{f}(\xi) = (i2\pi)^{|\alpha|} \xi_1^{\alpha_1} \cdots \xi_d^{\alpha_d} \hat{f}(\xi).$$

**Proof.** Consider the case  $d = 1$  and  $\alpha = 1$ . We want to show  $\widehat{f^{(1)}}(\xi) = i2\pi\xi \hat{f}(\xi)$ . Using integration by parts:

$$\begin{aligned} \widehat{f^{(1)}}(\xi) &= \int_{-\infty}^{\infty} f'(x) e^{-i2\pi x\xi} dx = \lim_{n \rightarrow \infty} \int_{-n}^n f'(x) e^{-i2\pi x\xi} dx \\ &= \lim_{n \rightarrow \infty} [f(x) e^{-i2\pi x\xi}]_{-n}^n - \int_{-n}^n f(x) (-i2\pi\xi e^{-i2\pi x\xi}) dx. \end{aligned}$$

Since  $f \in S$ ,  $\lim_{n \rightarrow \infty} f(\pm n) = 0$ , so the boundary terms vanish, leaving:

$$\widehat{f^{(1)}}(\xi) = 0 + i2\pi\xi \int_{-\infty}^{\infty} f(x) e^{-i2\pi x\xi} dx = i2\pi\xi \hat{f}(\xi).$$

The general result follows by iterating this process for each partial derivative. □

### Example

Let  $\Delta u(x) = \sum_{k=1}^d \frac{\partial^2}{\partial x_k^2} u(x)$  be the Laplacian of  $u \in S$ . Find  $\widehat{\Delta u}(\xi)$ .

Each operator  $\frac{\partial^2}{\partial x_k^2}$  matches a multi-index  $\alpha = (0, \dots, 2, \dots, 0)$  where 2 is in the  $k$ -th position. Applying Property 1:

$$\begin{aligned} \widehat{\Delta u}(\xi) &= \sum_{k=1}^d \widehat{\frac{\partial^2 u}{\partial x_k^2}}(\xi) = \sum_{k=1}^d (i2\pi\xi_k)^2 \hat{u}(\xi) \\ &= \hat{u}(\xi) \left( -4\pi^2 \sum_{k=1}^d \xi_k^2 \right) = -4\pi^2 |\xi|^2 \hat{u}(\xi). \end{aligned}$$

## Property 2

1. Let  $f \in S$  and  $y \in \mathbb{R}^d$ . Then shifts transform into modulation:

$$\widehat{f(\cdot + y)}(\xi) = e^{i2\pi\xi \cdot y} \hat{f}(\xi).$$

*Proof.* Using the substitution  $z = x + y$  with  $dz = dx$ :

$$\begin{aligned} \widehat{f(\cdot + y)}(\xi) &= \int_{\mathbb{R}^d} f(x + y) e^{-i2\pi x \cdot \xi} dx = \int_{\mathbb{R}^d} f(z) e^{-i2\pi(z-y) \cdot \xi} dz \\ &= e^{i2\pi y \cdot \xi} \int_{\mathbb{R}^d} f(z) e^{-i2\pi z \cdot \xi} dz = e^{i2\pi y \cdot \xi} \hat{f}(\xi). \end{aligned}$$

□

2. Let  $f \in S$  and  $\delta \in \mathbb{R} \setminus \{0\}$ . Dilation transforms inversely:

$$\widehat{f(\delta \cdot)}(\xi) = |\delta|^{-d} \hat{f}\left(\frac{\xi}{\delta}\right).$$

**Proof.** Using the change of coordinates  $z = \delta x$ , we have  $dz = |\delta|^d dx$ :

$$\widehat{f(\delta \cdot)}(\xi) = \int_{\mathbb{R}^d} f(\delta x) e^{-i2\pi x \cdot \xi} dx = \int_{\mathbb{R}^d} f(z) e^{-i2\pi \frac{z \cdot \xi}{\delta}} |\delta|^{-d} dz = |\delta|^{-d} \hat{f}\left(\frac{\xi}{\delta}\right).$$

□

## Example

Let  $f \in S$  such that  $\int_{\mathbb{R}^d} f(x) dx \neq 0$ . Show that:

$$\lim_{\delta \rightarrow 0} |\widehat{f(\delta \cdot)}(\xi)| = \begin{cases} 0, & \xi \neq 0, \\ +\infty, & \xi = 0. \end{cases}$$

For any  $\delta \neq 0$ , Property 2b gives  $|\widehat{f(\delta \cdot)}(\xi)| = |\delta|^{-d} \left| \hat{f}\left(\frac{\xi}{\delta}\right) \right|$ . Since  $\hat{f} \in S$ , it is bounded by its seminorms:

$$|\hat{f}(\xi)| \leq \frac{C}{(1 + |\xi|)^{d+1}}, \quad \text{where } C = |\hat{f}|_{d+1,0}.$$

When  $\xi \neq 0$ , we can manipulate the expression:

$$|\widehat{f(\delta \cdot)}(\xi)| = \frac{1}{|\xi|^d} \cdot \left( \frac{|\xi|}{|\delta|} \right)^d \cdot \left| \hat{f}\left(\frac{\xi}{\delta}\right) \right| \leq \frac{1}{|\xi|^d} \cdot \left( \frac{|\xi|}{|\delta|} \right)^d \cdot \frac{|\hat{f}|_{d+1,0}}{(1 + |\xi|/|\delta|)^{d+1}} \rightarrow 0 \quad \text{as } \delta \rightarrow 0.$$

When  $\xi = 0$ , the expression simplifies to:

$$|\widehat{f(\delta \cdot)}(0)| = |\delta|^{-d} |\hat{f}(0)| \rightarrow \infty \quad \text{as } \delta \rightarrow 0,$$

since  $\hat{f}(0) = \int_{\mathbb{R}^d} f(x) dx \neq 0$ . This highlights the reciprocal relation between physical scale and frequency scale under dilation.

### Property 3

If  $f \in S$ , then  $\hat{f} \in C^\infty(\mathbb{R}^d)$  with all derivatives bounded, satisfying:

$$\partial^\beta \hat{f}(\xi) = \hat{g}(\xi) \quad \text{where} \quad g(x) = (-i2\pi x)^\beta f(x) \in S, \quad \beta \in \mathbb{N}_0^d.$$

**Proof.** Differentiating under the integral sign (justified by the rapid decay of  $f$ ):

$$\partial^\beta \hat{f}(\xi) = \int_{\mathbb{R}^d} f(x) \frac{\partial^\beta}{\partial \xi^\beta} (e^{-i2\pi x \cdot \xi}) \, dx = \int_{\mathbb{R}^d} f(x) (-i2\pi x)^\beta e^{-i2\pi x \cdot \xi} \, dx = \hat{g}(\xi).$$

□

### Proposition 1

The Fourier transform maps the Schwartz space into itself continuously,  $\mathcal{F} : S \rightarrow S$ .

**Proof.** Let  $f \in S$ . By Property 3,  $\hat{f} \in C^\infty(\mathbb{R}^d)$ . To verify  $\hat{f} \in S$ , we must show that  $\sup_{\xi \in \mathbb{R}^d} |\xi^\alpha \partial^\beta \hat{f}(\xi)| < \infty$  for all  $\alpha, \beta \in \mathbb{N}_0^d$ . We write:

$$\begin{aligned} \xi^\alpha \partial^\beta \hat{f}(\xi) &= (i2\pi)^{-|\alpha|} (i2\pi \xi)^\alpha \partial^\beta \hat{f}(\xi) \\ &= (i2\pi)^{-|\alpha|} (i2\pi \xi)^\alpha \hat{g}(\xi) \quad \text{with } g(x) = (-i2\pi x)^\beta f(x) \in S. \end{aligned}$$

Applying Property 1 to the function  $g \in S$ , this becomes:

$$\xi^\alpha \partial^\beta \hat{f}(\xi) = (i2\pi)^{-|\alpha|} \widehat{\partial^\alpha g}(\xi),$$

Taking the supremum over all frequencies  $\xi \in \mathbb{R}^d$ :

$$\sup_{\xi \in \mathbb{R}^d} |\xi^\alpha \partial^\beta \hat{f}(\xi)| = (2\pi)^{-|\alpha|} \sup_{\xi \in \mathbb{R}^d} |\widehat{\partial^\alpha g}(\xi)| \leq (2\pi)^{-|\alpha|} \|\partial^\alpha g\|_{L^1} < \infty,$$

where the  $L^1$  norm is finite since  $\partial^\alpha g \in S \subset L^1(\mathbb{R}^d)$ . Thus,  $\hat{f} \in S$ . □

### Proposition 2

Let  $f, g \in S$ . Then  $\widehat{f * g}(\xi) = \hat{f}(\xi) \hat{g}(\xi)$ .

**Proof.** By expanding the definition and applying Fubini's Theorem:

$$\begin{aligned} \widehat{f * g}(\xi) &= \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} f(x - y) g(y) \, dy \, e^{-i2\pi x \cdot \xi} \, dx \\ &= \int_{\mathbb{R}^d} \left( \int_{\mathbb{R}^d} f(x - y) e^{-i2\pi(x-y) \cdot \xi} \, dx \right) g(y) e^{-i2\pi y \cdot \xi} \, dy. \end{aligned}$$

Setting  $z = x - y$  with  $dz = dx$  inside the inner integral yields:

$$\widehat{f * g}(\xi) = \int_{\mathbb{R}^d} \left( \int_{\mathbb{R}^d} f(z) e^{-i2\pi z \cdot \xi} \, dz \right) g(y) e^{-i2\pi y \cdot \xi} \, dy = \hat{f}(\xi) \cdot \hat{g}(\xi).$$

Fubini's theorem is fully applicable because:

$$\int_{\mathbb{R}^d} \int_{\mathbb{R}^d} |f(x - y)| |g(y)| \, dy \, dx = \|f\|_{L^1} \|g\|_{L^1} < \infty.$$

□

### Proposition 3

Let  $f, g \in S$ . Then the following dual identities hold:

$$\begin{aligned}\int_{\mathbb{R}^d} \hat{f}(z)g(z) \, dz &= \int_{\mathbb{R}^d} f(y)\hat{g}(y) \, dy, \\ \int_{\mathbb{R}^d} \mathcal{F}^{-1}(f)(z)g(z) \, dz &= \int_{\mathbb{R}^d} f(y)\mathcal{F}^{-1}(g)(y) \, dy.\end{aligned}$$

**Proof.** Using Fubini's Theorem to change the order of integration:

$$\begin{aligned}\int_{\mathbb{R}^d} \hat{f}(z)g(z) \, dz &= \int_{\mathbb{R}^d} \left( \int_{\mathbb{R}^d} f(y) e^{-i2\pi y \cdot z} \, dy \right) g(z) \, dz \\ &= \int_{\mathbb{R}^d} \left( \int_{\mathbb{R}^d} g(z) e^{-i2\pi y \cdot z} \, dz \right) f(y) \, dy = \int_{\mathbb{R}^d} \hat{g}(y)f(y) \, dy.\end{aligned}$$

This is valid since  $f, g \in S \subset L^1(\mathbb{R}^d)$ , meaning:

$$\int_{\mathbb{R}^d} \int_{\mathbb{R}^d} |f(y)||g(z)||e^{-i2\pi y \cdot z}| \, dy \, dz = \|f\|_{L^1} \|g\|_{L^1} < \infty.$$

An identical argument swapping  $e^{-i2\pi y \cdot z}$  for  $e^{i2\pi y \cdot z}$  establishes the  $\mathcal{F}^{-1}$  identity.  $\square$

### Inverse of $\mathcal{F}$

#### Theorem 1

Let  $f \in S$ . Then  $\mathcal{F}^{-1}\mathcal{F}(f)(x) = \mathcal{F}\mathcal{F}^{-1}(f)(x) = f(x)$  for all  $x \in \mathbb{R}^d$ . In particular,

$$\mathcal{F}^{-1}(\hat{f})(x) = \int_{\mathbb{R}^d} \hat{f}(\xi) e^{i2\pi x \cdot \xi} \, d\xi = f(x).$$

Consequently, both  $\mathcal{F}, \mathcal{F}^{-1} : S \rightarrow S$  are bijections.

**Proof.** Let  $f \in S$ . We first prove the inversion identity at the origin  $x = 0$ , i.e.,  $f(0) = \int_{\mathbb{R}^d} \hat{f}(\xi) \, d\xi$ . Let  $k(x) = e^{-\pi|x|^2}$  be the standard Gaussian, whose Fourier transform satisfies  $\hat{k}(y) = e^{-\pi|y|^2} = k(y)$ . For any  $\delta > 0$ , by the Lebesgue Dominated Convergence Theorem:

$$\int_{\mathbb{R}^d} f(\delta y) \hat{k}(y) \, dy \longrightarrow f(0) \int_{\mathbb{R}^d} \hat{k}(y) \, dy = f(0) \quad \text{as } \delta \rightarrow 0.$$

Applying Proposition 3 to the left-hand side:

$$\begin{aligned}\int_{\mathbb{R}^d} f(\delta y) \hat{k}(y) \, dy &= \int_{\mathbb{R}^d} \widehat{f(\delta \cdot)}(\xi) k(\xi) \, d\xi \\ &= \int_{\mathbb{R}^d} \delta^{-d} \hat{f}\left(\frac{\xi}{\delta}\right) k(\xi) \, d\xi \quad (\text{by Property 2b}) \\ &= \int_{\mathbb{R}^d} \delta^{-d} \hat{f}(z) k(\delta z) \delta^d \, dz \quad (\text{substituting } z = \frac{\xi}{\delta}, \, d\xi = \delta^d dz) \\ &= \int_{\mathbb{R}^d} \hat{f}(z) k(\delta z) \, dz.\end{aligned}$$



By the Dominated Convergence Theorem, as  $\delta \rightarrow 0$ ,  $k(\delta z) = e^{-\pi\delta^2|z|^2} \rightarrow k(0) = 1$  pointwise, so:

$$\lim_{\delta \rightarrow 0} \int_{\mathbb{R}^d} \hat{f}(z) k(\delta z) dz = \int_{\mathbb{R}^d} \hat{f}(z) \cdot 1 dz = \int_{\mathbb{R}^d} \hat{f}(z) dz.$$

Equating the two limits gives  $f(0) = \int_{\mathbb{R}^d} \hat{f}(z) dz$ .

To extend this to a general point  $y \in \mathbb{R}^d$ , we define a shifted function  $f_y(x) = f(x + y)$ , which also belongs to  $S$ . Applying the origin inversion formula to  $f_y$ :

$$\begin{aligned} f(y) &= f_y(0) = \int_{\mathbb{R}^d} \mathcal{F}(f_y)(\xi) d\xi \\ &= \int_{\mathbb{R}^d} \hat{f}(\xi) e^{i2\pi\xi \cdot y} d\xi \quad (\text{by Property 2a}) \\ &= \mathcal{F}^{-1}(\hat{f})(y). \end{aligned}$$

This establishes  $\mathcal{F}^{-1}\mathcal{F}(f) = f$ .

To prove  $\mathcal{F}\mathcal{F}^{-1}(f) = f$ , we recall that  $\mathcal{F}^{-1}g(x) = \mathcal{F}g(-x)$ . Let  $\tilde{f}(x) = f(-x)$ . By change of variables:

$$\mathcal{F}^{-1}f(x) = \int_{\mathbb{R}^d} f(\xi) e^{i2\pi x \cdot \xi} d\xi = \int_{\mathbb{R}^d} f(-z) e^{-i2\pi z \cdot x} dz = \mathcal{F}\tilde{f}(x).$$

Thus, substituting this structural relation:

$$\mathcal{F}(\mathcal{F}^{-1}(f))(x) = \mathcal{F}(\mathcal{F}\tilde{f})(x) = \mathcal{F}^{-1}(\mathcal{F}\tilde{f})(-x) = \tilde{f}(-x) = f(-(-x)) = f(x).$$

Therefore,  $\mathcal{F}\mathcal{F}^{-1}(f)(x) = f(x) = \mathcal{F}^{-1}\mathcal{F}(f)(x)$ , completing the proof.

## Analysis of the Dominated Convergence Theorem (DCT) Steps

### 1. First Use of DCT in the Proof.

$$\int_{\mathbb{R}^d} f(\delta y) \hat{k}(y) dy \longrightarrow f(0) \quad \text{as } \delta \rightarrow 0.$$

- *Pointwise Convergence:* Since  $f \in S$ , it is continuous everywhere, meaning  $f(\delta y) \rightarrow f(0)$  as  $\delta \rightarrow 0$  for each fixed vector  $y$ .
- *Dominating Function:* Because  $f \in S$ , it is bounded globally by a constant  $M = \|f\|_{L^\infty}$ . Thus,  $|f(\delta y) \hat{k}(y)| \leq M |\hat{k}(y)|$ . Since  $\hat{k}(y) = e^{-\pi|y|^2}$  is an integrable Gaussian function, it serves as a valid dominator.

### 2. Second Use of DCT in the Proof.

$$\int_{\mathbb{R}^d} \hat{f}(\xi) k(\delta \xi) d\xi \longrightarrow \int_{\mathbb{R}^d} \hat{f}(\xi) d\xi \quad \text{as } \delta \rightarrow 0.$$

- *Pointwise Convergence:*  $k(\delta \xi) = e^{-\pi\delta^2|\xi|^2} \rightarrow e^0 = 1$  as  $\delta \rightarrow 0$  for each  $\xi \in \mathbb{R}^d$ .

- *Dominating Function:* Since  $k$  is a Gaussian,  $|k(\delta\xi)| \leq 1$  for all  $\delta$ . This yields the bound  $|\hat{f}(\xi)k(\delta\xi)| \leq |\hat{f}(\xi)|$ . Since  $f \in S$ , Proposition 1 ensures that  $\hat{f} \in S \subset L^1(\mathbb{R}^d)$ , making  $\hat{f}$  its own integrable dominating function.

### 3. Significance of the Schwartz Space $S$ .

- Functions in  $S$  decay rapidly: Both  $f$  and  $\hat{f}$  decrease faster than any inverse polynomial, which natively guarantees absolute integrability over  $\mathbb{R}^d$ .
- Standard Gaussians  $k(x) = e^{-\pi|x|^2}$  act as perfect smoothing multipliers: they are smooth, self-dual under  $\mathcal{F}$ , and guarantee convergence margins in integral dualities. Without the rapid polynomial decay inherent to  $S$ , uniform integrable boundaries might fail.

□

## Plancherel's Equality

For  $f \in S$ ,  $\int_{\mathbb{R}^d} |f(x)|^2 dx = \int_{\mathbb{R}^d} |\hat{f}(\xi)|^2 d\xi$ , so  $\|f\|_{L^2} = \|\hat{f}\|_{L^2}$  where  $L^2 = L^2(\mathbb{R}^d)$ .

### Idea of the Proof

Use the property  $f(0) = \int_{\mathbb{R}^d} \hat{f}(\xi) d\xi$  and the Fourier transform of the convolution. Let  $(f * g)(x) = \int_{\mathbb{R}^d} f(x-y)g(y) dy$  and  $\mathcal{F}(f * g)(\xi) = \hat{f}(\xi)\hat{g}(\xi)$ . Applying the inversion formula at  $x = 0$ , we obtain:

$$(f * g)(0) = \int_{\mathbb{R}^d} g(y)f(-y) dy = \int_{\mathbb{R}^d} \mathcal{F}(f * g)(\xi) d\xi = \int_{\mathbb{R}^d} \hat{f}(\xi)\hat{g}(\xi) d\xi \quad (\#).$$

In  $(\#)$  above, we choose an appropriate substitution for  $f$  and  $g$  to handle the reflection  $f(-y)$  and introduce the complex conjugate.

### Proposition 4 (Plancherel's Equality)

For any  $f, g \in S$ ,

$$\int_{\mathbb{R}^d} g(y)\overline{f(y)} dy = \int_{\mathbb{R}^d} \hat{g}(\xi)\overline{\hat{f}(\xi)} d\xi.$$

Thus,  $\langle g, f \rangle_{L^2} = \langle \hat{g}, \hat{f} \rangle_{L^2}$ . In particular,  $\|f\|_{L^2} = \|\hat{f}\|_{L^2}$  when  $f = g$ .

**Proof.** Let  $f, g \in S$ . Define  $\tilde{f}(y) = \overline{f(-y)}$  for  $y \in \mathbb{R}^d$ , and note that  $\tilde{f} \in S$ . By identity  $(\#)$ , we have:

$$\int_{\mathbb{R}^d} g(y)\overline{f(y)} dy = \int_{\mathbb{R}^d} g(y)\tilde{f}(-y) dy = \int_{\mathbb{R}^d} \hat{g}(\xi)\mathcal{F}(\tilde{f})(\xi) d\xi.$$

Now, we compute  $\mathcal{F}(\tilde{f})(\xi)$ :

$$\begin{aligned}\mathcal{F}(\tilde{f})(\xi) &= \int_{\mathbb{R}^d} \overline{f(-y)} e^{-i2\pi y \cdot \xi} dy \\ &= \int_{\mathbb{R}^d} \overline{f(x)} e^{i2\pi x \cdot \xi} dx \quad (\text{substituting } x = -y, dx = dy, |J| = 1) \\ &= \overline{\int_{\mathbb{R}^d} f(x) e^{-i2\pi x \cdot \xi} dx} = \overline{\hat{f}(\xi)}.\end{aligned}$$

Hence,  $\int_{\mathbb{R}^d} g(y) \overline{f(y)} dy = \int_{\mathbb{R}^d} \hat{g}(\xi) \overline{\hat{f}(\xi)} d\xi$ , which implies  $\langle g, f \rangle_{L^2} = \langle \hat{g}, \hat{f} \rangle_{L^2}$ .  $\square$

**Remark.** The operators  $\mathcal{F}$  and  $\mathcal{F}^{-1}$  can be uniquely extended to unitary maps on  $L^2(\mathbb{R}^d)$ . For any  $f \in L^2$ , there exists a sequence  $f_n \in S$  such that  $f_n \rightarrow f$  in  $L^2$ , since  $C_c^\infty \subseteq S$  and  $C_c^\infty$  is dense in  $L^2$ . By Proposition 4, if  $\|f_n - f_m\|_{L^2} \rightarrow 0$  as  $m, n \rightarrow \infty$ , then:

$$\|f_n - f_m\|_{L^2} = \|\hat{f}_n - \hat{f}_m\|_{L^2} = \|\mathcal{F}^{-1}(f_n) - \mathcal{F}^{-1}(f_m)\|_{L^2}.$$

This allows us to define  $\mathcal{F}f = \lim_{n \rightarrow \infty} \mathcal{F}(f_n)$  and  $\mathcal{F}^{-1}f = \lim_{n \rightarrow \infty} \mathcal{F}^{-1}(f_n)$  via Cauchy completeness. This definition is independent of the chosen sequence. Indeed, if  $g_n \in S$  is another sequence with  $g_n \rightarrow f$  in  $L^2$ , then  $\|g_n - f_n\|_{L^2} \rightarrow 0$  as  $n \rightarrow \infty$ . By Proposition 4,

$$\|g_n - f_n\|_{L^2} = \|\hat{g}_n - \hat{f}_n\|_{L^2} = \|\mathcal{F}^{-1}g_n - \mathcal{F}^{-1}f_n\|_{L^2} \rightarrow 0,$$

which implies  $\hat{g}_n \rightarrow \mathcal{F}f$  and  $\mathcal{F}^{-1}g_n \rightarrow \mathcal{F}^{-1}f$  in  $L^2$ . Hence,  $\mathcal{F} : L^2 \rightarrow L^2$  is a unitary operator satisfying  $\mathcal{F}^* = \mathcal{F}^{-1}$ .

### Corollary 1

The maps  $\mathcal{F}, \mathcal{F}^{-1} : S \rightarrow S$  uniquely extend to bounded linear operators  $\mathcal{F}, \mathcal{F}^{-1} : L^2 \rightarrow L^2$  via:

$$\mathcal{F}f = \hat{f} = \lim_{n \rightarrow \infty} \mathcal{F}f_n, \quad \mathcal{F}^{-1}f = \lim_{n \rightarrow \infty} \mathcal{F}^{-1}f_n \quad (\text{in } L^2),$$

where  $f_n \in S$  and  $f_n \rightarrow f$  in  $L^2$ . Furthermore,  $\mathcal{F}\mathcal{F}^{-1}f = \mathcal{F}^{-1}\mathcal{F}f = f$  and

$$\|\mathcal{F}f\|_{L^2} = \|\mathcal{F}^{-1}f\|_{L^2} = \|f\|_{L^2} \quad \text{for all } f \in L^2.$$

### Proposition 6

Let  $f \in L^1 \cap L^2$ . Let  $\hat{f} = \lim_{n \rightarrow \infty} \hat{f}_n$  and  $\mathcal{F}^{-1}f = \lim_{n \rightarrow \infty} \mathcal{F}^{-1}f_n$  in  $L^2$ , where  $f_n \in S$  and  $f_n \rightarrow f$  in  $L^2$ . Then the standard integral representations hold pointwise almost everywhere:

$$\hat{f}(\xi) = \int_{\mathbb{R}^d} f(x) e^{-i2\pi x \cdot \xi} dx \quad \text{and} \quad \mathcal{F}^{-1}f(x) = \int_{\mathbb{R}^d} f(\xi) e^{i2\pi x \cdot \xi} d\xi.$$

**Proof.** By standard properties of the Fourier transform on  $S$ , for any test function  $h \in S$ , we have:

$$\int_{\mathbb{R}^d} \hat{f}_n h dx = \int_{\mathbb{R}^d} f_n \hat{h} dx \quad \text{and} \quad \int_{\mathbb{R}^d} \mathcal{F}^{-1}(f_n) h dx = \int_{\mathbb{R}^d} f_n \mathcal{F}^{-1}(h) dx.$$

Taking the limit as  $n \rightarrow \infty$ ,  $L^2$  convergence implies weak convergence against  $h \in S$ :

$$\int_{\mathbb{R}^d} \hat{f}_n h \rightarrow \int_{\mathbb{R}^d} \hat{f} h, \quad \int_{\mathbb{R}^d} f_n \hat{h} \rightarrow \int_{\mathbb{R}^d} f \hat{h},$$

yielding  $\int_{\mathbb{R}^d} \hat{f} h = \int_{\mathbb{R}^d} f \hat{h}$  (and similarly for  $\mathcal{F}^{-1}$ ). Now, for any  $h \in S$ :

$$\begin{aligned} \int_{\mathbb{R}^d} \hat{f}(y) h(y) \, dy &= \int_{\mathbb{R}^d} f(x) \hat{h}(x) \, dx \\ &= \int_{\mathbb{R}^d} f(x) \left( \int_{\mathbb{R}^d} h(y) e^{-i2\pi x \cdot y} \, dy \right) \, dx \\ &= \int_{\mathbb{R}^d} \left( \int_{\mathbb{R}^d} f(x) e^{-i2\pi x \cdot y} \, dx \right) h(y) \, dy \quad (\text{by Fubini's Theorem}). \end{aligned}$$

Since this holds for all  $h \in S$ , by the fundamental lemma of the calculus of variations, we conclude:

$$\hat{f}(y) = \int_{\mathbb{R}^d} f(x) e^{-i2\pi x \cdot y} \, dx \quad \text{a.e.}$$

Fubini's theorem is justified since  $\int_{\mathbb{R}^d} \int_{\mathbb{R}^d} |f(x)| \cdot |h(y)| \, dx \, dy = \|f\|_{L^1} \|h\|_{L^1} < \infty$ . An analogous proof holds for  $\mathcal{F}^{-1}$ .  $\square$

## Corollary 2

Let  $f \in L^2(\mathbb{R}^d)$ . Then the Fourier transform can be evaluated as a principal value limit:

$$\begin{aligned} \mathcal{F}f(\xi) &= \hat{f}(\xi) = \lim_{n \rightarrow \infty} \int_{|x| \leq n} f(x) e^{-i2\pi x \cdot \xi} \, dx \quad (\text{in } L^2), \\ \mathcal{F}^{-1}f(x) &= \lim_{n \rightarrow \infty} \int_{|\xi| \leq n} f(\xi) e^{i2\pi x \cdot \xi} \, d\xi \quad (\text{in } L^2). \end{aligned}$$

**Proof.** Let  $f \in L^2(\mathbb{R}^d)$  and define  $f_n = \chi_{\{|x| \leq n\}} f$ . By Hölder's inequality,  $\int_{\mathbb{R}^d} |f_n| \leq \|f\|_{L^2} \|\chi_{\{|x| \leq n\}}\|_{L^2} < \infty$ , so  $f_n \in L^1 \cap L^2$ . Furthermore,  $f_n \rightarrow f$  in  $L^2$  by the Dominated Convergence Theorem, since  $|f_n| \leq |f|$  and  $|f|^2 \in L^1$ .

By Proposition 6, since  $f_n \in L^1 \cap L^2$ :

$$\hat{f}_n(\xi) = \int_{\mathbb{R}^d} f_n(x) e^{-i2\pi x \cdot \xi} \, dx = \int_{|x| \leq n} f(x) e^{-i2\pi x \cdot \xi} \, dx.$$

Since  $\mathcal{F}$  is unitary on  $L^2$ ,  $\|\hat{f}_n - \hat{f}\|_{L^2} = \|f_n - f\|_{L^2} \rightarrow 0$ , meaning  $\hat{f}_n \rightarrow \hat{f}$  in  $L^2$ . Hence,

$$\hat{f}(\xi) = \lim_{n \rightarrow \infty} \int_{|x| \leq n} f(x) e^{-i2\pi x \cdot \xi} \, dx \quad \text{in } L^2.$$

An analogous argument using  $g_n = \chi_{\{|\xi| \leq n\}} f$  yields the formula for  $\mathcal{F}^{-1}$ .  $\square$

# Some Applications of the Fourier Transform

## 1. Heisenberg Uncertainty Principle

Let  $\psi \in S(\mathbb{R})$  be a normalized wave function, satisfying  $\int_{\mathbb{R}} |\psi(x)|^2 dx = 1$ . In quantum mechanics,  $|\psi(x)|^2$  represents the probability density function (pdf) of a particle's position. By Plancherel's theorem:

$$\int_{\mathbb{R}} |\hat{\psi}(\xi)|^2 d\xi = \int_{\mathbb{R}} |\psi(x)|^2 dx = 1,$$

meaning  $|\hat{\psi}(\xi)|^2$  serves as the pdf of the particle's momentum. If  $x_0$  and  $\xi_0$  denote the expected position and momentum respectively, their corresponding variances (mean square errors) are:

$$(\Delta x)^2 = \int_{\mathbb{R}} |x - x_0|^2 |\psi(x)|^2 dx, \quad (\Delta \xi)^2 = \int_{\mathbb{R}} |\xi - \xi_0|^2 |\hat{\psi}(\xi)|^2 d\xi.$$

### Theorem 2 (Heisenberg Uncertainty Principle).

Let  $\psi \in S(\mathbb{R})$  with  $\int_{\mathbb{R}} |\psi(x)|^2 dx = 1$ . Then:

$$(\Delta x)^2 (\Delta \xi)^2 \geq \frac{1}{16\pi^2}.$$

**Proof.** First, consider the case where  $x_0 = \xi_0 = 0$ , giving  $(\Delta x)^2 = \int_{\mathbb{R}} x^2 |\psi(x)|^2 dx$  and  $(\Delta \xi)^2 = \int_{\mathbb{R}} \xi^2 |\hat{\psi}(\xi)|^2 d\xi$ . Since  $\psi \in S(\mathbb{R})$ , it decays faster than any polynomial, implying  $x|\psi(x)|^2 \rightarrow 0$  as  $|x| \rightarrow \infty$ . Integration by parts yields:

$$\begin{aligned} 1 &= \int_{-\infty}^{\infty} |\psi(x)|^2 dx = \left[ x |\psi(x)|^2 \right]_{-\infty}^{\infty} - \int_{-\infty}^{\infty} x (\psi'(x) \overline{\psi(x)} + \psi(x) \overline{\psi'(x)}) dx \\ &= -2 \operatorname{Re} \int_{-\infty}^{\infty} x \psi'(x) \overline{\psi(x)} dx. \end{aligned}$$

Applying the Cauchy–Schwarz inequality:

$$\begin{aligned} 1 &\leq 2 \int_{-\infty}^{\infty} |x \psi(x)| |\psi'(x)| dx \\ &\leq 2 \left( \int_{-\infty}^{\infty} x^2 |\psi(x)|^2 dx \right)^{1/2} \left( \int_{-\infty}^{\infty} |\psi'(x)|^2 dx \right)^{1/2} \\ &= 2 (\Delta x) \left( \int_{-\infty}^{\infty} |\psi'|^2 dx \right)^{1/2}. \end{aligned}$$

By Plancherel's equality and derivative properties,  $\widehat{\psi'}(\xi) = i2\pi\xi \hat{\psi}(\xi)$ , which implies:

$$\int_{-\infty}^{\infty} |\psi'(x)|^2 dx = \int_{-\infty}^{\infty} |\widehat{\psi'}(\xi)|^2 d\xi = \int_{-\infty}^{\infty} (2\pi|\xi|)^2 |\hat{\psi}(\xi)|^2 d\xi = 4\pi^2 (\Delta \xi)^2.$$

Combining these gives  $1 \leq 2(\Delta x) \cdot 2\pi(\Delta \xi) = 4\pi(\Delta x)(\Delta \xi)$ , squaring both sides results in:

$$(\Delta x)^2(\Delta \xi)^2 \geq \frac{1}{16\pi^2}.$$

For general  $x_0, \xi_0 \in \mathbb{R}$ , apply the above result to the shifted wave function  $\varphi(x) = e^{-i2\pi\xi_0 x} \psi(x + x_0)$ . It easily follows that  $\int_{\mathbb{R}} |\varphi|^2 = 1$  and  $|\hat{\varphi}(\xi)|^2 = |\hat{\psi}(\xi + \xi_0)|^2$ . A standard change of variables confirms that the variances transform accordingly, completing the proof.  $\square$

## 2. Helmholtz Equation in $S(\mathbb{R}^d)$

Consider the Helmholtz-type equation:

$$u(x) - \Delta u(x) = f(x), \quad x \in \mathbb{R}^d, \quad f \in S(\mathbb{R}^d), \quad (\text{H})$$

where  $\Delta = \sum_{j=1}^d \frac{\partial^2}{\partial x_j^2}$  denotes the Laplacian operator. Taking the Fourier transform gives:

$$\widehat{\Delta u}(\xi) = -4\pi^2|\xi|^2 \hat{u}(\xi).$$

### Proposition 7

For any  $f \in S(\mathbb{R}^d)$ , there exists a unique solution  $u \in S(\mathbb{R}^d)$  solving (H). That is, the operator  $(I - \Delta) : S \rightarrow S$  is a bijection.

**Proof.** Let  $f \in S$ . Then  $\hat{f} \in S$ . Since the polynomial  $1 + 4\pi^2|\xi|^2$  is smooth, bounded below by 1, and has bounded derivatives of all orders, its reciprocal  $\frac{1}{1+4\pi^2|\xi|^2}$  serves as a multiplier on  $S$ . Thus,  $\frac{\hat{f}(\xi)}{1+4\pi^2|\xi|^2} \in S$ .

(1) *Uniqueness.* Suppose  $u \in S$  satisfies  $u - \Delta u = f$ . Taking the Fourier transform on both sides:

$$\widehat{u - \Delta u}(\xi) = \hat{f}(\xi) \implies (1 + 4\pi^2|\xi|^2) \hat{u}(\xi) = \hat{f}(\xi) \implies \hat{u}(\xi) = \frac{\hat{f}(\xi)}{1 + 4\pi^2|\xi|^2}.$$

Thus  $\hat{u}$  is uniquely determined. Since  $\mathcal{F} : S \rightarrow S$  is bijective,  $u$  is unique.

(2) *Existence.* Given  $f \in S$ , define  $g(\xi) = \frac{\hat{f}(\xi)}{1+4\pi^2|\xi|^2} \in S$ . Setting  $u(x) = (\mathcal{F}^{-1}g)(x) \in S$ , we check that it solves (H):

$$\widehat{u - \Delta u}(\xi) = (1 + 4\pi^2|\xi|^2) \hat{u}(\xi) = (1 + 4\pi^2|\xi|^2) g(\xi) = \hat{f}(\xi).$$

By the injectivity of  $\mathcal{F}$ , we conclude  $u - \Delta u = f$ .  $\square$

## Tempered Distributions

### Fundamental Definitions and Recall

**Notation.** Let  $\mathcal{S}' = \mathcal{S}'(\mathbb{R}^d)$  denote the dual space containing all linear continuous maps  $T : \mathcal{S}(\mathbb{R}^d) \rightarrow \mathbb{C}$ . The elements of  $\mathcal{S}'$  are called *tempered distributions*, and  $\mathcal{S}$  represents the Schwartz space of rapidly decreasing functions.

## Continuity in $\mathcal{S}$

A linear map  $T : \mathcal{S} \rightarrow \mathbb{C}$  is continuous if and only if it is bounded with respect to the topology generated by the Schwartz semi-norms. Specifically, there exist constants  $m \geq 1$ ,  $C \geq 0$ , multi-indices  $\alpha_1, \dots, \alpha_m \in \mathbb{N}_0^d$ , and integers  $N_1, \dots, N_m \in \mathbb{N}_0$  such that:

$$|T(f)| \leq C \sum_{k=1}^m |f|_{N_k, \alpha_k} \quad \forall f \in \mathcal{S}.$$

Equivalently, using the directed system of semi-norms  $|f|_{(N)} = \max_{|\alpha|, \beta \leq N} \sup_x |x^\beta \partial^\alpha f(x)|$ , continuity is satisfied if there exist  $N \in \mathbb{N}_0$  and  $C \geq 0$  such that:

$$|T(f)| \leq C |f|_{(N)} \quad \forall f \in \mathcal{S}.$$

## Convergence in $\mathcal{S}'$

Let  $\{T_n\}_{n=1}^\infty \subset \mathcal{S}'$  and  $T \in \mathcal{S}'$ . We say that  $T_n \rightarrow T$  weakly-\* in  $\mathcal{S}'$  if:

$$T_n(f) \rightarrow T(f) \quad \forall f \in \mathcal{S}.$$

## Example: The Space of Smooth Functions of Slow Growth ( $\mathcal{O}_M$ )

Let  $\mathcal{O}_M(\mathbb{R}^d)$  be the space of all smooth functions  $g \in C^\infty(\mathbb{R}^d)$  whose derivatives grow at most polynomially. That is, for every multi-index  $\alpha \in \mathbb{N}_0^d$ , there exist constants  $C = C(\alpha) \geq 0$  and  $N = N(\alpha) \in \mathbb{N}_0$  such that:

$$|\partial^\alpha g(x)| \leq C(1 + |x|)^N \quad \forall x \in \mathbb{R}^d.$$

- a. *Regular Distribution Mapping.* Every  $g \in \mathcal{O}_M$  induces a well-defined regular tempered distribution  $T_g \in \mathcal{S}'$  via the canonical pairing:

$$T_g(f) = \int_{\mathbb{R}^d} g(x) f(x) dx, \quad f \in \mathcal{S}.$$

**Proof.** Let  $\alpha = 0$ . By definition, there exist  $C(0) \geq 0$  and  $N \in \mathbb{N}_0$  such that  $|g(x)| \leq C(0)(1 + |x|)^N$ . For any  $f \in \mathcal{S}$ , we can ensure convergence by controlling the polynomial growth with a sufficiently large algebraic weight:

$$\begin{aligned} \int_{\mathbb{R}^d} |g(x)| \cdot |f(x)| dx &\leq C(0) \int_{\mathbb{R}^d} (1 + |x|)^N |f(x)| dx \\ &= C(0) \int_{\mathbb{R}^d} (1 + |x|)^{N+d+1} |f(x)| \cdot \frac{dx}{(1 + |x|)^{d+1}} \\ &\leq C(0) |f|_{N+d+1, 0} \int_{\mathbb{R}^d} \frac{dx}{(1 + |x|)^{d+1}}. \end{aligned}$$

Setting  $C = C(0) \int_{\mathbb{R}^d} (1 + |x|)^{-(d+1)} dx < \infty$ , we obtain:

$$|T_g(f)| \leq \left| \int_{\mathbb{R}^d} g(x) f(x) dx \right| \leq C |f|_{N+d+1, 0}.$$

This verifies that  $T_g$  is bounded relative to the Schwartz semi-norms, so  $T_g \in \mathcal{S}'$ .  $\square$

b. *Injectivity.* The linear mapping  $\mathcal{O}_M \ni g \mapsto T_g \in \mathcal{S}'$  is injective.

**Proof.** Let  $g_1, g_2 \in \mathcal{O}_M$  such that  $T_{g_1}(f) = T_{g_2}(f)$  for all  $f \in \mathcal{S}$ . It follows that  $\int_{\mathbb{R}^d} (g_1(x) - g_2(x))f(x) dx = 0$ . Since  $C_c^\infty(\mathbb{R}^d) \subset \mathcal{S}$  is dense in  $L_{\text{loc}}^1(\mathbb{R}^d)$ , and smooth functions are locally integrable, it follows by the fundamental lemma of the calculus of variations that  $g_1 = g_2$  almost everywhere. By continuity,  $g_1(x) = g_2(x)$  identically for all  $x \in \mathbb{R}^d$ .  $\square$

c. *Action on Derivatives.* Let  $\alpha \in \mathbb{N}_0^d$ . If  $g \in \mathcal{O}_M$ , then  $\partial^\alpha g \in \mathcal{O}_M$ . Since functions  $f \in \mathcal{S}$  decay rapidly at infinity, boundary terms vanish under integration by parts, yielding:

$$\langle \partial^\alpha g, f \rangle = \int_{\mathbb{R}^d} \partial^\alpha g(x) \cdot f(x) dx = (-1)^{|\alpha|} \int_{\mathbb{R}^d} g(x) \cdot \partial^\alpha f(x) dx = (-1)^{|\alpha|} T_g(\partial^\alpha f).$$

*Summary of Embeddings.* We establish the nested chain of topological vector spaces:

$$C_c^\infty(\mathbb{R}^d) \subseteq \mathcal{S}(\mathbb{R}^d) \subseteq \mathcal{O}_M(\mathbb{R}^d) \subseteq \mathcal{S}'(\mathbb{R}^d).$$

Moving forward, we simplify notation by writing  $T_g \equiv g$  and  $T_g(f) \equiv \langle g, f \rangle$ .

## Polynomial growth

A measurable function  $g : \mathbb{R}^d \rightarrow \mathbb{C}$  is of *polynomial growth* if there exist constants  $C > 0$  and  $N \geq 1$  such that:

$$|g(x)| \leq C(1 + |x|)^N \quad \forall x \in \mathbb{R}^d.$$

**Remark.** Any such function defines a regular tempered distribution  $T_g \in \mathcal{S}'$  with the continuous control bound given by  $|T_g(f)| \leq C|f|_{N+d+1,0}$ .

## Example: Functions of Polynomial Growth & Distributional Derivatives

Using duality, for any general tempered distribution  $T \in \mathcal{S}'$ , we define its *distributional derivative*  $\partial^\alpha T \in \mathcal{S}'$  by shifting the differential operator onto the test function:

$$(\partial^\alpha T)(f) = (-1)^{|\alpha|} T(\partial^\alpha f) \quad \forall f \in \mathcal{S}. \quad (\# \text{ micro})$$

Formally, as an operator composition, this reads:

$$\partial^\alpha T = (-1)^{|\alpha|} T \circ \partial^\alpha, \quad \mathcal{S} \xrightarrow{\partial^\alpha} \mathcal{S} \xrightarrow{T} \mathbb{C}.$$

*The Ramp Functions.* Consider the one-dimensional ramp function  $g : \mathbb{R} \rightarrow \mathbb{R}$ :

$$g(x) = \max(0, x) = \begin{cases} x, & x \geq 0 \\ 0, & x < 0 \end{cases}$$



- a. *First Distributional Derivative* ( $g'$ ). For any  $f \in \mathcal{S}(\mathbb{R})$ , applying definition (*#micro*) yields:

$$\begin{aligned}\left\langle \frac{d}{dx}g, f \right\rangle &= - \int_{-\infty}^{\infty} g(x)f'(x) dx = - \int_0^{\infty} x f'(x) dx \\ &= - \left[ x f(x) \right]_0^{\infty} + \int_0^{\infty} f(x) dx.\end{aligned}$$

The boundary term vanishes at  $\infty$  because  $f \in \mathcal{S}$  decays faster than any polynomial. Thus:

$$\left\langle \frac{d}{dx}g, f \right\rangle = \int_0^{\infty} f(x) dx = \int_{-\infty}^{\infty} H(x)f(x) dx,$$

where  $H(x)$  is the Heaviside step function. Consequently,  $g' = H$  in the distributional sense:

$$H(x) = \begin{cases} 1, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

- b. *Second Distributional Derivative* ( $H'$ ). Differentiating the Heaviside distribution again:

$$\begin{aligned}\left\langle \frac{dH}{dx}, f \right\rangle &= - \int_{-\infty}^{\infty} H(x)f'(x) dx = - \int_0^{\infty} f'(x) dx \\ &= - \left[ f(x) \right]_0^{\infty} = f(0).\end{aligned}$$

This maps to the evaluation at the origin, which corresponds to the Dirac delta distribution  $\delta_0$ :

$$\langle \delta_0, f \rangle = f(0) \implies \frac{dH}{dx} = \delta_0.$$

Formally, this relation highlights the link between distributional derivatives and Lebesgue-Stieltjes integration:  $\int_{-\infty}^{\infty} f(x)H'(x) dx = \int_{-\infty}^{\infty} f(x) dH(x)$ .

### Example: $L^p$ Spaces as Tempered Distributions

Let  $g \in L^p(\mathbb{R}^d)$  with  $p \geq 1$ . Then  $g$  induces a regular tempered distribution  $T_g \in \mathcal{S}'$ .

**Proof.** Let  $q$  be the conjugate exponent matching  $p$  ( $\frac{1}{p} + \frac{1}{q} = 1$ ). Applying Hölder's inequality:

$$\int_{\mathbb{R}^d} |g(x)| \cdot |f(x)| dx \leq \|g\|_{L^p} \|f\|_{L^q}.$$

To dominate the  $L^q$  norm of the test function  $f \in \mathcal{S}$  using standard semi-norms, choose an integer index  $m = \lceil \frac{d+1}{q} \rceil$ . This selection ensures polynomial convergence:

$$\begin{aligned}\|f\|_{L^q}^q &= \int_{\mathbb{R}^d} |f(x)|^q dx \\ &= \int_{\mathbb{R}^d} \left( |f(x)| (1 + |x|)^{\frac{d+1}{q}} \right)^q \cdot \frac{dx}{(1 + |x|)^{d+1}} \\ &\leq |f|_{m,0}^q \int_{\mathbb{R}^d} \frac{dx}{(1 + |x|)^{d+1}}.\end{aligned}$$

Since  $\int_{\mathbb{R}^d} (1 + |x|)^{-(d+1)} dx = C < \infty$ , taking the  $q$ -th root gives  $\|f\|_{L^q} \leq C^{1/q} |f|_{m,0}$ . Combining these results:

$$|T_g(f)| \leq \|g\|_{L^p} \|f\|_{L^q} \leq C^{1/q} \|g\|_{L^p} |f|_{m,0} < \infty.$$

Thus,  $T_g$  is a continuous linear functional on  $\mathcal{S}$ , meaning  $L^p(\mathbb{R}^d) \subset \mathcal{S}'(\mathbb{R}^d)$ . Injectivity follows via a standard density argument analogous to the one used in part b of the preceding example.  $\square$

**Recall.** We embed  $L^p(\mathbb{R}^d) \subseteq \mathcal{S}'(\mathbb{R}^d)$  for  $p \geq 1$  by identifying  $g \in L^p$  with its regular tempered distribution  $T_g \equiv g$ , defined via:

$$\langle g, f \rangle = \int_{\mathbb{R}^d} g(x) f(x) dx, \quad f \in \mathcal{S}(\mathbb{R}^d).$$

In particular,  $L^2(\mathbb{R}^d) \subseteq \mathcal{S}'(\mathbb{R}^d)$ . Recall that if  $g \in L^2$ , its Fourier transform  $\mathcal{F}g = \hat{g}$  and inverse Fourier transform  $\mathcal{F}^{-1}g$  also belong to  $L^2$ . By the Parseval-Plancherel identities on  $\mathcal{S}$ , we have:

$$\int_{\mathbb{R}^d} \hat{g}(x) f(x) dx = \int_{\mathbb{R}^d} g(x) \hat{f}(x) dx, \quad \int_{\mathbb{R}^d} (\mathcal{F}^{-1}g)(x) f(x) dx = \int_{\mathbb{R}^d} g(x) (\mathcal{F}^{-1}f)(x) dx,$$

which holds by density for all  $f \in L^2$ . This motivates the general definitions of basic operations on  $\mathcal{S}'(\mathbb{R}^d)$  via duality.

### Main Operations on $\mathcal{S}'(\mathbb{R}^d)$

1. *Differentiation.* Given  $T \in \mathcal{S}'(\mathbb{R}^d)$  and a multi-index  $\alpha \in \mathbb{N}_0^d$ , the distributional derivative  $\partial^\alpha T \in \mathcal{S}'(\mathbb{R}^d)$  is defined by:

$$\langle \partial^\alpha T, f \rangle = (-1)^{|\alpha|} \langle T, \partial^\alpha f \rangle \quad \forall f \in \mathcal{S}(\mathbb{R}^d).$$

2. *Fourier Transform.* For any  $T \in \mathcal{S}'(\mathbb{R}^d)$ , we define its Fourier transform  $\mathcal{F}T = \hat{T} \in \mathcal{S}'(\mathbb{R}^d)$  and inverse Fourier transform  $\mathcal{F}^{-1}T \in \mathcal{S}'(\mathbb{R}^d)$  via:

$$\langle \mathcal{F}T, f \rangle = \langle T, \mathcal{F}f \rangle, \quad \langle \mathcal{F}^{-1}T, f \rangle = \langle T, \mathcal{F}^{-1}f \rangle \quad \forall f \in \mathcal{S}(\mathbb{R}^d).$$

**Note.** Operationally,  $\mathcal{F}T = T \circ \mathcal{F}$ , corresponding to the composition  $\mathcal{S} \xrightarrow{\mathcal{F}} \mathcal{S} \xrightarrow{T} \mathbb{C}$ .

3. *Multiplication by Smooth Functions of Polynomial Growth ( $\mathcal{O}_M$ ).* Given  $T \in \mathcal{S}'(\mathbb{R}^d)$  and  $F \in \mathcal{O}_M(\mathbb{R}^d)$ , the product  $FT \in \mathcal{S}'(\mathbb{R}^d)$  is defined by:

$$\langle FT, f \rangle = \langle T, Ff \rangle \quad \forall f \in \mathcal{S}(\mathbb{R}^d),$$

which is well-defined since the map  $f \mapsto Ff$  is continuous from  $\mathcal{S}$  into itself.

## The Space of Moderate Measures ( $\mathcal{M}_{\text{mod}}$ )

Let  $\mathcal{M}_{\text{mod}}(\mathbb{R}^d)$  denote the space of signed (or complex) Borel measures  $\mu$  on  $\mathbb{R}^d$  of *moderate growth*, meaning there exists an integer  $N \geq 1$  such that:

$$\int_{\mathbb{R}^d} (1 + |x|)^{-N} d|\mu|(x) < \infty. \quad (\# \text{ mod})$$

The Lebesgue measure on  $\mathbb{R}^d$  belongs to  $\mathcal{M}_{\text{mod}}$  since  $(\# \text{ mod})$  converges for any integer  $N \geq d + 1$ . Every  $\mu \in \mathcal{M}_{\text{mod}}$  induces a linear functional  $T_\mu \in \mathcal{S}'$  via integration:

$$T_\mu(f) = \int_{\mathbb{R}^d} f(x) d\mu(x), \quad f \in \mathcal{S}.$$

$T_\mu$  is a continuous linear functional on  $\mathcal{S}$ .

**Proof.** Linearity is immediate. It remains to prove continuity. Let  $f \in \mathcal{S}$ . Using the moderate growth bound  $(\# \text{ mod})$ :

$$\int_{\mathbb{R}^d} |f(x)| d|\mu|(x) = \int_{\mathbb{R}^d} |f(x)|(1 + |x|)^N \cdot (1 + |x|)^{-N} d|\mu|(x) \leq |f|_{N,0} \int_{\mathbb{R}^d} (1 + |x|)^{-N} d|\mu|(x).$$

Setting  $C = \int_{\mathbb{R}^d} (1 + |x|)^{-N} d|\mu|(x) < \infty$ , we obtain  $|T_\mu(f)| \leq C|f|_{N,0}$ . This verifies that  $T_\mu \in \mathcal{S}'$ .  $\square$

The mapping  $\mu \mapsto T_\mu$  is injective, allowing the identification  $\mathcal{M}_{\text{mod}} \subseteq \mathcal{S}'$ . We write  $\langle \mu, f \rangle = \int_{\mathbb{R}^d} f d\mu$ . Its operational extensions to derivatives and Fourier transforms follow by duality:

$$\langle \partial^\alpha \mu, f \rangle = (-1)^{|\alpha|} \langle \mu, \partial^\alpha f \rangle, \quad \langle \mathcal{F}\mu, f \rangle = \langle \mu, \mathcal{F}f \rangle, \quad \langle \mathcal{F}^{-1}\mu, f \rangle = \langle \mu, \mathcal{F}^{-1}f \rangle.$$

Now assume that  $\mu$  is a finite signed Borel measure. By Fubini's theorem, we can compute its inverse Fourier transform explicitly:

$$\begin{aligned} \langle \mathcal{F}^{-1}\mu, f \rangle &= \langle \mu, \mathcal{F}^{-1}f \rangle = \int_{\mathbb{R}^d} (\mathcal{F}^{-1}f)(x) d\mu(x) \\ &= \int_{\mathbb{R}^d} \left( \int_{\mathbb{R}^d} f(\xi) e^{i2\pi x \cdot \xi} d\xi \right) d\mu(x) \\ &= \int_{\mathbb{R}^d} f(\xi) \left( \int_{\mathbb{R}^d} e^{i2\pi x \cdot \xi} d\mu(x) \right) d\xi = \int_{\mathbb{R}^d} f(\xi) \phi(\xi) d\xi, \end{aligned}$$

where  $\phi(\xi) = \int_{\mathbb{R}^d} e^{i2\pi x \cdot \xi} d\mu(x) = \mathbb{E}[e^{i2\pi X \cdot \xi}]$  is the characteristic function of the measure  $\mu$  (equivalently, its Fourier–Stieltjes transform). Therefore, in the sense of  $\mathcal{S}'$  we have  $\mathcal{F}^{-1}\mu = \phi$ , so the inverse transform is the regular distribution induced by the bounded continuous function  $\phi$ .

### Proposition 1

The operators  $\mathcal{F}, \mathcal{F}^{-1} : \mathcal{S}'(\mathbb{R}^d) \rightarrow \mathcal{S}'(\mathbb{R}^d)$  are linear bijections.

**Proof.** By the inversion theorem on  $\mathcal{S}$ , we know  $\mathcal{F}\mathcal{F}^{-1} = \mathcal{F}^{-1}\mathcal{F} = \text{Id}_{\mathcal{S}}$ . Testing the distributional composition on an arbitrary  $f \in \mathcal{S}$ :

$$\langle \mathcal{F}^{-1}\mathcal{F}T, f \rangle = \langle \mathcal{F}T, \mathcal{F}^{-1}f \rangle = \langle T, \mathcal{F}\mathcal{F}^{-1}f \rangle = \langle T, f \rangle.$$

Similarly,  $\langle \mathcal{F}\mathcal{F}^{-1}T, f \rangle = \langle T, f \rangle$ . Thus  $\mathcal{F}^{-1}\mathcal{F}T = \mathcal{F}\mathcal{F}^{-1}T = T$ , completing the proof.  $\square$

### Derivative-Polynomial Duality

Let  $T \in \mathcal{S}'(\mathbb{R}^d)$  and  $\alpha \in \mathbb{N}_0^d$ . Prove that  $\widehat{\partial^\alpha T} = (i2\pi\xi)^\alpha \hat{T}$ .

**Proof.** Since  $(i2\pi\xi)^\alpha$  has smooth coefficients growing polynomially, it belongs to  $\mathcal{O}_M$ . For any  $f \in \mathcal{S}$ :

$$\langle \widehat{\partial^\alpha T}, f \rangle = \langle \partial^\alpha T, \hat{f} \rangle = (-1)^{|\alpha|} \langle T, \partial^\alpha \hat{f} \rangle.$$

Using the standard identity  $\partial^\alpha \hat{f} = \mathcal{F}((-i2\pi x)^\alpha f)$ , we set  $g(x) = (-i2\pi x)^\alpha f(x) \in \mathcal{S}$ . Continuing the evaluation:

$$\begin{aligned} (-1)^{|\alpha|} \langle T, \hat{g} \rangle &= (-1)^{|\alpha|} \langle \hat{T}, g \rangle = (-1)^{|\alpha|} \int_{\mathbb{R}^d} \hat{T}(\xi) (-i2\pi\xi)^\alpha f(\xi) d\xi \\ &= \int_{\mathbb{R}^d} (i2\pi\xi)^\alpha \hat{T}(\xi) f(\xi) d\xi = \langle (i2\pi\xi)^\alpha \hat{T}, f \rangle. \end{aligned}$$

Thus,  $\widehat{\partial^\alpha T} = (i2\pi\xi)^\alpha \hat{T}$  holds in the sense of distributions.  $\square$

### Theorem 1 (Structure Theorem for Tempered Distributions)

For every  $T \in \mathcal{S}'(\mathbb{R}^d)$ , there exists a continuous function  $g$  of polynomial growth and a multi-index  $\alpha \in \mathbb{N}_0^d$  such that  $T = \partial^\alpha g$ . That is,

$$\langle T, f \rangle = (-1)^{|\alpha|} \int_{\mathbb{R}^d} g(x) \partial^\alpha f(x) dx \quad \forall f \in \mathcal{S}(\mathbb{R}^d).$$

**Note on Polynomial Growth.** Recall that a function  $g : \mathbb{R}^d \rightarrow \mathbb{C}$  is of *polynomial growth* if  $|g(x)| \leq C(1 + |x|)^N$  for all  $x \in \mathbb{R}^d$ , for some  $C > 0$  and integer  $N \geq 1$ . As an example, the function  $g(x) = \max\{x, 0\}$  satisfies  $g \in \mathcal{S}'(\mathbb{R})$  and its second derivative yields the Dirac delta distribution,  $\frac{d^2}{dx^2}g = \delta_0$ .

**Proof.** Case  $d = 1$ . We first establish an integration lemma.

**Lemma 1**

Let  $h : \mathbb{R} \rightarrow \mathbb{C}$  be a measurable function of polynomial growth. Then there exists a continuous function  $H$  of polynomial growth such that:

$$\int_{-\infty}^{\infty} h(x) f^{(n)}(x) dx = \int_{-\infty}^{\infty} H(x) f^{(n+1)}(x) dx \quad \forall f \in \mathcal{S}(\mathbb{R}), n \in \mathbb{N}_0.$$

**Proof.** Define the primitive  $G(x) = \int_0^x h(t) dt$ . Since  $h$  is of polynomial growth, there are constants such that  $|h(x)| \leq C_1(1 + |x|)^N$ . Using  $|x| \leq \frac{|x|^2+1}{2}$ , we can also bound it as  $|h(x)| \leq C_2(1 + |x|^2)^N \leq C_3(1 + |x|^{2N})$  (since  $|x|^k \leq \frac{|x|^{2N}}{C_4} + C_5$ ). For  $x \geq 0$ :

$$|G(x)| \leq \int_0^x |h(t)| dt \leq C_6 \int_0^x (1 + t^{2N}) dt = C_6 x + C_6 \frac{x^{2N+1}}{2N+1}.$$

A symmetric argument holds for  $x < 0$ , where  $|G(x)| \leq \int_x^0 |h(t)| dt \leq C_6|x| + C_6 \frac{|x|^{2N+1}}{2N+1}$ ; hence  $|G(x)| \leq C_7(1 + |x|)^{2N+1}$ , meaning  $G$  is continuous and grows at most polynomially. Integrating by parts:

$$\int_{-\infty}^{\infty} G(x) f^{(n+1)}(x) dx = \left[ G(x) f^{(n)}(x) \right]_{-\infty}^{\infty} - \int_{-\infty}^{\infty} h(x) f^{(n)}(x) dx.$$

Because  $f \in \mathcal{S}$ ,  $f^{(n)}(x)$  decays faster than any power of  $x$ , causing the boundary term to vanish. Setting  $H = -G$  yields  $\langle H, f^{(n+1)} \rangle = 0 - \langle h, f^{(n)} \rangle$ , which completes the proof of the lemma.  $\square$

**Proof of Theorem 1 Continued.** Let  $T \in \mathcal{S}'(\mathbb{R})$ . There exist constants  $C > 0$  and  $N \in \mathbb{N}_0$  such that:

$$|\langle T, f \rangle| \leq C|f|_{(N)} \leq C \sum_{k=0}^N \sup_{x \in \mathbb{R}} (1 + |x|)^N |f^{(k)}(x)| \leq C_1 \sum_{k=0}^N \sup_{x \in \mathbb{R}} (1 + |x|^2)^N |f^{(k)}(x)|. \quad (1)$$

Introduce the algebraic dampening weight  $\varphi(x) = (1 + |x|^2)^{-N}$ . Note that  $\varphi \in \mathcal{O}_M$  and  $|\varphi^{(\ell)}(x)| \leq C_\ell \varphi(x)$  for all  $\ell \in \mathbb{N}_0$ . We analyze the modulated distribution  $\varphi T$ , defined by  $\langle \varphi T, f \rangle = \langle T, \varphi f \rangle$ .

**Claim 1.** The tempered distribution  $\varphi T$  satisfies the bound:

$$|\langle \varphi T, f \rangle| \leq C_2 \sum_{k=0}^N \sup_{x \in \mathbb{R}} |f^{(k)}(x)| \quad \forall f \in \mathcal{S}.$$

**Proof.** Using inequality (1) on the product  $\varphi f$ :

$$|\langle \varphi T, f \rangle| = |\langle T, \varphi f \rangle| \leq C_1 \sum_{k=0}^N \sup_{x \in \mathbb{R}} (1 + |x|^2)^N |(\varphi f)^{(k)}(x)|.$$

By the Leibniz rule,  $|(\varphi f)^{(k)}(x)| \leq \sum_{j=0}^k \binom{k}{j} |f^{(j)}(x)| |\varphi^{(k-j)}(x)| \leq \sum_{j=0}^k C'_{k,j} |f^{(j)}(x)| |\varphi(x)|$ , where  $C'_{k,j} = \binom{k}{j} C_{k-j}$ . Since  $(1 + |x|^2)^N \varphi(x) = 1$ , we see:

$$|\langle \varphi T, f \rangle| \leq C_1 \sum_{k=0}^N \sum_{j=0}^k C'_{k,j} \sup_{x \in \mathbb{R}} |f^{(j)}(x)| \leq C_2 \sum_{k=0}^N \sup_{x \in \mathbb{R}} |f^{(k)}(x)|.$$

The weight cancels out, yielding the desired uniform derivative bound.  $\square$

**Claim 2.** There exist finite complex Borel measures  $\mu_0, \mu_1, \dots, \mu_N$  on  $\mathbb{R}$  such that:

$$\langle \varphi T, f \rangle = \sum_{k=0}^N \int_{\mathbb{R}} f^{(k)}(x) d\mu_k(x) \quad \forall f \in \mathcal{S}.$$

**Proof.** Let  $C_0(\mathbb{R})$  be the Banach space of continuous functions vanishing at infinity, equipped with the supremum norm  $\|g\|_u = \sup_x |g(x)|$ . Form the product Banach space  $E = \prod_{k=0}^N C_0(\mathbb{R})$  with the sum norm  $\|(g_0, \dots, g_N)\|_u = \sum_{k=0}^N \|g_k\|_u$ . Define the linear subspace:

$$M = \{(f, f', \dots, f^{(N)}) : f \in \mathcal{S}\} \subseteq E.$$

The functional  $\Lambda : M \rightarrow \mathbb{C}$  given by  $\Lambda(f, \dots, f^{(N)}) = \langle \varphi T, f \rangle$  is bounded under  $\|\cdot\|_u$  by Claim 1:

$$|\Lambda(f, \dots, f^{(N)})| = |\langle \varphi T, f \rangle| \leq C_2 \|(f, \dots, f^{(N)})\|_u.$$

By the Hahn-Banach theorem,  $\Lambda$  extends to a bounded linear functional on all of  $E$  with  $\|\Lambda\|_{E^*} \leq C_2$ . Since  $\Lambda(g_0, 0, \dots, 0)$  defines a bounded functional on  $C_0(\mathbb{R})$ , the Riesz Representation Theorem implies that there exists a finite complex Borel measure  $\mu_0$  such that  $\Lambda(g_0, 0, \dots, 0) = \int_{\mathbb{R}} g_0 d\mu_0$ . Proceeding slot by slot (using linearity  $\Lambda(g_0, g_1) = \Lambda(g_0, 0) + \Lambda(0, g_1)$ ), we get components  $\mu_1, \dots, \mu_N$  satisfying  $\Lambda(g_0, \dots, g_N) = \sum_{k=0}^N \int_{\mathbb{R}} g_k(x) d\mu_k(x)$ , yielding the stated integral representation for  $\langle \varphi T, f \rangle$ .  $\square$

**Claim 3.** There exist bounded measurable functions  $F_0, F_1, \dots, F_N$  such that:

$$\langle \varphi T, f \rangle = \sum_{k=0}^N \int_{\mathbb{R}} f^{(k+1)}(x) F_k(x) dx \quad \forall f \in \mathcal{S}.$$

**Proof.** Let  $G_k(x) = \mu_k((-\infty, x])$  be the cumulative distribution functions. Since the measures are finite, each  $G_k$  is bounded and of bounded variation. Integrating by parts against the smooth test function  $f^{(k)}$ :

$$\langle \varphi T, f \rangle = \sum_{k=0}^N \int_{-\infty}^{\infty} f^{(k)}(x) dG_k(x) = \sum_{k=0}^N \left[ f^{(k)}(x) G_k(x) \right]_{-\infty}^{\infty} - \sum_{k=0}^N \int_{-\infty}^{\infty} f^{(k+1)}(x) G_k(x) dx.$$

Because  $f^{(k)} \in \mathcal{S}$  and  $G_k$  is bounded, the boundary terms vanish, leaving  $F_k = -G_k$  as the required bounded functions.  $\square$

**Proof of Theorem 1 Continued.** We reconstruct  $T$  by writing  $T = \frac{1}{\varphi}(\varphi T)$ . Because  $\frac{1}{\varphi}(x) = (1 + |x|^2)^N \in \mathcal{O}_M$ , the test function  $\frac{f}{\varphi}$  belongs to  $\mathcal{S}$ :

$$\langle T, f \rangle = \left\langle \varphi T, \frac{f}{\varphi} \right\rangle = \sum_{k=0}^N \int_{\mathbb{R}} F_k(x) \left( \frac{f}{\varphi} \right)^{(k+1)}(x) dx.$$

Expanding  $(f/\varphi)^{(k+1)}$  via the Leibniz rule gives  $\left( \frac{1}{\varphi} f \right)^{(k+1)} = \sum_{\ell=0}^{k+1} \binom{k+1}{\ell} f^{(\ell)} \left( \frac{1}{\varphi} \right)^{(k+1-\ell)}$ . This yields a finite sum of terms of the form:

$$\langle T, f \rangle = \sum_{k=0}^N \sum_{\ell=0}^{k+1} \int_{\mathbb{R}} f^{(\ell)}(x) H_{k,\ell}(x) dx, \quad \text{where } H_{k,\ell} = \binom{k+1}{\ell} F_k \left( \frac{1}{\varphi} \right)^{(k+1-\ell)}.$$

Each  $H_{k,\ell}$  is a bounded function multiplied by a derivative of  $(1 + |x|^2)^N$ , which is measurable and of polynomial growth.

Applying Lemma 1 repeatedly to each term allows us to shift all derivatives up to a uniform high order  $N + 2$ , converting each component to  $\int_{\mathbb{R}} f^{(N+2)}(x) \widetilde{H}_{k,\ell}(x) dx$ , where  $\widetilde{H}_{k,\ell}$  are continuous and of polynomial growth. Summing these up, we define:

$$g(x) = (-1)^{N+2} \sum_{k=0}^N \sum_{\ell=0}^{k+1} \widetilde{H}_{k,\ell}(x),$$

which is continuous and of polynomial growth. This leaves us with

$$\langle T, f \rangle = (-1)^{N+2} \int_{\mathbb{R}} g(x) f^{(N+2)}(x) dx = \langle \partial^{N+2} g, f \rangle,$$

confirming  $T = \partial^{N+2} g$  (with  $\alpha = N + 2$ ). □

## The Helmholtz Equation in $\mathcal{S}'(\mathbb{R}^d)$

We investigate the generalized Helmholtz equation on tempered distributions:

$$u - \Delta u = g \quad \text{in } \mathcal{S}'(\mathbb{R}^d), \tag{H}$$

where  $g \in \mathcal{S}'(\mathbb{R}^d)$  is given and  $u \in \mathcal{S}'(\mathbb{R}^d)$  is the unknown distribution.

### Proposition 2

For any  $g \in \mathcal{S}'(\mathbb{R}^d)$ , there exists a unique solution  $u \in \mathcal{S}'(\mathbb{R}^d)$  to (H). Thus, the operator  $(I - \Delta) : \mathcal{S}'(\mathbb{R}^d) \rightarrow \mathcal{S}'(\mathbb{R}^d)$  is a bijection, and the solution is given explicitly by:

$$u = \mathcal{F}^{-1} \left( \frac{1}{1 + 4\pi^2 |\xi|^2} \hat{g} \right).$$

**Proof.** The polynomial factor  $p(\xi) = 1 + 4\pi^2|\xi|^2$  and its reciprocal  $p(\xi)^{-1} = \frac{1}{1+4\pi^2|\xi|^2}$  both belong to  $\mathcal{O}_M(\mathbb{R}^d)$ . Applying the Fourier transform to both sides of (H) converts differentiation into algebraic multiplication (recalling  $\widehat{\partial^\alpha T} = (i2\pi\xi)^\alpha \hat{T}$ , which means  $\widehat{\Delta u} = -4\pi^2|\xi|^2 \hat{u}$ ):

$$u - \Delta u = g \text{ in } \mathcal{S}' \iff \widehat{u - \Delta u} = \hat{g} \text{ in } \mathcal{S}' \iff (1 + 4\pi^2|\xi|^2)\hat{u} = \hat{g} \text{ in } \mathcal{S}'.$$

Since multiplication by  $p(\xi)^{-1}$  is a valid and invertible operation within  $\mathcal{S}'(\mathbb{R}^d)$ , we solve uniquely for  $\hat{u}$ :

$$\hat{u} = \frac{1}{1 + 4\pi^2|\xi|^2} \hat{g} \text{ in } \mathcal{S}'.$$

Applying  $\mathcal{F}^{-1}$  to both sides yields  $u = \mathcal{F}^{-1}\left(\frac{1}{1+4\pi^2|\xi|^2} \hat{g}\right)$ . This uniquely determines  $u \in \mathcal{S}'$ , completing the bijection proof.  $\square$

## The $L^2$ Setting and Sobolev Spaces

Let  $s \in \mathbb{N}_0$ . We define the classical Sobolev space  $H^s(\mathbb{R}^d)$  as:

$$H^s(\mathbb{R}^d) = \{u \in \mathcal{S}'(\mathbb{R}^d) : \partial^\alpha u \in L^2(\mathbb{R}^d) \text{ for all } |\alpha| \leq s\}.$$

This space is a Hilbert space under the norm:

$$\|u\|_{H^s} = \left( \sum_{|\alpha| \leq s} \|\partial^\alpha u\|_{L^2}^2 \right)^{1/2}.$$

Operationally,  $u \in H^s(\mathbb{R}^d)$  implies that  $u \in L^2(\mathbb{R}^d)$  (since  $H^s \subseteq L^2$ ) and its distributional derivatives up to order  $s$  can be represented by  $L^2$  functions satisfying  $\int_{\mathbb{R}^d} \partial^\alpha u f \, dx = (-1)^{|\alpha|} \int_{\mathbb{R}^d} u \partial^\alpha f \, dx$  for all  $f \in \mathcal{S}$ .

### Proposition 3

For every  $g \in L^2(\mathbb{R}^d)$ , the unique solution  $u$  to (H) belongs to  $H^2(\mathbb{R}^d)$ . It satisfies  $u - \Delta u = g$  almost everywhere, and obeys the structural stability estimate:

$$\|u\|_{H^2} \leq C \|g\|_{L^2},$$

where  $C$  is a constant independent of  $g$ . Thus,  $(I - \Delta) : H^2(\mathbb{R}^d) \rightarrow L^2(\mathbb{R}^d)$  is a bijection.

**Proof.** Let  $g \in L^2(\mathbb{R}^d)$ , which implies  $\hat{g} \in L^2(\mathbb{R}^d)$ . By Proposition 2, the unique distribution solution is  $u = \mathcal{F}^{-1}\left(\frac{1}{1+4\pi^2|\xi|^2} \hat{g}\right)$ , which belongs to  $L^2(\mathbb{R}^d)$  since its transform is bounded by  $|\hat{g}|$ . Taking the Fourier transform representation, for any multi-index  $|\alpha| \leq 2$ :

$$\widehat{\partial^\alpha u}(\xi) = (i2\pi\xi)^\alpha \hat{u}(\xi) = \frac{(i2\pi\xi)^\alpha}{1 + 4\pi^2|\xi|^2} \hat{g}(\xi).$$



Because  $|(i2\pi\xi)^\alpha| \leq (2\pi|\xi|)^{|\alpha|} \leq 1 + 4\pi^2|\xi|^2$  holds for any  $|\alpha| \leq 2$ , the multiplier is uniformly bounded by 1. Consequently,  $|\widehat{\partial^\alpha u}(\xi)| \leq |\widehat{g}(\xi)|$ . By Plancherel's theorem,  $\partial^\alpha u \in L^2(\mathbb{R}^d)$  and satisfies:

$$\|\partial^\alpha u\|_{L^2} = \|\widehat{\partial^\alpha u}\|_{L^2} \leq \|\widehat{g}\|_{L^2} = \|g\|_{L^2}.$$

Summing over all  $|\alpha| \leq 2$  yields the norm stability estimate  $\|u\|_{H^2} \leq C\|g\|_{L^2}$ . Uniqueness is guaranteed by the overarching uniqueness in  $\mathcal{S}'$  (Proposition 2). Thus,  $(I - \Delta) : H^2(\mathbb{R}^d) \rightarrow L^2(\mathbb{R}^d)$  is a bijection.  $\square$

**Approximation via Smooth Limits.** Alternatively, we can resolve the equation for  $g \in L^2$  as a limit of smooth solutions. Choose a sequence  $g_n \in \mathcal{S}$  such that  $g_n \rightarrow g$  in  $L^2$ . Since  $(I - \Delta) : \mathcal{S} \rightarrow \mathcal{S}$  is an isomorphism, there correspond unique smooth solutions  $u_n \in \mathcal{S}$ . Linearity implies  $(u_n - u_m) - \Delta(u_n - u_m) = g_n - g_m$ , and applying the stability estimate from Proposition 3:

$$\|\partial^\alpha u_n - \partial^\alpha u_m\|_{L^2} \leq C\|g_n - g_m\|_{L^2} \rightarrow 0 \quad \text{as } n, m \rightarrow \infty,$$

for all  $|\alpha| \leq 2$ . By the completeness of  $L^2$ , each derivative sequence  $\partial^\alpha u_n$  converges to some function  $u_\alpha \in L^2$  (with  $u \equiv u_0$ ). For any test function  $f \in \mathcal{S}$ , the standard pairing satisfies  $\int_{\mathbb{R}^d} \partial^\alpha u_n f \, dx = (-1)^{|\alpha|} \int_{\mathbb{R}^d} u_n \partial^\alpha f \, dx$ . Passing to the limit  $n \rightarrow \infty$  inside the integrals gives:

$$\int_{\mathbb{R}^d} u_\alpha f \, dx = (-1)^{|\alpha|} \int_{\mathbb{R}^d} u \partial^\alpha f \, dx,$$

confirming that  $u_\alpha = \partial^\alpha u$  in the distributional sense. Hence  $u \in H^2(\mathbb{R}^d)$  and satisfies  $u - \Delta u = g$  almost everywhere.

#### Proposition 4

For any source term  $g \in H^s(\mathbb{R}^d)$  ( $s \in \mathbb{N}_0$ ), the unique solution  $u$  to (H) satisfies  $u \in H^{s+2}(\mathbb{R}^d)$ . The operator  $(I - \Delta) : H^{s+2}(\mathbb{R}^d) \rightarrow H^s(\mathbb{R}^d)$  is a bijection.

**Proof.** Differentiating the equation (H) distributionally yields  $\partial^\alpha u - \Delta(\partial^\alpha u) = \partial^\alpha g$  for any multi-index  $\alpha$ . If  $g \in H^s(\mathbb{R}^d)$ , then for all  $|\alpha| \leq s$ , the derivative term satisfies  $\partial^\alpha g \in L^2(\mathbb{R}^d)$ . Applying Proposition 3 to this differentiated equation, we find that  $\partial^\alpha u \in H^2(\mathbb{R}^d)$ , which implies that for all  $|\beta| \leq 2$ :

$$\|\partial^{\beta+\alpha} u\|_{L^2} = \|\partial^\beta(\partial^\alpha u)\|_{L^2} \leq C\|\partial^\alpha g\|_{L^2}.$$

Any multi-index  $\gamma$  with  $|\gamma| \leq s + 2$  can be split into components  $\gamma = \beta + \alpha$  where  $|\alpha| \leq s$  and  $|\beta| \leq 2$ . Accumulating these bounds implies:

$$\|\partial^\gamma u\|_{L^2} \leq C\|g\|_{H^s} \quad \text{for all } |\gamma| \leq s + 2.$$

Hence,  $u \in H^{s+2}(\mathbb{R}^d)$  solves (H), and uniqueness follows directly from the broad uniqueness property in  $\mathcal{S}'$  established in Proposition 2. Thus,  $(I - \Delta) : H^{s+2}(\mathbb{R}^d) \rightarrow H^s(\mathbb{R}^d)$  is a bijection.  $\square$

## Weak Convergence

### Motivation and Fundamental Definition

Let  $E$  be a normed vector space and  $E^* = \mathcal{L}(E, \mathbb{R})$  denote its continuous topological dual space. Consider a sequence  $\{x_n\}_{n=1}^\infty \subset E$  approximating an element  $x \in E$ .

By a fundamental consequence of the Hahn–Banach theorem, the standard norm topology characterizes strong convergence via uniform control over the dual unit ball:

$$\|x_n - x\|_E = \sup_{\|\ell\|_{E^*} \leq 1} |\ell(x_n - x)| \longrightarrow 0 \quad \text{as } n \rightarrow \infty.$$

Equivalently, norm convergence dictates that  $\ell(x_n) \rightarrow \ell(x)$  *uniformly* over bounded subsets of functionals. *Weak convergence* relaxes this topological requirement by demanding only pointwise (element-by-element) convergence across the dual space.

### Weak Convergence

We say that a sequence  $\{x_n\}_{n=1}^\infty \subset E$  converges *weakly* to  $x \in E$ , denoted by  $x_n \rightharpoonup x$ , if:

$$\ell(x_n) \longrightarrow \ell(x) \quad \text{as } n \rightarrow \infty \quad \forall \ell \in E^*.$$

Using duality pairing notation, this is expressed as:

$$\langle \ell, x_n \rangle_{E^*, E} \longrightarrow \langle \ell, x \rangle_{E^*, E} \quad \text{as } n \rightarrow \infty \quad \forall \ell \in E^*.$$

### Concrete Realizations in Classical Spaces

- *The  $L^p$  Setting.* Let  $E = L^p(\Omega, \mu)$  with  $p \in [1, \infty)$ . Under standard  $\sigma$ -finite conditions, the dual space is isomorphic to  $L^q(\Omega, \mu)$  where  $\frac{1}{p} + \frac{1}{q} = 1$ . Consequently, a sequence  $f_n \rightharpoonup f$  weakly in  $L^p$  if and only if:

$$\int_{\Omega} f_n(x)g(x) \, d\mu(x) \longrightarrow \int_{\Omega} f(x)g(x) \, d\mu(x) \quad \forall g \in L^q(\Omega, \mu).$$

- *The Space of Continuous Functions  $C(X)$ .* Let  $X$  be a compact metric space and  $E = C(X)$  equipped with the uniform supremum norm  $\|f\|_u = \sup_{x \in X} |f(x)|$ . By the Riesz–Markov–Kakutani representation theorem, the dual space  $[C(X)]^*$  is isometric to  $M(X)$ , the space of finite signed Radon measures on  $X$ . Thus,  $f_n \rightharpoonup f$  weakly in  $C(X)$  if and only if:

$$\int_X f_n(x) \, d\mu(x) \longrightarrow \int_X f(x) \, d\mu(x) \quad \forall \mu \in M(X).$$

- *Hilbert Spaces.* Let  $E = H$  be a Hilbert space. By the Riesz representation theorem,  $H^* \cong H$ , meaning weak convergence  $x_n \rightharpoonup x$  is fully characterized by the inner product:

$$\langle x_n, z \rangle_H \longrightarrow \langle x, z \rangle_H \quad \text{as } n \rightarrow \infty \quad \forall z \in H.$$

*Connection to strong convergence:*  $x_n \rightarrow x$  strongly in  $H$  if and only if  $x_n \rightharpoonup x$  weakly and  $\limsup_{n \rightarrow \infty} \|x_n\|_H \leq \|x\|_H$ .

## The Finite-Dimensional Special Case

### Proposition 1

Let  $E$  be a finite-dimensional normed vector space (e.g.,  $E = \mathbb{R}^d$ ). Then the weak and strong topologies coincide; that is,

$$x_n \rightarrow x \iff x_n \rightharpoonup x.$$

**Proof.**

- ( $\Rightarrow$ ) Assume  $x_n \rightarrow x$  strongly. For each continuous linear functional  $\ell \in E^*$ , boundedness implies:

$$|\ell(x_n) - \ell(x)| = |\ell(x_n - x)| \leq \|\ell\|_{E^*} \|x_n - x\|_E \longrightarrow 0 \quad \text{as } n \rightarrow \infty,$$

which proves  $x_n \rightharpoonup x$  weakly. This direction holds universally in any normed space.

- ( $\Leftarrow$ ) Suppose  $\dim E = d < \infty$  and  $x_n \rightharpoonup x$ . Let  $\{v_1, \dots, v_d\}$  be a basis for  $E$ , and let  $\{\ell_1, \dots, \ell_d\} \subset E^*$  represent the corresponding dual basis satisfying  $\ell_i(v_j) = \delta_{ij}$ . Every element  $y \in E$  can be decomposed as  $y = \sum_{i=1}^d \ell_i(y) v_i$ . By the definition of weak convergence, we obtain coordinate-wise convergence:  $\ell_i(x_n) \rightarrow \ell_i(x)$  in  $\mathbb{R}$  for each  $i \in \{1, \dots, d\}$ . Because all norms on a finite-dimensional space are equivalent, we can bound the standard norm by the  $\ell^1$ -coordinate norm:

$$\|x_n - x\|_E \leq C \sum_{i=1}^d |\ell_i(x_n) - \ell_i(x)| \longrightarrow 0 \quad \text{as } n \rightarrow \infty,$$

confirming strong norm convergence. □

*Example.* For a sequence of vectors  $\{x^{(n)}\}_{n=1}^\infty \subset \mathbb{R}^d$ , strong convergence  $x^{(n)} \rightarrow x$  is precisely equivalent to component-wise convergence:  $x_i^{(n)} \rightarrow x_i$  for every coordinate index  $i = 1, 2, \dots, d$ .

### The Infinite-Dimensional Counterexample

In general infinite-dimensional spaces, weak convergence does *not* imply strong convergence. Consider an infinite-dimensional Hilbert space  $H$  containing a complete orthonormal system (CONS)  $\{e_n\}_{n=1}^\infty$ . For any  $n \neq m$ , Pythagoras' theorem dictates:

$$\|e_n - e_m\|_H^2 = \|e_n\|^2 - 2\langle e_n, e_m \rangle_H + \|e_m\|^2 = 1 - 0 + 1 = 2.$$

Because the distance between any two distinct elements is always  $\sqrt{2}$ , the sequence  $\{e_n\}$  cannot contain any strongly convergent subsequence. However,  $e_n \rightharpoonup 0$  weakly. Indeed, for every element  $x \in H$ , Bessel's inequality / Parseval's identity guarantees that:

$$\sum_{n=1}^{\infty} |\langle x, e_n \rangle_H|^2 = \|x\|_H^2 < \infty \implies \lim_{n \rightarrow \infty} \langle e_n, x \rangle_H = 0 = \langle 0, x \rangle_H.$$

## The Canonical Embedding and Weakly Cauchy Sequences

The canonical evaluation map  $J : E \rightarrow E^{**}$  into the bidual space  $[E^*]^*$ . Every element  $x \in E$  acts as a bounded functional  $T_x \in E^{**}$  via  $T_x(\ell) = \ell(x)$  for all  $\ell \in E^*$ . The Hahn–Banach theorem guarantees this map is a linear isometry:

$$\|T_x\|_{E^{**}} = \sup_{\|\ell\|_{E^*} \leq 1} |T_x(\ell)| = \sup_{\|\ell\|_{E^*} \leq 1} |\ell(x)| = \|x\|_E.$$

## Weakly Cauchy Sequence

A sequence  $\{x_n\}_{n=1}^\infty \subset E$  is called *weakly Cauchy* if the scalar sequence  $\{\ell(x_n)\}_{n=1}^\infty$  is Cauchy in  $\mathbb{R}$  (or  $\mathbb{C}$ ) for every continuous linear functional  $\ell \in E^*$ .

**Note.** Every weakly convergent sequence is fundamentally weakly Cauchy because any convergent sequence in  $\mathbb{R}$  is a Cauchy sequence.

## Proposition 2

Let  $E$  be a normed vector space. Then the following properties hold:

- a. If  $x_n \rightharpoonup x$ , then the norm is sequentially lower semicontinuous:

$$\|x\|_E \leq \liminf_{n \rightarrow \infty} \|x_n\|_E.$$

- b. If  $\{x_n\}$  is weakly Cauchy, then it is bounded in norm:  $\sup_n \|x_n\|_E < \infty$ .
- c. If  $E$  is a reflexive Banach space, then the space is weakly sequentially complete; i.e., every weakly Cauchy sequence converges weakly to an element in  $E$ .

## Proof.

- a. *Lower Semicontinuity.* By the Hahn–Banach theorem, choose a normalized functional  $\ell \in E^*$  such that  $\|\ell\|_{E^*} = 1$  and  $\ell(x) = \|x\|_E$ . Evaluating the limit under weak convergence:

$$\|x\|_E = \ell(x) = \lim_{n \rightarrow \infty} \ell(x_n) = \liminf_{n \rightarrow \infty} \ell(x_n).$$

Since  $\ell(x_n) \leq \|\ell\|_{E^*} \|x_n\|_E = \|x_n\|_E$  holds uniformly for all  $n$ , taking the lower limit yields:

$$\|x\|_E = \liminf_{n \rightarrow \infty} \ell(x_n) \leq \liminf_{n \rightarrow \infty} \|x_n\|_E.$$

□

- b. *Uniform Boundedness.* Let  $\{x_n\}$  be weakly Cauchy. For each individual functional  $\ell \in E^*$ , the sequence  $\{\ell(x_n)\}$  converges in  $\mathbb{R}$ , which implies it is bounded:  $\sup_n |\ell(x_n)| < \infty$ . By setting  $T_{x_n}(\ell) = \ell(x_n)$ . The sequence of operators  $\{T_{x_n}\} \subset E^{**}$  satisfies a pointwise boundedness condition on the Banach space  $E^*$ :  $\sup_n |T_{x_n}(\ell)| < \infty$  for all  $\ell \in E^*$ . Applying the Banach–Steinhaus Uniform Boundedness Principle yields a uniform bound on the operator norms:

$$\sup_n \|x_n\|_E = \sup_n \|T_{x_n}\|_{E^{**}} = \sup_n \sup_{\|\ell\|_{E^*} \leq 1} |T_{x_n}(\ell)| < \infty.$$

□

- c. *Reflexive Completeness.* Let  $\{x_n\}$  be weakly Cauchy inside a reflexive space ( $E = E^{**}$ ). By part b, there exists a uniform bound  $M = \sup_n \|x_n\|_E < \infty$ . Define a functional  $T : E^* \rightarrow \mathbb{R}$  point-by-point via the pointwise limit:

$$T(\ell) = \lim_{n \rightarrow \infty} \ell(x_n).$$

Linearity of  $T$  follows directly from the linearity of the limit operation. Taking boundaries, the inequality  $|\ell(x_n)| \leq M\|\ell\|_{E^*}$  implies  $|T(\ell)| \leq M\|\ell\|_{E^*}$ , hence  $T \in E^{**}$ . Because  $E$  is reflexive, the canonical mapping is surjective, ensuring there exists a unique element  $x \in E$  such that  $T(\ell) = \ell(x)$  for all  $\ell \in E^*$ . Substituting this back gives  $\ell(x) = \lim_{n \rightarrow \infty} \ell(x_n)$ , which matches the definition of  $x_n \rightharpoonup x$ . □

## Weak Compactness

### Solving $Lx = g$ by Approximation

Solving a bounded linear equation

$$Lx = g, \quad L \in \mathcal{L}(E, F),$$

together with a family of *approximating* problems

$$L_n x_n = g_n, \quad \text{with the a priori bound} \quad \sup_n \|x_n\|_E < \infty.$$

Two questions govern whether the approximate solutions  $\{x_n\}$  recover an exact solution of the original equation.

- *Extraction of a limit.* Does there exist a subsequence and a candidate limit  $x \in E$  with

$$x_{n_k} \rightharpoonup x \quad (\text{weakly}) \quad ?$$

- *Identification of the limit.* Once such an  $x$  exists, can we pass to the limit and conclude  $Lx = g$ ? Using the dual (adjoint) operator  $L'$  and the duality pairing  $\langle \cdot, \cdot \rangle_{E^*, E}$ ,

$$\langle (L_n)' \ell, x_n \rangle = \langle \ell, L_n x_n \rangle = \langle \ell, g_n \rangle \xrightarrow{n \rightarrow \infty} \langle L' \ell, x \rangle = \langle \ell, g \rangle, \quad \forall \ell \in F^*.$$

Since  $\langle L' \ell, x \rangle = \langle \ell, Lx \rangle$ , this yields  $\langle \ell, Lx \rangle = \langle \ell, g \rangle$  for all  $\ell \in F^*$ , hence  $Lx = g$ , the range of  $L$  being a closed subspace of the Banach space  $F$ .

### Lemma 1

Reflexivity passes to closed subspaces: if  $E^{**} = E$ , then for every closed subspace  $M \subseteq E$  one has  $M^{**} = M$ ; that is,  $M$  is itself reflexive.

### Lemma 2

Separability is inherited from the dual: if  $E^*$  is separable, then  $E$  is separable.

## Weak Compactness Theorem

Let  $E$  be a reflexive Banach space. Then every bounded sequence in  $E$  has a weakly convergent subsequence. Explicitly, if

$$\|x_n\|_E \leq C \quad \forall n \quad \text{for some } C > 0,$$

then there exists a subsequence  $\{x_{n_k}\}$  and an element  $x \in E$  such that

$$x_{n_k} \rightharpoonup x \quad \text{as } k \rightarrow \infty.$$

**Proof.** Let  $\{x_n\} \subset E$  satisfy  $\|x_n\|_E \leq C$  for all  $n$ . Since every weakly Cauchy sequence in a reflexive Banach space possesses a weak limit, it suffices to produce a subsequence  $\{x_{n_k}\}$  that is *weakly Cauchy*, i.e., such that

$$\{\ell(x_{n_k})\}_{k=1}^\infty \text{ is Cauchy in } \mathbb{R} \quad \forall \ell \in E^*.$$

*A separable, reflexive ambient subspace.* Let  $M$  denote the closure of the linear span of the sequence,

$$M = \overline{\text{span}}\{x_n : n \geq 1\}.$$

Then  $M$  is *separable*: rational (finite) linear combinations of the  $x_n$  form a countable dense subset of  $M$ . Moreover,  $M$  is a closed subspace of the reflexive space  $E$ , hence *reflexive* by Lemma 1. Consequently  $(M^*)^* = M$  is separable, so applying Lemma 2 to  $M^*$  shows that  $M^*$  is separable.

*A countable dense family of functionals.* Choose a countable dense set

$$D = \{\ell_m : m \geq 1\} \subseteq M^*, \quad \overline{D} = M^*,$$

which exists by separability of  $M^*$ . For each fixed  $m$ , the scalar sequence  $\{\ell_m(x_n)\}_n$  is bounded:

$$|\ell_m(x_n)| \leq \|\ell_m\|_{M^*} \|x_n\|_E \leq C \|\ell_m\|_{M^*} < \infty, \quad \|\ell_m\|_{M^*} = \sup_{\substack{x \in M \\ \|x\|_E = 1}} |\ell_m(x)|,$$

so that  $\sup_n |\ell_m(x_n)| < \infty$ .

*Diagonal extraction.* By Bolzano–Weierstrass, each bounded sequence  $\{\ell_m(x_n)\}_n$  admits a convergent subsequence. Applying this successively for  $m = 1, 2, 3, \dots$  and passing to a diagonal subsequence yields a single subsequence  $\{x_{n_k}\}$  for which

$$\lim_{k \rightarrow \infty} \ell_m(x_{n_k}) \quad \text{exists for every } m \geq 1.$$

The same subsequence works simultaneously for all  $m$ .

*Weak Cauchy property.* Fix an arbitrary  $\ell \in E^*$  and let  $R\ell := \ell|_M \in M^*$  denote its restriction. For any  $m \geq 1$ , inserting and removing  $\ell_m$  gives

$$\begin{aligned} |\ell(x_{n_k}) - \ell(x_{n_j})| &= |R\ell(x_{n_k}) - R\ell(x_{n_j})| \\ &\leq |R\ell(x_{n_k}) - \ell_m(x_{n_k})| + |\ell_m(x_{n_k}) - \ell_m(x_{n_j})| + |\ell_m(x_{n_j}) - R\ell(x_{n_j})| \\ &\leq \|R\ell - \ell_m\|_{M^*} \|x_{n_k}\|_E + |\ell_m(x_{n_k}) - \ell_m(x_{n_j})| + \|R\ell - \ell_m\|_{M^*} \|x_{n_j}\|_E \\ &\leq 2C \|R\ell - \ell_m\|_{M^*} + |\ell_m(x_{n_k}) - \ell_m(x_{n_j})|. \end{aligned}$$

Taking the limit superior over  $k, j \rightarrow \infty$ ,

$$\limsup_{k, j \rightarrow \infty} |\ell(x_{n_k}) - \ell(x_{n_j})| \leq 2C\|R\ell - \ell_m\|_{M^*} + \underbrace{\limsup_{k, j \rightarrow \infty} |\ell_m(x_{n_k}) - \ell_m(x_{n_j})|}_{=0},$$

the bracketed term vanishing because  $\lim_k \ell_m(x_{n_k})$  exists for every  $m \geq 1$ .

*Density closes the argument.* Since  $\{\ell_m\}_{m \geq 1}$  is dense in  $M^*$ , the quantity  $2C\|R\ell - \ell_m\|_{M^*}$  can be made arbitrarily small by a suitable choice of  $m$ . Therefore

$$\limsup_{k, j \rightarrow \infty} |\ell(x_{n_k}) - \ell(x_{n_j})| = 0,$$

i.e.,  $\{\ell(x_{n_k})\}_k$  is Cauchy for every  $\ell \in E^*$ . Hence  $\{x_{n_k}\}$  is weakly Cauchy and, by reflexivity, admits a weak limit  $x \in E$  with  $x_{n_k} \rightharpoonup x$ .  $\square$

## Weak-\* Convergence and Weakly-\* Compact Sets

Let  $E$  be a normed vector space with continuous dual  $E^* = \mathcal{L}(E, \mathbb{R})$ . Whereas weak convergence on  $E^*$  tests against the full bidual  $E^{**}$ , weak-\* convergence tests only against the original space  $E$ .

### Weak-\* Convergence

Let  $\ell_n, \ell \in E^*$ . We say that  $\ell_n$  converges *weakly-\** to  $\ell$ , denoted  $\ell_n \xrightarrow{*} \ell$ , in  $E^*$  if

$$\ell_n(x) \longrightarrow \ell(x) \quad \text{as } n \rightarrow \infty \quad \forall x \in E.$$

**Note.** On the dual space  $E^*$  there are three distinct modes of convergence:

1. *Norm (strong) convergence* in  $E^*$ .  $\|\ell_n - \ell\|_{E^*} \rightarrow 0$ .
2. *Weak convergence*.  $\rightharpoonup$  tested against every functional in the bidual  $E^{**}$ .
3. *Weak-\* convergence*.  $\xrightarrow{*}$  tested only against the evaluations  $x \in E \subseteq E^{**}$ .

### Remarks.

- a. *Hierarchy of the three modes.* Recall the canonical embedding  $E \subseteq (E^*)^* = E^{**}$ . Because weak-\* convergence tests against the smaller collection  $E \subseteq E^{**}$ , each stronger mode implies the next:

$$\|\ell_n - \ell\|_{E^*} \rightarrow 0 \implies \ell_n \rightharpoonup \ell \implies \ell_n \xrightarrow{*} \ell.$$

The first implication is the general fact that norm convergence implies weak convergence; concretely, for any  $x \in E^{**}$ ,

$$|x(\ell_n) - x(\ell)| \leq \|x\|_{E^{**}} \|\ell_n - \ell\|_{E^*} \longrightarrow 0.$$

- b. *Coincidence under reflexivity.* If  $E$  is a reflexive Banach space, so that  $(E^*)^* = E$ , then the bidual reduces to  $E$  and weak and weak-\* convergence *coincide*:

$$\ell_n \rightharpoonup \ell \iff \ell_n \xrightarrow{*} \ell.$$

Consequently, weak-\* convergence is genuinely interesting only when  $E$  is *not* reflexive.

- c. *The space  $L^1$ .* Let  $E = L^1(\mu)$  with  $\mu$  a  $\sigma$ -finite measure. Then  $E^* = L^\infty(\mu)$ , while  $E$  is *not* reflexive since  $(L^\infty)^* \neq L^1$ . Here, for  $f_n, f \in L^\infty$ ,

$$f_n \xrightarrow{*} f \text{ in } L^\infty \iff \int f_n g \, d\mu \longrightarrow \int f g \, d\mu \quad \forall g \in L^1(\mu).$$

The dualities  $(L^p)^* = L^q$ ; in particular  $\sigma$ -finiteness gives  $(L^1)^* = L^\infty$ , but  $(L^\infty)^* \neq L^1$ .

- d. *The space  $C(X)$ .* Let  $X$  be a compact metric space and  $E = C(X)$  with the uniform norm  $\|f\|_u$ . Then  $E^* = M(X)$ , the space of finite signed (real) Borel measures on  $X$ , and for  $\mu_n, \mu \in M(X)$ ,

$$\mu_n \xrightarrow{*} \mu \text{ in } M(X) \iff \int_X f \, d\mu_n \longrightarrow \int_X f \, d\mu \quad \forall f \in C(X).$$

## Weak-\* Sequential Compactness

Recall that for functionals  $\ell_n, \ell \in E^*$ , weak-\* convergence  $\ell_n \xrightarrow{*} \ell$  in  $E^*$  means

$$\ell_n(x) \longrightarrow \ell(x) \quad \text{as } n \rightarrow \infty, \quad \text{for every } x \in E.$$

We first record an example showing that weak-\* convergence is strictly weaker than weak convergence, and then prove the sequential form of the Banach–Alaoglu theorem.

### Example: Weak-\* but Not Weak Convergence

Let

$$E = c_0 = \left\{ x = (x_n)_{n \geq 1} \in \ell^\infty : \lim_{n \rightarrow \infty} x_n = 0 \right\},$$

equipped with the  $\ell^\infty$ -norm  $\|x\|_\infty = \sup_n |x_n|$ . One can show that the dual is

$$E^* = \ell^1, \quad \text{and hence} \quad (E^*)^* = (\ell^1)^* = \ell^\infty \supsetneq c_0.$$

In particular  $c_0$  is *not* reflexive: the bidual  $\ell^\infty$  is strictly larger than  $c_0$ .

For each  $n \geq 1$ , let  $\ell_n \in \ell^1 = E^*$  be the sequence with a single 1 in the  $n$ -th position and 0 elsewhere:

$$\ell_n = (\underbrace{0, \dots, 0}_{n-1}, 1, 0, \dots).$$

As a functional on  $c_0$ ,  $\ell_n$  acts on  $x = (x_k) \in c_0$  by

$$\ell_n(x) = \sum_{k=1}^{\infty} (\ell_n)_k x_k = x_n.$$



$\ell_n \xrightarrow{*} 0$ . Fix any  $x = (x_k) \in c_0$ . By the definition of  $c_0$  we have  $x_n \rightarrow 0$  as  $n \rightarrow \infty$ , so

$$\ell_n(x) = x_n \rightarrow 0 = \mathbf{0}(x) \quad \text{for every } x \in c_0.$$

This is exactly the statement that  $\ell_n \xrightarrow{*} 0$ , weak-\* convergence to the zero functional.

*Claim:*  $\ell_n$  has no weak limit in  $E^* = \ell^1$ . Weak convergence of  $\{\ell_n\}$  in  $\ell^1$  would require  $\Lambda(\ell_n)$  to converge for every  $\Lambda \in (\ell^1)^* = \ell^\infty$ . Take the specific element

$$z = (0, 1, 0, 1, \dots) \in \ell^\infty \quad (\text{entries alternating between 0 and 1}),$$

regarded as the functional on  $\ell^1$  given by  $z(\ell) = \sum_{k=1}^{\infty} z_k \ell_k$ . Then

$$z(\ell_n) = \sum_{k=1}^{\infty} z_k (\ell_n)_k = z_n = \begin{cases} 0, & n \text{ odd}, \\ 1, & n \text{ even}, \end{cases}$$

and  $\lim_{n \rightarrow \infty} z_n$  does not exist. So  $\Lambda(\ell_n)$  fails to converge for this  $\Lambda = z$ , and  $\{\ell_n\}$  has no weak limit, even though it has the weak-\* limit 0. This is why weak-\* convergence is interesting precisely when  $E$  is not reflexive.

### Banach–Alaoglu Theorem (sequential form)

Let  $E$  be a separable normed vector space. Then every bounded sequence in  $E^*$  has a weak-\* convergent subsequence. Explicitly: if  $\{\ell_n\} \subset E^*$  satisfies

$$\|\ell_n\|_{E^*} \leq C \quad \text{for all } n, \text{ for some } C > 0,$$

then there exist a subsequence  $\{\ell_{n_k}\}$  and a functional  $\ell \in E^*$  such that  $\ell_{n_k} \xrightarrow{*} \ell$ .

**Proof.** Let  $\{\ell_n\} \subset E^*$  with  $\|\ell_n\|_{E^*} \leq C$  for all  $n$ . The proof has five steps.

*A countable dense set in  $E$ .* Since  $E$  is separable, we may choose a countable set

$$\{x_m : m \geq 1\} \subseteq E \quad \text{that is dense in } E.$$

*Each scalar sequence is bounded.* Fix  $m$ . For every  $n$ ,

$$|\ell_n(x_m)| \leq \|\ell_n\|_{E^*} \|x_m\|_E \leq C \|x_m\|_E.$$

The right-hand side does not depend on  $n$ , so the real sequence  $\{\ell_n(x_m)\}_{n \geq 1}$  is bounded:

$$\sup_n |\ell_n(x_m)| \leq C \|x_m\|_E < \infty.$$

*Diagonal extraction of a subsequence.* We extract one subsequence that makes *all* of these scalar sequences converge at once.

- For  $m = 1$ :  $\{\ell_n(x_1)\}_n$  is a bounded real sequence, so by Bolzano–Weierstrass it has a convergent subsequence. Write its indices as  $\{n_k^{(1)}\}_{k \geq 1}$ .

- For  $m = 2$ : along the previous subsequence,  $\{\ell_{n_k^{(1)}}(x_2)\}_k$  is again bounded, so it has a further convergent subsequence  $\{n_k^{(2)}\} \subseteq \{n_k^{(1)}\}$ .
- Continuing inductively, at stage  $m$  we obtain  $\{n_k^{(m)}\} \subseteq \{n_k^{(m-1)}\}$  along which  $\ell_{n_k^{(m)}}(x_j)$  converges as  $k \rightarrow \infty$  for each  $j = 1, \dots, m$ .

Now set the *diagonal* sequence  $n_k := n_k^{(k)}$ . For each fixed  $m$ , the tail  $\{n_k : k \geq m\}$  is a subsequence of  $\{n_k^{(m)}\}$ , along which  $\ell_{n_k}(x_m)$  converges. Therefore

$$\lim_{k \rightarrow \infty} \ell_{n_k}(x_m) \quad \text{exists for every } m \geq 1.$$

$\{\ell_{n_k}(x)\}_k$  is Cauchy for every  $x \in E$ . Fix  $x \in E$  and  $\varepsilon > 0$ . Since  $\{x_m\}$  is dense, choose  $m$  with  $\|x - x_m\|_E < \varepsilon$ . For any indices  $j, k$ , insert and subtract the values at  $x_m$  and use the triangle inequality:

$$|\ell_{n_k}(x) - \ell_{n_j}(x)| \leq \underbrace{|\ell_{n_k}(x) - \ell_{n_k}(x_m)|}_{\text{(I)}} + \underbrace{|\ell_{n_k}(x_m) - \ell_{n_j}(x_m)|}_{\text{(II)}} + \underbrace{|\ell_{n_j}(x_m) - \ell_{n_j}(x)|}_{\text{(III)}}.$$

For term (I), use linearity of  $\ell_{n_k}$  and the bound  $\|\ell_{n_k}\|_{E^*} \leq C$ :

$$\text{(I)} = |\ell_{n_k}(x - x_m)| \leq \|\ell_{n_k}\|_{E^*} \|x - x_m\|_E \leq C \varepsilon.$$

By the same reasoning,  $\text{(III)} \leq C \varepsilon$ . Hence

$$|\ell_{n_k}(x) - \ell_{n_j}(x)| \leq 2C\varepsilon + |\ell_{n_k}(x_m) - \ell_{n_j}(x_m)|.$$

By diagonal extraction of a subsequence,  $\{\ell_{n_k}(x_m)\}_k$  converges, so it is Cauchy; thus there is  $K$  such that

$$|\ell_{n_k}(x_m) - \ell_{n_j}(x_m)| < \varepsilon \quad \text{for all } j, k \geq K.$$

For such  $j, k$  we get  $|\ell_{n_k}(x) - \ell_{n_j}(x)| \leq 2C\varepsilon + \varepsilon = (2C + 1)\varepsilon$ . Since  $\varepsilon > 0$  was arbitrary, the sequence  $\{\ell_{n_k}(x)\}_k$  is Cauchy in  $\mathbb{R}$ .

*The limit is a bounded linear functional.* Since  $\mathbb{R}$  is complete, for each  $x \in E$  the limit

$$\ell(x) := \lim_{k \rightarrow \infty} \ell_{n_k}(x) \quad \text{exists in } \mathbb{R}.$$

*Linearity:* for  $x, y \in E$  and scalars  $a, b$ , using linearity of each  $\ell_{n_k}$  and of the limit,

$$\ell(ax + by) = \lim_k \ell_{n_k}(ax + by) = \lim_k [a \ell_{n_k}(x) + b \ell_{n_k}(y)] = a \ell(x) + b \ell(y).$$

*Boundedness:* taking the limit in  $|\ell_{n_k}(x)| \leq C\|x\|_E$  gives

$$|\ell(x)| = \lim_{k \rightarrow \infty} |\ell_{n_k}(x)| \leq C\|x\|_E \quad \text{for every } x \in E,$$

so  $\ell \in E^*$  with  $\|\ell\|_{E^*} \leq C$ . Finally, by the definition of  $\ell$  we have  $\ell_{n_k}(x) \rightarrow \ell(x)$  for every  $x \in E$ , which is exactly  $\ell_{n_k} \xrightarrow{*} \ell$ .  $\square$

## Strong and Weak Operator Convergence

Throughout,  $E$  and  $F$  are normed vector spaces, and  $\mathcal{L}(E, F)$  is the space of bounded linear operators  $T : E \rightarrow F$  with the operator norm  $\|T\| = \sup_{\|x\|_E \leq 1} \|T(x)\|_F$ .

### Strong and Weak Operator Convergence

Let  $T_n, T \in \mathcal{L}(E, F)$ .

a. *Strong operator convergence.* We say  $T_n \rightarrow T$  *strongly*, written  $T_n \xrightarrow{s} T$ , if

$$\|T_n(x) - T(x)\|_F \rightarrow 0 \quad \text{as } n \rightarrow \infty, \quad \text{for every } x \in E.$$

b. *Weak operator convergence.* We say  $T_n \rightarrow T$  *weakly*, written  $T_n \xrightarrow{w} T$ , if  $T_n(x) \rightarrow T(x)$  weakly in  $F$  for every  $x \in E$ ; that is,

$$\ell(T_n(x)) \rightarrow \ell(T(x)) \quad \text{as } n \rightarrow \infty, \quad \text{for every } \ell \in F^* \text{ and every } x \in E.$$

**Note.** On  $\mathcal{L}(E, F)$  there are three types of convergence, ordered from strongest to weakest:

$$\|T_n - T\| \rightarrow 0 \implies T_n \xrightarrow{s} T \implies T_n \xrightarrow{w} T,$$

where  $\|T_n - T\| \rightarrow 0$  is convergence in operator norm (uniform convergence). The first implication holds because, for each fixed  $x$ ,

$$\|T_n(x) - T(x)\|_F \leq \|T_n - T\| \|x\|_E \rightarrow 0.$$

The second holds because norm convergence in  $F$  implies weak convergence in  $F$ : if  $T_n(x) \rightarrow T(x)$  in  $F$ , then  $T_n(x) \rightarrow T(x)$  in  $F$ . The converses fail in general, as the next example shows.

### Example: Shift Operators on $\ell^2$

Let  $E = F = \ell^2$ . For each  $n \geq 1$  define the *right shift by  $n$* ,  $T_n \in \mathcal{L}(\ell^2, \ell^2)$ , and the *left shift by  $n$* ,  $T_n^* \in \mathcal{L}(\ell^2, \ell^2)$ , by

$$T_n(x) = (\underbrace{0, \dots, 0}_{n \text{ zeros}}, x_1, x_2, \dots), \quad T_n^*(x) = (x_{n+1}, x_{n+2}, \dots), \quad x = (x_k) \in \ell^2.$$

i.  $T_n$  *preserves the norm.* Right-shifting only inserts zeros, so

$$\|T_n(x)\|_{\ell^2}^2 = \sum_{k=1}^{\infty} |x_k|^2 = \|x\|_{\ell^2}^2,$$

hence  $\|T_n(x)\|_{\ell^2} = \|x\|_{\ell^2}$  for all  $x$ , and in particular  $\|T_n\| = 1$ .

ii.  $T_n^*$  *is the adjoint of  $T_n$ .* For  $x, y \in \ell^2$ , the only nonzero entries of  $T_n(x)$  are  $(T_n x)_{n+j} = x_j$ , so

$$\langle T_n(x), y \rangle = \sum_{k=1}^{\infty} (T_n x)_k y_k = \sum_{j=1}^{\infty} x_j y_{n+j} = \sum_{j=1}^{\infty} x_j (T_n^* y)_j = \langle x, T_n^*(y) \rangle.$$

iii.  $T_n^* \xrightarrow{s} 0$  but  $\|T_n^*\| = 1$ . For a fixed  $x = (x_k) \in \ell^2$ ,

$$\|T_n^*(x)\|_{\ell^2}^2 = \sum_{k=n+1}^{\infty} |x_k|^2 \longrightarrow 0 \quad \text{as } n \rightarrow \infty,$$

since it is the tail of the convergent series  $\sum_{k=1}^{\infty} |x_k|^2 = \|x\|_{\ell^2}^2 < \infty$ . So  $T_n^* \xrightarrow{s} 0$  (strong operator convergence). However  $\|T_n^*\| = 1$  for every  $n$ : applying  $T_n^*$  to the basis vector  $e_{n+1} = (0, \dots, 0, 1, 0, \dots)$  gives  $T_n^*(e_{n+1}) = e_1$ , which has norm 1 while  $\|e_{n+1}\| = 1$ . Thus  $T_n^* \not\xrightarrow{s} 0$  in operator norm. *Strong operator convergence does not imply norm convergence.*

iv.  $T_n \xrightarrow{w} 0$  but  $T_n \not\xrightarrow{s} 0$ . Since  $F = \ell^2$  is a Hilbert space, every functional on  $\ell^2$  is  $\langle \cdot, y \rangle$  for some  $y \in \ell^2$ , so weak operator convergence  $T_n \xrightarrow{w} 0$  means  $\langle T_n(x), y \rangle \rightarrow 0$  for all  $x, y \in \ell^2$ . Using part ii and the Cauchy–Schwarz inequality,

$$|\langle T_n(x), y \rangle| = |\langle x, T_n^*(y) \rangle| \leq \|x\|_{\ell^2} \|T_n^*(y)\|_{\ell^2} \longrightarrow 0,$$

where the limit is 0 by part iii. Hence  $T_n \xrightarrow{w} 0$ . But  $T_n$  does *not* converge strongly to 0, because by part i

$$\|T_n(x) - 0\|_{\ell^2} = \|T_n(x)\|_{\ell^2} = \|x\|_{\ell^2} \not\xrightarrow{s} 0 \quad (x \neq 0).$$

*Weak operator convergence does not imply strong operator convergence.*

**Remark.** If  $F = \mathbb{R}$ , then  $\mathcal{L}(E, \mathbb{R}) = E^*$  and the operators  $T_n$  are just functionals. In this case strong operator convergence  $T_n \xrightarrow{s} T$  means  $|T_n(x) - T(x)| \rightarrow 0$  for every  $x \in E$ , i.e.  $T_n(x) \rightarrow T(x)$  for every  $x \in E$ , which is exactly weak-\* convergence  $T_n \xrightarrow{*} T$  in  $E^*$ . So for scalar-valued operators the two notions coincide.

### Proposition 3

Let  $T \in \mathcal{L}(E, F)$  and suppose  $x_n \rightharpoonup x$  weakly in  $E$ . Then  $T(x_n) \rightharpoonup T(x)$  weakly in  $F$ .

**Proof.** We must show  $\ell(T(x_n)) \rightarrow \ell(T(x))$  for every  $\ell \in F^*$ . Fix  $\ell \in F^*$ . The composition  $\ell \circ T : E \rightarrow \mathbb{R}$  is linear (a composition of linear maps) and bounded, since for every  $x \in E$ ,

$$|(\ell \circ T)(x)| = |\ell(T(x))| \leq \|\ell\|_{F^*} \|T(x)\|_F \leq \|\ell\|_{F^*} \|T\| \|x\|_E.$$

Therefore  $\ell \circ T \in E^*$ . Applying the definition of weak convergence  $x_n \rightharpoonup x$  to this particular functional  $\ell \circ T$ ,

$$\ell(T(x_n)) = (\ell \circ T)(x_n) \longrightarrow (\ell \circ T)(x) = \ell(T(x)).$$

Since  $\ell \in F^*$  was arbitrary,  $T(x_n) \rightharpoonup T(x)$ . □

### Proposition 4

Let  $T_n, T \in \mathcal{L}(E, F)$ .

a. If  $T_n \xrightarrow{s} T$ , then  $\|T\| \leq \liminf_{n \rightarrow \infty} \|T_n\|$ .

- b. Suppose  $\sup_n \|T_n\| < \infty$  and  $T_n(y) \rightarrow T(y)$  in  $F$  for every  $y$  in a dense subset  $D \subseteq E$ . Then  $T_n \xrightarrow{s} T$ .

**Proof.**

- a. Fix  $x \in E$ . Strong convergence gives  $T_n(x) \rightarrow T(x)$  in  $F$ , so by continuity of the norm  $\|T_n(x)\|_F \rightarrow \|T(x)\|_F$ ; in particular the limit equals the lower limit:

$$\|T(x)\|_F = \lim_{n \rightarrow \infty} \|T_n(x)\|_F = \liminf_{n \rightarrow \infty} \|T_n(x)\|_F.$$

For each  $n$  we have  $\|T_n(x)\|_F \leq \|T_n\| \|x\|_E$ . Taking  $\liminf$  of both sides (the factor  $\|x\|_E$  is a nonnegative constant),

$$\|T(x)\|_F = \liminf_{n \rightarrow \infty} \|T_n(x)\|_F \leq \liminf_{n \rightarrow \infty} (\|T_n\| \|x\|_E) = \left( \liminf_{n \rightarrow \infty} \|T_n\| \right) \|x\|_E.$$

This holds for all  $x \in E$ . Dividing by  $\|x\|_E$  for  $x \neq 0$  and taking the supremum over such  $x$ ,

$$\|T\| = \sup_{x \neq 0} \frac{\|T(x)\|_F}{\|x\|_E} \leq \liminf_{n \rightarrow \infty} \|T_n\|. \quad \square$$

- b. Write  $M := \sup_k \|T_k\| < \infty$ . Fix  $x \in E$  and let  $\varepsilon > 0$ . For any  $y \in D$ , the triangle inequality gives

$$\|T(x) - T_n(x)\|_F \leq \|T(x) - T(y)\|_F + \|T(y) - T_n(y)\|_F + \|T_n(y) - T_n(x)\|_F.$$

Bound the first and third terms using boundedness of  $T$  and of each  $T_n$ :

$$\|T(x) - T(y)\|_F = \|T(x - y)\|_F \leq \|T\| \|x - y\|_E,$$

$$\|T_n(y) - T_n(x)\|_F = \|T_n(y - x)\|_F \leq \|T_n\| \|x - y\|_E \leq M \|x - y\|_E.$$

Writing  $A := \|T\| \|x - y\|_E$ ,  $B := M \|x - y\|_E$ , and  $C := \|T(y) - T_n(y)\|_F$ , we obtain

$$\|T(x) - T_n(x)\|_F \leq \underbrace{(\|T\| + M) \|x - y\|_E}_{A+B} + \underbrace{\|T(y) - T_n(y)\|_F}_C.$$

Now choose the two pieces small. Since  $D$  is dense in  $E$ , pick  $y \in D$  with

$$\|x - y\|_E \leq \frac{\varepsilon}{2(\|T\| + M + 1)}, \quad \text{so that} \quad A + B = (\|T\| + M) \|x - y\|_E \leq \frac{\varepsilon}{2}.$$

Fix this  $y$ . By hypothesis  $T_n(y) \rightarrow T(y)$  in  $F$  (because  $y \in D$ ), so there is  $N$  with

$$C = \|T(y) - T_n(y)\|_F \leq \frac{\varepsilon}{2} \quad \text{for all } n \geq N.$$

Therefore, for all  $n \geq N$ ,

$$\|T(x) - T_n(x)\|_F \leq (A + B) + C \leq \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Since  $\varepsilon > 0$  was arbitrary,  $\|T_n(x) - T(x)\|_F \rightarrow 0$ ; and since  $x \in E$  was arbitrary,  $T_n \xrightarrow{s} T$ .  $\square$

### Theorem 3

Let  $E$  be a Banach space and  $F$  a reflexive Banach space. Let  $T_n \in \mathcal{L}(E, F)$  be such that for every  $x \in E$  the sequence  $\{T_n(x) : n \geq 1\}$  is weakly Cauchy in  $F$ . Then there exists  $T \in \mathcal{L}(E, F)$  such that  $T_n \xrightarrow{w} T$ .

**Proof.** Define  $T$  pointwise. Fix  $x \in E$ . By assumption  $\{T_n(x)\}_n$  is weakly Cauchy in  $F$ . Since  $F$  is a reflexive Banach space, every weakly Cauchy sequence in a reflexive Banach space has a weak limit, which gives an element of  $F$ , which we call  $T(x)$ , with

$$T_n(x) \rightharpoonup T(x) \quad \text{weakly in } F.$$

The weak limit is unique, so  $T(x)$  is well defined. This defines a map  $T : E \rightarrow F$ .

*T is linear.* Let  $x, x' \in E$  and let  $a, b$  be scalars. Each  $T_n$  is linear, so  $T_n(ax + bx') = aT_n(x) + bT_n(x')$ . Weak limits respect linear combinations: for any  $\ell \in F^*$ ,

$$\ell(aT_n(x) + bT_n(x')) = a\ell(T_n(x)) + b\ell(T_n(x')) \longrightarrow a\ell(T(x)) + b\ell(T(x')) = \ell(aT(x) + bT(x')).$$

Thus  $T_n(ax + bx') \rightharpoonup aT(x) + bT(x')$ , and since weak limits are unique,  $T(ax + bx') = aT(x) + bT(x')$ .

*T is bounded.* First, each weakly Cauchy sequence  $\{T_n(x)\}_n$  is bounded in  $F$  by Proposition 2(b):

$$\sup_n \|T_n(x)\|_F < \infty \quad \text{for every } x \in E.$$

Since  $E$  is a Banach space and each  $T_n \in \mathcal{L}(E, F)$ , the family  $\{T_n\}$  is pointwise bounded on  $E$ . By the Uniform Boundedness Principle (Banach–Steinhaus),

$$\sup_n \|T_n\| =: C < \infty.$$

Next, applying weak lower semicontinuity of the norm to  $T_n(x) \rightharpoonup T(x)$ , then using  $\|T_n(x)\|_F \leq \|T_n\| \|x\|_E$ ,

$$\|T(x)\|_F \leq \liminf_{n \rightarrow \infty} \|T_n(x)\|_F \leq \liminf_{n \rightarrow \infty} (\|T_n\| \|x\|_E) \leq \left( \sup_k \|T_k\| \right) \|x\|_E = C \|x\|_E.$$

So  $T$  is bounded with  $\|T\| \leq C$ , that is,  $T \in \mathcal{L}(E, F)$ .

From above  $T_n(x) \rightharpoonup T(x)$  weakly in  $F$  for every  $x \in E$ , which is exactly  $T_n \xrightarrow{w} T$ .  $\square$

## Bounded Operators

### The Banach Algebra of Bounded Operators

Throughout this section,  $E$  is a Banach space over the complex field  $\mathbb{C}$ . Consider the space of bounded linear operators from  $E$  to itself,

$$\mathcal{B} := \mathcal{L}(E) = \mathcal{L}(E, E).$$

This  $\mathcal{B}$  is a Banach space under the operator norm  $\|A\| = \sup_{\|x\|_E \leq 1} \|A(x)\|_E$ , and it also carries a *multiplication*, namely composition of operators,

$$AB := A \circ B, \quad \text{which satisfies} \quad \|AB\| \leq \|A\| \|B\| \quad (A, B \in \mathcal{B}).$$

The inequality  $\|AB\| \leq \|A\| \|B\|$  is called submultiplicativity. A Banach space with such a multiplication is a *Banach algebra*. Moreover  $\mathcal{B}$  has a unit, the identity map  $I \in \mathcal{B}$ :

$$IA = AI = A \quad \text{for every } A \in \mathcal{B}.$$

### Invertible Operator

We say  $T \in \mathcal{B}$  is *invertible* if there exists  $S \in \mathcal{B}$  with

$$ST = TS = I.$$

Such an  $S$  is unique; we denote it  $S = T^{-1}$ .

### Example: Neumann Series

Let  $T \in \mathcal{B}$  with  $\|T\| < 1$ . Then  $I - T$  is invertible, and

$$(I - T)^{-1} = \sum_{n=0}^{\infty} T^n \quad (\text{with the convention } T^0 := I).$$

**Proof.** *The series converges in  $\mathcal{B}$ .* By submultiplicativity,  $\|T^n\| \leq \|T\|^n$  for every  $n \geq 0$  (and  $\|T^0\| = \|I\| = 1 = \|T\|^0$ ). Since  $\|T\| < 1$ , the geometric series of norms converges:

$$\sum_{n=0}^{\infty} \|T^n\| \leq \sum_{n=0}^{\infty} \|T\|^n = \frac{1}{1 - \|T\|} < \infty.$$

Let  $S_n := \sum_{k=0}^n T^k$  be the partial sums. For  $m > n$ ,

$$\|S_m - S_n\| = \left\| \sum_{k=n+1}^m T^k \right\| \leq \sum_{k=n+1}^m \|T^k\| \leq \sum_{k=n+1}^m \|T\|^k \xrightarrow{n, m \rightarrow \infty} 0,$$

because the right-hand side is a tail of a convergent series. So  $\{S_n\}$  is Cauchy in  $\mathcal{B}$ . Since  $E$  is a Banach space,  $\mathcal{B} = \mathcal{L}(E)$  is also complete, so  $\{S_n\}$  converges to some

$$S := \sum_{n=0}^{\infty} T^n \in \mathcal{B}.$$

$S$  is a two-sided inverse of  $I - T$ . Compute the product  $S_n(I - T)$  and watch the terms telescope:

$$S_n(I - T) = \sum_{k=0}^n T^k(I - T) = \sum_{k=0}^n (T^k - T^{k+1}) = \underbrace{\sum_{k=0}^n T^k}_{I + T + \dots + T^n} - \underbrace{\sum_{k=0}^n T^{k+1}}_{T + T^2 + \dots + T^{n+1}} = I - T^{n+1}.$$

Now let  $n \rightarrow \infty$  in the equation  $S_n(I - T) = I - T^{n+1}$ .

- *Right-hand side.*  $\|T^{n+1}\| \leq \|T\|^{n+1} \rightarrow 0$ , so  $T^{n+1} \rightarrow 0$  in  $\mathcal{B}$ , and therefore  $I - T^{n+1} \rightarrow I$ .
- *Left-hand side.* Multiplication is continuous: if  $A_n \rightarrow A$  in  $\mathcal{B}$ , then for any fixed  $B$ ,

$$\|A_n B - AB\| = \|(A_n - A)B\| \leq \|A_n - A\| \|B\| \rightarrow 0,$$

so  $A_n B \rightarrow AB$ . Taking  $A_n = S_n$ ,  $A = S$ , and  $B = I - T$  gives  $S_n(I - T) \rightarrow S(I - T)$ .

Both sides converge, so their limits agree:  $S(I - T) = I$ . The same telescoping with the factor on the left gives  $(I - T)S_n = I - T^{n+1} \rightarrow I$ , and by continuity  $(I - T)S_n \rightarrow (I - T)S$ , so  $(I - T)S = I$ . Therefore  $S = (I - T)^{-1}$ .  $\square$

**Remark.** In this computation  $S_n$  and  $I - T$  commute, because both are polynomials in  $T$ ; so the left and right products are equal and a single calculation handles both sides. In general, to prove an operator is invertible one must check *both*  $ST = I$  and  $TS = I$  separately.

## Spectrum and Resolvent

We now study, for a fixed  $T \in \mathcal{B}$  and a scalar  $\lambda \in \mathbb{C}$ , the invertibility of

$$\lambda I - T := \lambda I - T \in \mathcal{B}.$$

**Finite dimensions.** Let  $E = \mathbb{C}^d$ , so  $T$  is a  $d \times d$  complex matrix. Then

$$\lambda I - T \text{ is invertible} \iff \det(\lambda I - T) \neq 0 \iff \lambda \text{ is not an eigenvalue of } T,$$

and  $\lambda$  is an eigenvalue exactly when  $\det(\lambda I - T) = 0$ . In infinite dimensions there is no determinant, so we keep invertibility of  $\lambda I - T$  as the basic notion and make the following definitions.

## Resolvent Set, Resolvent, Spectrum

Let  $T \in \mathcal{B} = \mathcal{L}(E)$ .

- The *resolvent set* of  $T$  is

$$\rho(T) = \{\lambda \in \mathbb{C} : \lambda I - T \text{ is invertible in } \mathcal{B}\}.$$

For  $\lambda \in \rho(T)$ , the operator

$$R_\lambda(T) := (\lambda I - T)^{-1} \in \mathcal{B}$$

is called the *resolvent* of  $T$  at  $\lambda$ .

- The *spectrum* of  $T$  is the complement

$$\sigma(T) = \mathbb{C} \setminus \rho(T) = \{\lambda \in \mathbb{C} : \lambda I - T \text{ is not invertible}\}.$$



## Eigenvalue, Eigenvector

A *nonzero* vector  $x \in E$ ,  $x \neq 0$ , is an *eigenvector* of  $T$  if  $T(x) = \lambda x$  for some  $\lambda \in \mathbb{C}$ ; this  $\lambda$  is then an *eigenvalue* of  $T$ . If  $\lambda$  is an eigenvalue, then  $(\lambda I - T)(x) = 0$  with  $x \neq 0$ , so  $\lambda I - T$  is not injective, hence not invertible, and therefore  $\lambda \in \sigma(T)$ . Thus every eigenvalue lies in the spectrum.

## The Resolvent Identity and Commuting

Let  $\lambda, \mu \in \rho(T)$ . The idea is to rewrite each resolvent by inserting the other one. Since  $(\mu I - T)R_\mu(T) = I$  and  $R_\lambda(T)(\lambda I - T) = I$ ,

$$R_\lambda(T) = R_\lambda(T) \underbrace{(\mu I - T)R_\mu(T)}_{=I}, \quad R_\mu(T) = \underbrace{R_\lambda(T)(\lambda I - T)}_{=I} R_\mu(T).$$

Subtracting these two expressions and factoring out  $R_\lambda(T)$  on the left and  $R_\mu(T)$  on the right,

$$R_\lambda(T) - R_\mu(T) = R_\lambda(T)[(\mu I - T) - (\lambda I - T)]R_\mu(T).$$

The middle bracket simplifies:  $(\mu I - T) - (\lambda I - T) = \mu I - \lambda I = (\mu - \lambda)I$ . Therefore

$$R_\lambda(T) - R_\mu(T) = (\mu - \lambda) R_\lambda(T) R_\mu(T). \quad (1)$$

Now swap the names  $\lambda$  and  $\mu$  in (1):

$$R_\mu(T) - R_\lambda(T) = (\lambda - \mu) R_\mu(T) R_\lambda(T), \quad \text{i.e.} \quad R_\lambda(T) - R_\mu(T) = (\mu - \lambda) R_\mu(T) R_\lambda(T). \quad (2)$$

Comparing (1) and (2) gives  $(\mu - \lambda)R_\lambda(T)R_\mu(T) = (\mu - \lambda)R_\mu(T)R_\lambda(T)$ . When  $\lambda \neq \mu$  we cancel the nonzero scalar  $(\mu - \lambda)$  to obtain

$$R_\lambda(T)R_\mu(T) = R_\mu(T)R_\lambda(T)$$

and for  $\lambda = \mu$  this is trivial. Using this commutativity in (1) also yields the stated form  $R_\lambda(T) - R_\mu(T) = (\mu - \lambda)R_\mu(T)R_\lambda(T)$ .

## Theorem 1

Let  $T \in \mathcal{B} = \mathcal{L}(E)$ , with  $E$  a Banach space over  $\mathbb{C}$ . Then:

- a. The resolvent set  $\rho(T) \subseteq \mathbb{C}$  is open, and the map  $\lambda \mapsto R_\lambda(T)$  is a  $\mathcal{B}$ -valued analytic function on  $\rho(T)$ : for each  $\lambda_0 \in \rho(T)$  there are coefficients  $c_n \in \mathcal{B}$  with

$$R_\lambda(T) = \sum_{n=0}^{\infty} c_n (\lambda - \lambda_0)^n \quad \text{for all } \lambda \text{ sufficiently close to } \lambda_0.$$

Moreover, for all  $\lambda, \mu \in \rho(T)$  the *resolvent identity* holds,

$$R_\lambda(T) - R_\mu(T) = (\mu - \lambda) R_\mu(T) R_\lambda(T),$$

and the resolvents commute:  $R_\mu(T) R_\lambda(T) = R_\lambda(T) R_\mu(T)$ .

b. The spectrum  $\sigma(T) \subseteq \mathbb{C}$  is a nonempty compact set, and its *spectral radius* satisfies

$$r(T) := \sup\{|\lambda| : \lambda \in \sigma(T)\} \leq \|T\|.$$

**Proof of Part a.** *Motivation (treat everything as numbers).* Fix  $\lambda_0 \in \rho(T)$ . If  $\lambda$ ,  $\lambda_0$ , and  $T$  were ordinary numbers, we would write, for  $\lambda$  near  $\lambda_0$ ,

$$\frac{1}{\lambda - T} = \frac{1}{(\lambda_0 - T) - (\lambda_0 - \lambda)} = \frac{1}{\lambda_0 - T} \cdot \frac{1}{1 - \frac{\lambda_0 - \lambda}{\lambda_0 - T}} = \frac{1}{\lambda_0 - T} \sum_{n=0}^{\infty} \left( \frac{\lambda_0 - \lambda}{\lambda_0 - T} \right)^n,$$

the geometric series being valid when  $\left| \frac{\lambda_0 - \lambda}{\lambda_0 - T} \right| < 1$ . This is only a guide; we now do the same thing rigorously with operators, replacing  $\frac{1}{\lambda_0 - T}$  by the operator  $R_{\lambda_0}(T)$ .

*Build a candidate inverse.* Assume

$$|\lambda - \lambda_0| < \frac{1}{\|R_{\lambda_0}(T)\|}, \quad \text{equivalently} \quad |\lambda - \lambda_0| \|R_{\lambda_0}(T)\| < 1.$$

$R_{\lambda_0}(T)$  is invertible, so it is not the zero operator and  $\|R_{\lambda_0}(T)\| > 0$ . Then the operator  $(\lambda_0 - \lambda)R_{\lambda_0}(T)$  has norm

$$\|(\lambda_0 - \lambda)R_{\lambda_0}(T)\| = |\lambda_0 - \lambda| \|R_{\lambda_0}(T)\| < 1,$$

so by the Neumann Series Example,  $I - (\lambda_0 - \lambda)R_{\lambda_0}(T)$  is invertible and

$$[I - (\lambda_0 - \lambda)R_{\lambda_0}(T)]^{-1} = \sum_{n=0}^{\infty} (\lambda_0 - \lambda)^n [R_{\lambda_0}(T)]^n. \quad (*)$$

Define the candidate operator

$$\tilde{R}_{\lambda}(T) := R_{\lambda_0}(T) [I - (\lambda_0 - \lambda)R_{\lambda_0}(T)]^{-1} \in \mathcal{B}.$$

*A commuting fact and a rewriting.* By (\*), the operator  $[I - (\lambda_0 - \lambda)R_{\lambda_0}(T)]^{-1}$  is a norm-convergent series of powers of  $R_{\lambda_0}(T)$ . Since  $R_{\lambda_0}(T)$  commutes with each of its own powers, and multiplication is continuous,  $R_{\lambda_0}(T)$  commutes with this inverse:

$$R_{\lambda_0}(T) [I - (\lambda_0 - \lambda)R_{\lambda_0}(T)]^{-1} = [I - (\lambda_0 - \lambda)R_{\lambda_0}(T)]^{-1} R_{\lambda_0}(T). \quad (**)$$

Also write  $\lambda I - T$  in terms of  $\lambda_0$ :

$$\lambda I - T = (\lambda_0 I - T) - (\lambda_0 - \lambda)I, \quad \text{since } \lambda_0 I - (\lambda_0 - \lambda)I = \lambda I.$$

*Check that  $\tilde{R}_{\lambda}(T)$  inverts  $\lambda I - T$ .* Using the rewriting and then (\*\*) to move the inverse to the front,

$$\begin{aligned} \tilde{R}_{\lambda}(T) (\lambda I - T) &= R_{\lambda_0}(T) [I - (\lambda_0 - \lambda)R_{\lambda_0}(T)]^{-1} [(\lambda_0 I - T) - (\lambda_0 - \lambda)I] \\ &= [I - (\lambda_0 - \lambda)R_{\lambda_0}(T)]^{-1} R_{\lambda_0}(T) [(\lambda_0 I - T) - (\lambda_0 - \lambda)I] \\ &= [I - (\lambda_0 - \lambda)R_{\lambda_0}(T)]^{-1} [R_{\lambda_0}(T)(\lambda_0 I - T) - (\lambda_0 - \lambda)R_{\lambda_0}(T)]. \end{aligned}$$

Now  $R_{\lambda_0}(T)(\lambda_0 I - T) = (\lambda_0 I - T)^{-1}(\lambda_0 I - T) = I$ , so the bracket simplifies to

$$R_{\lambda_0}(T)(\lambda_0 I - T) - (\lambda_0 - \lambda)R_{\lambda_0}(T) = I - (\lambda_0 - \lambda)R_{\lambda_0}(T),$$

and therefore

$$\tilde{R}_\lambda(T)(\lambda I - T) = [I - (\lambda_0 - \lambda)R_{\lambda_0}(T)]^{-1}[I - (\lambda_0 - \lambda)R_{\lambda_0}(T)] = I.$$

The identical computation with the two factors in the opposite order (again using (\*\*)) gives  $(\lambda I - T)\tilde{R}_\lambda(T) = I$ . Hence  $\lambda I - T$  is invertible, with inverse  $\tilde{R}_\lambda(T)$ .

*Openness and the power series.* We have shown: whenever  $|\lambda - \lambda_0| < 1/\|R_{\lambda_0}(T)\|$ , the point  $\lambda$  lies in  $\rho(T)$ , and

$$R_\lambda(T) = \tilde{R}_\lambda(T) = R_{\lambda_0}(T) \sum_{n=0}^{\infty} (\lambda_0 - \lambda)^n [R_{\lambda_0}(T)]^n = \sum_{n=0}^{\infty} (-1)^n [R_{\lambda_0}(T)]^{n+1} (\lambda - \lambda_0)^n,$$

where we used  $(\lambda_0 - \lambda)^n = (-1)^n (\lambda - \lambda_0)^n$ . So every  $\lambda_0 \in \rho(T)$  is the center of an open disk of radius  $1/\|R_{\lambda_0}(T)\|$  contained in  $\rho(T)$ ; this proves  $\rho(T)$  is open. On that disk,  $R_\lambda(T)$  is given by a norm-convergent power series in  $(\lambda - \lambda_0)$  with coefficients  $c_n = (-1)^n [R_{\lambda_0}(T)]^{n+1} \in \mathcal{B}$ , so  $R_\lambda(T)$  is analytic on  $\rho(T)$ .

*The resolvent identity and commuting.* Let  $\lambda, \mu \in \rho(T)$ . Start from the elementary identity

$$(\mu I - T) - (\lambda I - T) = (\mu - \lambda)I.$$

Multiply on the left by  $R_\mu(T) = (\mu I - T)^{-1}$  and on the right by  $R_\lambda(T) = (\lambda I - T)^{-1}$ :

$$R_\mu(T)[(\mu I - T) - (\lambda I - T)]R_\lambda(T) = (\mu - \lambda)R_\mu(T)R_\lambda(T).$$

Expand the left-hand side into two terms and simplify each:

$$\underbrace{R_\mu(T)(\mu I - T)}_{=I} R_\lambda(T) - R_\mu(T) \underbrace{(\lambda I - T)R_\lambda(T)}_{=I} = R_\lambda(T) - R_\mu(T).$$

This gives the resolvent identity

$$R_\lambda(T) - R_\mu(T) = (\mu - \lambda)R_\mu(T)R_\lambda(T).$$

Swapping the roles of  $\lambda$  and  $\mu$  in this identity gives  $R_\mu(T) - R_\lambda(T) = (\lambda - \mu)R_\lambda(T)R_\mu(T)$ , that is,  $R_\lambda(T) - R_\mu(T) = (\mu - \lambda)R_\lambda(T)R_\mu(T)$ . Comparing the two expressions for  $R_\lambda(T) - R_\mu(T)$ , if  $\mu \neq \lambda$  we may cancel the nonzero factor  $(\mu - \lambda)$  to get  $R_\mu(T)R_\lambda(T) = R_\lambda(T)R_\mu(T)$ ; for  $\mu = \lambda$  this is trivial. So the resolvents commute.  $\square$

**Proof of Part b.** *Large  $\lambda$  lie in  $\rho(T)$ .* As motivation, if  $\lambda$  and  $T$  were numbers we would write, for  $|\lambda| > |T|$ ,

$$\frac{1}{\lambda - T} = \frac{1}{\lambda} \cdot \frac{1}{1 - T/\lambda} = \frac{1}{\lambda} \sum_{n=0}^{\infty} \frac{T^n}{\lambda^n}.$$

For operators, fix  $\lambda \in \mathbb{C}$  with  $|\lambda| > \|T\|$ . Then  $\|\frac{1}{\lambda}T\| = \|T\|/|\lambda| < 1$ , so by the Neumann Series Example  $I - \frac{1}{\lambda}T$  is invertible. Set

$$\tilde{R}_\lambda(T) := \frac{1}{\lambda} \left( I - \frac{1}{\lambda}T \right)^{-1}.$$

Since  $\lambda I - T = \lambda(I - \frac{1}{\lambda}T)$ , a direct check gives

$$\tilde{R}_\lambda(T)(\lambda I - T) = \frac{1}{\lambda} \left( I - \frac{1}{\lambda}T \right)^{-1} \lambda \left( I - \frac{1}{\lambda}T \right) = \left( I - \frac{1}{\lambda}T \right)^{-1} \left( I - \frac{1}{\lambda}T \right) = I,$$

and the same on the other side. So  $\lambda \in \rho(T)$  and  $R_\lambda(T) = \tilde{R}_\lambda(T)$  for every  $|\lambda| > \|T\|$ .

$\sigma(T)$  is compact, and  $r(T) \leq \|T\|$ . Previous part says every  $\lambda$  with  $|\lambda| > \|T\|$  is in  $\rho(T)$ . Equivalently,

$$\sigma(T) \subseteq \{z \in \mathbb{C} : |z| \leq \|T\|\},$$

so  $\sigma(T)$  is bounded and  $r(T) = \sup\{|\lambda| : \lambda \in \sigma(T)\} \leq \|T\|$ . Also  $\sigma(T) = \mathbb{C} \setminus \rho(T)$  is closed, being the complement of the open set  $\rho(T)$ . A closed and bounded subset of  $\mathbb{C}$  is compact (Heine–Borel), so  $\sigma(T)$  is compact.

$\sigma(T)$  is nonempty. Suppose, for contradiction, that  $\sigma(T) = \emptyset$ , i.e.  $\rho(T) = \mathbb{C}$ . Then  $\lambda \mapsto R_\lambda(T)$  is analytic on all of  $\mathbb{C}$ . For  $|\lambda| > \|T\|$ , expanding the Neumann series from Previous part,

$$R_\lambda(T) = \frac{1}{\lambda} \left( I - \frac{1}{\lambda}T \right)^{-1} = \frac{1}{\lambda} \sum_{n=0}^{\infty} \frac{T^n}{\lambda^n},$$

$$\|R_\lambda(T)\| \leq \frac{1}{|\lambda|} \sum_{n=0}^{\infty} \frac{\|T\|^n}{|\lambda|^n} = \frac{1}{|\lambda|} \cdot \frac{1}{1 - \|T\|/|\lambda|} = \frac{1}{|\lambda| - \|T\|} \xrightarrow{|\lambda| \rightarrow \infty} 0.$$

Thus  $R_\lambda(T)$  is an entire  $\mathcal{B}$ -valued function that is bounded on  $\mathbb{C}$  (it is continuous, hence bounded on the disk  $|\lambda| \leq \|T\| + 1$ , and small outside) and tends to 0 at infinity. By Liouville's theorem (for Banach-space-valued analytic functions) a bounded entire function is constant; since it also tends to 0, that constant is 0, so  $R_\lambda(T) \equiv 0$ . But  $R_\lambda(T) = (\lambda I - T)^{-1}$  means  $(\lambda I - T)R_\lambda(T) = I$ ; if  $R_\lambda(T) = 0$  then  $(\lambda I - T) \cdot 0 = 0 \neq I$ , a contradiction. Hence  $\sigma(T) \neq \emptyset$ .  $\square$

## Spectral Radius Formula

Let  $T \in \mathcal{L}(E)$ . Then the limit  $\lim_{n \rightarrow \infty} \|T^n\|^{1/n}$  exists and equals the spectral radius:

$$r(T) = \lim_{n \rightarrow \infty} \|T^n\|^{1/n} \quad (\leq \|T\|).$$

The bound  $\|T^n\|^{1/n} \leq \|T\|$  holds for every  $n$ , because  $\|T^n\| \leq \|T\|^n$  by submultiplicativity; hence the limit is  $\leq \|T\|$ , which refines the estimate  $r(T) \leq \|T\|$  from Theorem 1(b). The equality  $r(T) = \lim_n \|T^n\|^{1/n}$  is proved using the analyticity of the resolvent from Theorem 1; we take it as stated here.

### Example: Invertible Operators

Let  $T \in \mathcal{L}(E)$  be invertible. Then  $0 \notin \sigma(T)$ , since  $0 \cdot I - T = -T$  is invertible. Then:

- a.  $\sigma(T^{-1}) = \{1/\lambda : \lambda \in \sigma(T)\}$ .
- b. If  $S \in \mathcal{L}(E)$  commutes with  $T$ , that is  $ST = TS$ , then  $S$  commutes with  $T^{-1}$ :  $ST^{-1} = T^{-1}S$ .

**Proof.**

- a. We want to show  $\lambda \in \sigma(T^{-1})$  if and only if  $1/\lambda \in \sigma(T)$ . Equivalently, we want to show  $\lambda \in \rho(T^{-1})$  if and only if  $1/\lambda \in \rho(T)$ . Since  $T$  and  $T^{-1}$  are invertible,  $0 \notin \sigma(T)$  and  $0 \notin \sigma(T^{-1})$ , so we only consider  $\lambda \neq 0$ . For such  $\lambda$ , factor

$$\lambda I - T^{-1} = -\lambda T^{-1} \left( \frac{1}{\lambda} I - T \right).$$

Indeed, expanding the right side:  $-\lambda T^{-1} \left( \frac{1}{\lambda} I - T \right) = -\lambda T^{-1} \cdot \frac{1}{\lambda} I + \lambda T^{-1} T = -T^{-1} + \lambda I = \lambda I - T^{-1}$ . The factor  $-\lambda T^{-1}$  is invertible (a nonzero scalar times an invertible operator). Therefore  $\lambda I - T^{-1}$  is invertible if and only if  $\frac{1}{\lambda} I - T$  is invertible:

$$\lambda \in \rho(T^{-1}) \iff \frac{1}{\lambda} \in \rho(T).$$

Taking complements,  $\lambda \in \sigma(T^{-1}) \iff \frac{1}{\lambda} \in \sigma(T)$ , which is exactly  $\sigma(T^{-1}) = \{1/\lambda : \lambda \in \sigma(T)\}$ .

- b. We want  $ST^{-1} = T^{-1}S$ . Since  $T$  is invertible, it is left-cancellable, so it is enough to show  $T(ST^{-1}) = T(T^{-1}S)$ . Compute both sides using  $TS = ST$ :

$$T(ST^{-1}) = (TS)T^{-1} = (ST)T^{-1} = S(TT^{-1}) = S, \quad T(T^{-1}S) = (TT^{-1})S = S.$$

Both equal  $S$ , so  $T(ST^{-1}) = T(T^{-1}S)$ ; multiplying on the left by  $T^{-1}$  gives  $ST^{-1} = T^{-1}S$ .  $\square$

We extend the polynomial calculus  $P \mapsto P(T)$  to entire functions. Recall that an *entire* function  $f : \mathbb{C} \rightarrow \mathbb{C}$  is given by a power series

$$f(z) = \sum_{n=0}^{\infty} c_n z^n, \quad z \in \mathbb{C},$$

with infinite radius of convergence; equivalently,  $\sum_{n=0}^{\infty} |c_n| r^n < \infty$  for every  $r > 0$ .

### Definition ( $f(T)$ )

Given an entire function  $f(z) = \sum_{n=0}^{\infty} c_n z^n$  and  $T \in \mathcal{L}(E)$ , set

$$f(T) := \sum_{n=0}^{\infty} c_n T^n \in \mathcal{L}(E).$$

This series converges absolutely in  $\mathcal{L}(E)$ : taking  $r = \|T\|$  in the convergence condition,

$$\sum_{n=0}^{\infty} \|c_n T^n\| \leq \sum_{n=0}^{\infty} |c_n| \|T\|^n < \infty,$$

and since  $\mathcal{L}(E)$  is complete, the partial sums converge.

## Spectral Mapping Theorem

Let  $T \in \mathcal{L}(E)$  and let  $f : \mathbb{C} \rightarrow \mathbb{C}$  be entire. Then

$$\sigma(f(T)) = f(\sigma(T)) = \{f(\lambda) : \lambda \in \sigma(T)\}.$$

**Proof.** For  $f$  a polynomial, we prove the two parts. The general entire case is the same idea, with extra care about convergence of the infinite series. Let  $f(z) = \sum_{n=0}^N c_n z^n$  be a polynomial of degree  $N$ . Then  $f(T) = \sum_{n=0}^N c_n T^n$  is a polynomial in  $T$ . We show  $\sigma(f(T)) = f(\sigma(T))$ .

1.  $f(\sigma(T)) \subseteq \sigma(f(T))$ . Let  $\alpha \in \sigma(T)$ ; we show  $f(\alpha) \in \sigma(f(T))$ . View  $f(\alpha) - f(z)$  as a polynomial in  $z$ . It vanishes at  $z = \alpha$ , so  $(z - \alpha)$  divides it; writing the linear factor as  $(\alpha - z)$ , there is a polynomial  $g$  with

$$f(\alpha) - f(z) = (\alpha - z)g(z).$$

Substituting the operator  $T$  for  $z$  (a constant  $f(\alpha)$  becomes  $f(\alpha)I$ ), the polynomial calculus gives

$$f(\alpha)I - f(T) = (\alpha I - T)g(T).$$

Suppose, for contradiction, that  $G := f(\alpha)I - f(T)$  is invertible. Then

$$I = G^{-1}G = G^{-1}(\alpha I - T)g(T).$$

All of  $G^{-1}$ ,  $g(T)$ , and  $\alpha I - T$  are polynomials in  $T$ , so they commute with one another. Rearranging,

$$I = (\alpha I - T)[g(T)G^{-1}] = [g(T)G^{-1}](\alpha I - T),$$

which exhibits  $g(T)G^{-1}$  as a two-sided inverse of  $\alpha I - T$ . So  $\alpha \in \rho(T)$ , contradicting  $\alpha \in \sigma(T)$ . Hence  $f(\alpha)I - f(T)$  is not invertible, i.e.  $f(\alpha) \in \sigma(f(T))$ .

2.  $\sigma(f(T)) \subseteq f(\sigma(T))$ . Suppose, for contradiction, there is  $\mu \in \sigma(f(T))$  with  $\mu \notin f(\sigma(T))$ . Factor the polynomial  $\mu - f(z)$  (of degree  $N \geq 1$ , with leading coefficient  $c \neq 0$ ) into linear factors over  $\mathbb{C}$ :

$$\mu - f(z) = c \prod_{k=1}^N (z_k - z),$$

where  $z_1, \dots, z_N$  are its roots, so  $f(z_k) = \mu$  for each  $k$ . None of these roots lies in  $\sigma(T)$ : if some  $z_k \in \sigma(T)$ , then  $\mu = f(z_k) \in f(\sigma(T))$ , contradicting  $\mu \notin f(\sigma(T))$ . Hence every  $z_k \in \rho(T)$ , so each  $z_k I - T$  is invertible. Substituting  $T$  for  $z$ ,

$$\mu I - f(T) = c \prod_{k=1}^N (z_k I - T).$$

This is a nonzero scalar times a product of invertible operators, which all commute, being polynomials in  $T$ , so it is invertible. Therefore  $\mu \in \rho(f(T))$ , contradicting  $\mu \in \sigma(f(T))$ .

The two parts give  $\sigma(f(T)) = f(\sigma(T))$ . □

**Example: Spectra of  $ST$  and  $TS$** 

Let  $S, T \in \mathcal{L}(E)$ . Then  $\sigma(ST)$  and  $\sigma(TS)$  can differ only at the point 0:

$$\sigma(ST) \setminus \{0\} = \sigma(TS) \setminus \{0\}.$$

**Proof.** It suffices to show that for  $\lambda \neq 0$ , if  $\lambda \in \rho(TS)$  then  $\lambda \in \rho(ST)$ ; the reverse inclusion follows by swapping the roles of  $S$  and  $T$ . So fix  $\lambda \neq 0$  with  $\lambda \in \rho(TS)$ , and put

$$A := (\lambda I - TS)^{-1}, \quad \text{so} \quad (\lambda I - TS)A = A(\lambda I - TS) = I.$$

We claim that  $\frac{1}{\lambda}(I + SAT)$  is the inverse of  $\lambda I - ST$ . Compute

$$(\lambda I - ST)(I + SAT) = \lambda I + \lambda SAT - ST - STSAT = \lambda I - ST + [\lambda SAT - S(TS)AT].$$

The bracketed terms combine, using  $\lambda A - (TS)A = (\lambda I - TS)A = I$ :

$$\lambda SAT - S(TS)AT = S[\lambda A - (TS)A]T = S[(\lambda I - TS)A]T = S I T = ST.$$

Therefore  $(\lambda I - ST)(I + SAT) = \lambda I - ST + ST = \lambda I$ . The analogous computation (using  $A(\lambda I - TS) = I$ ) gives  $(I + SAT)(\lambda I - ST) = \lambda I$ . Hence

$$(\lambda I - ST) \cdot \frac{1}{\lambda}(I + SAT) = \frac{1}{\lambda}(I + SAT) \cdot (\lambda I - ST) = I,$$

so  $\lambda I - ST$  is invertible and  $\lambda \in \rho(ST)$ . This proves  $\rho(TS) \setminus \{0\} \subseteq \rho(ST)$ ; by symmetry the two agree off 0, and so do their complements.  $\square$